Project 3 Report

ME 249

Developing and Applying Custom-Loss Physics-Informed Neural Networks for Steady State Heat Transfer Modeling

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Abstract: This project applies and extends the concept of custom loss functions within Physics-Informed Neural Networks (PINNs) to solve steady-state heat conduction problems. We first build familiarity with data preprocessing, normalization, and conventional neural-network training using photovoltaic system data. After constructing baseline models and evaluating their prediction accuracy, we transition to designing a PINN framework where a custom loss function enforces both temperature boundary conditions and the Laplacian residual across interior points. The loss integrates squared boundary errors, temperature-limit penalties, and PDE consistency through κ-weighted Laplacian terms.

Using this custom PINN, we trained models over rectangular and hexagonal domains, exploring the effects of network size and κ-values on convergence. The improved model architecture achieved a minimum loss of 0.00010—significantly outperforming the base configuration—while closely matching Gauss–Seidel finite-difference solutions. Higher κ values slowed convergence, and the hexagonal domain produced a higher minimum loss (0.03044), though its final temperature field remained visually consistent with the benchmark solution. Overall, the project demonstrates both conceptual and practical understanding of custom loss construction and highlights the effectiveness of PINNs in solving PDE-based heat-transfer problems.

This project was completed collaboratively by Md Elius and Parker Yang, with responsibilities shared across computation and report preparation. In both Part 1 and Part 2, Parker contributed to the initial data processing and setup, while the PINN model development, training, and extended computational runs were carried out on Elius's computing system and conda environment. The project report was prepared jointly, with each member contributing to the written sections and both participating in the final editing, formatting, and organization of the report.

Task 3.1.1

(a) We converted the original dataset by transforming air temperature from Celsius to Kelvin, calculated the median value of each input and output parameter, and normalized the full dataset by dividing every parameter value by its median.

(b) We randomly divided the normalized dataset into two groups: a training set containing 2/3 of the data and a validation set containing the remaining 1/3, ensuring an unbiased and randomized split.

(c) We inserted the normalized training data into the provided skeleton code and constructed a Keras Sequential neural network with:

a RandomUniform initializer,

an input layer of 6 ELU neurons,

three hidden layers with 6, 12, and 6 ELU neurons,

an output layer with 2 neurons,

and the RMSprop optimizer.
This fully matched the required architecture.

(d) We trained the model using backpropagation and early stopping. We experimented with training length and learning rate to improve convergence and achieved a mean absolute error close to 0.0222 for RMSprop 0.003.

(e) We evaluated the trained model on the training dataset, computed the mean absolute error, and generated a log–log plot (Figure 1) comparing predicted vs. actual power output, including a reference.

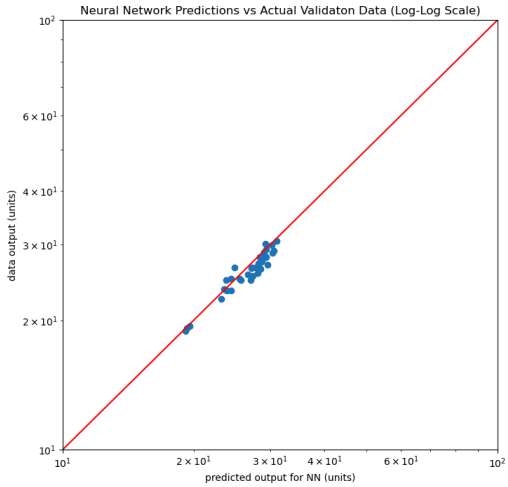


Figure 1. Predicted vs. Actual Power output (training data) log-log plot

(f) We repeated the evaluation for the validation dataset, reporting the mean absolute error and producing another log–log performance plot for comparison (Figure 2).

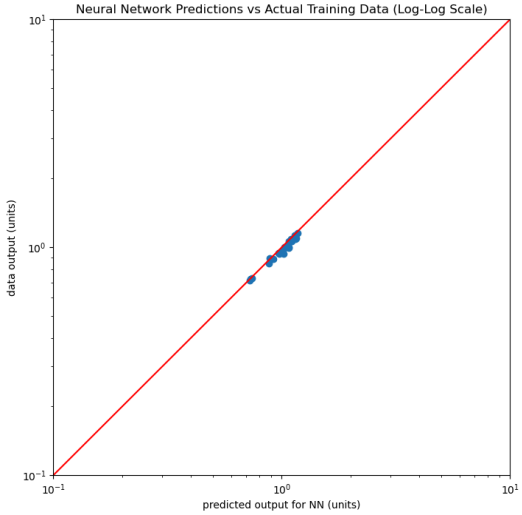


Figure 2. Predicted vs. Actual Power output (validation data) log-log plot

(g) With the air temperature fixed at 20°C, we used the trained model to predict power output across the ranges 4–8 Ω (load resistance) and 500–1200 W/m² (irradiance).

We generated a 3D surface plot, a contour plot (Figure 3), and summary statistics to visualize how solar power output varies with these two parameters.

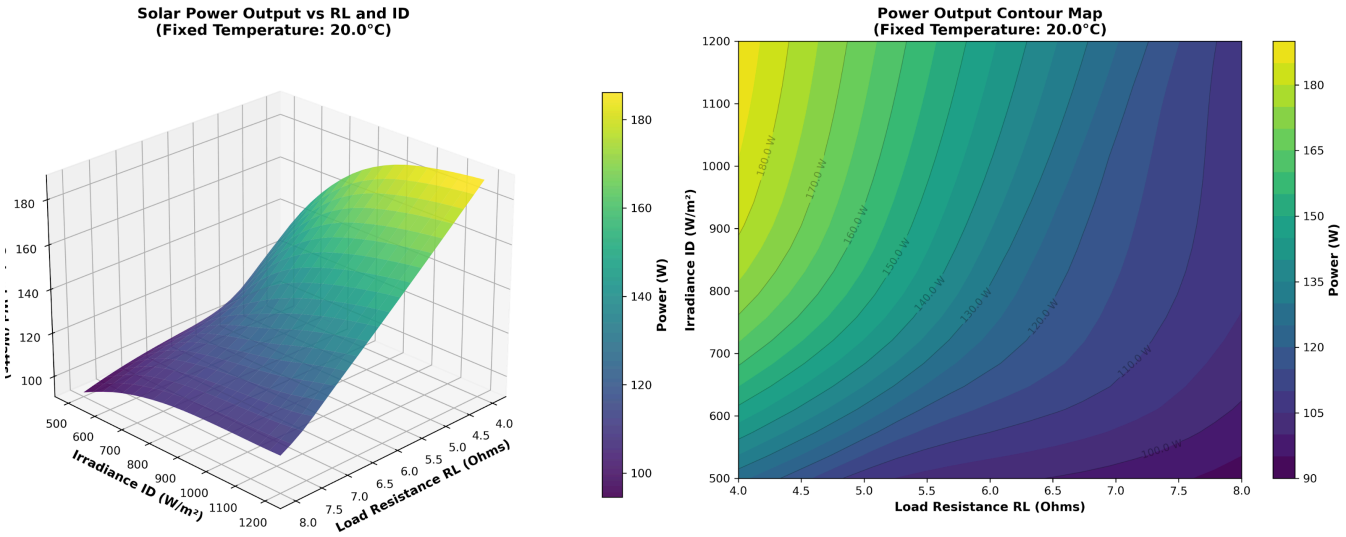


Figure 3. Solar Power output surface plot & contour map

Key observations were as followed:

Higher irradiance consistently increases power output, showing the expected dependence on available solar energy.

For any fixed irradiance level, power is highest at lower load resistance (near 4 Ω) and gradually decreases as RL increases.

The surface flattens slightly at high irradiance, indicating diminishing returns—power increases less rapidly as irradiance approaches 1200 W/m².

Overall, the model captures the realistic behavior that solar panels deliver more power at high irradiance and at load resistances closer to the panel's internal optimum.

Task 3.1.2

a) The instructed changes were made to the model shown in the Table 1 below.

b)

Loss	Loss at end of 6000 epochs	Best MSE of 6000 epochs	RMS Error = sqrt(best MSE)
MSE loss baseline	0.000481256283819 67545	8.137484110193327 e-05 (At 5777 epochs)	0.00902
Custom M4E Loss	Loss at end of 6000 epochs	Best M4E of 6000 epochs	r4s Error = sqrt(sqrt(best M4E))
	7.751622455032248 e-09	7.750137420714509 e-09 (At 5997 epochs)	0.00938

Table 1. Result from the trained models

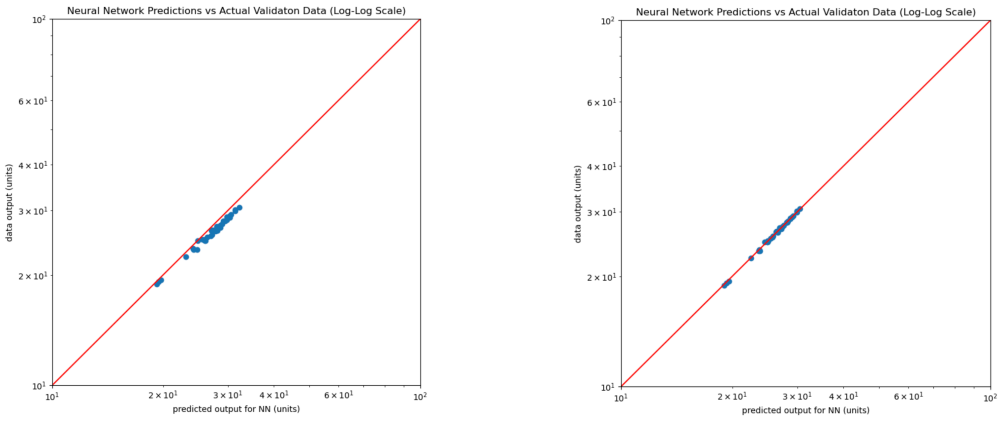


Figure 4. Log-log plot of predicted power output vs. data value power output

c) Provide an assessment of whether this loss function appears to offer any training advantages over a conventional MSE loss function

In this experiment, the custom 'Mean Double Squared Error' (M4E) loss function did not appear to offer training advantages over the conventional MSE loss function. The custom loss function did not provide an advantage in this run seeing that the conventional model (0.00902) has a slightly lower error than the custom model (0.00938).

Task 3.2.1

We completed the customization of the PINN loss function by adding all the required spatial points used for enforcing the boundary and interior physics constraints. First, we fully expanded the pointsBoundaryChk array to include all 32 boundary [x,y] coordinates specified in the assignment figure. Next, we completed the points array by adding all 45 interior grid points where the Laplace residual is evaluated. After verifying these arrays, we copied the entire updated custom loss function from CodeP3.3F25.ipynb and inserted it into CodeP3.4F25.ipynb at the designated location under the #CUSTOM LOSS FUNCTION HERE comment. Once installed, we executed the first four code cells and confirmed that the model successfully trained for 100 epochs, indicating that all required modifications were implemented correctly.

Task 3.2.2

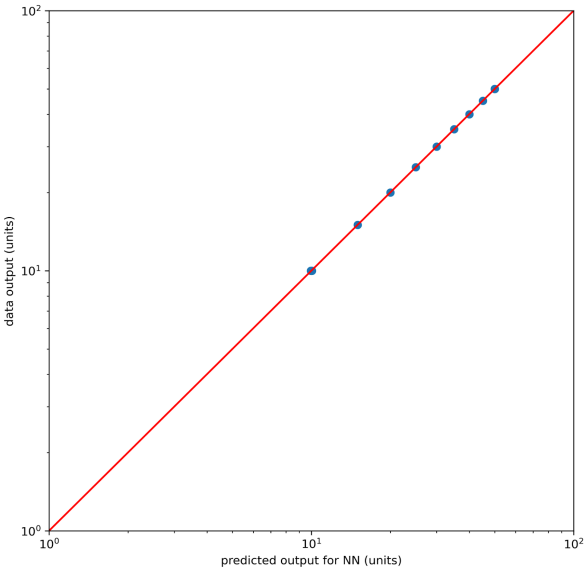
KAPPA = 1.3

Last 6 epochs are shown here from the base model run

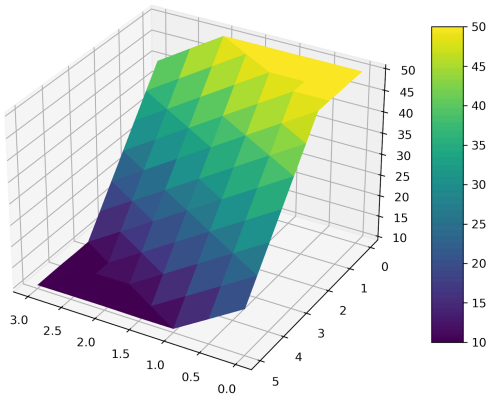
1/1 ——————————————————————— 0s 34ms/step - loss: 7.4115e-04
Epoch 99995/100000
1/1 ——————————————————————— 0s 32ms/step - loss: 7.8920e-04
Epoch 99996/100000
1/1 ——————————————————————— 0s 34ms/step - loss: 6.7390e-04
Epoch 99997/100000
1/1 ——————————————————————— 0s 34ms/step - loss: 4.9243e-04
Epoch 99998/100000
1/1 ——————————————————————— 0s 33ms/step - loss: 3.8249e-04
Epoch 99999/100000
1/1 ——————————————————————— 0s 34ms/step - loss: 4.0477e-04
Epoch 100000/100000
1/1 ——————————————————————— 0s 34ms/step - loss: 5.0481e-04

best epoch = 99954

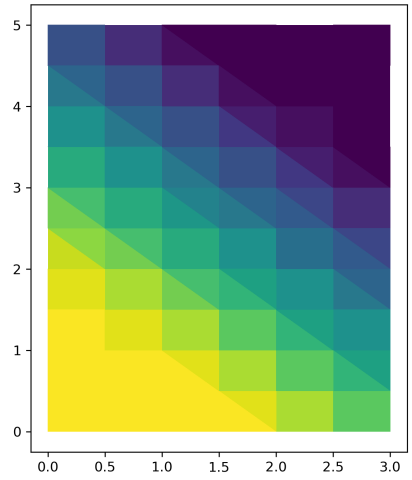
smallest loss = 0.00037596558104269207



Log-log plot

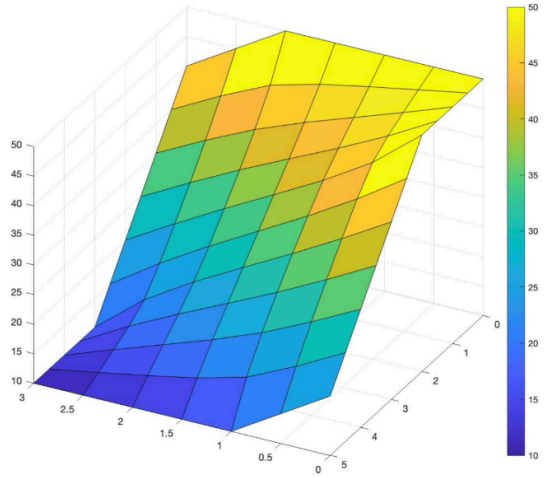
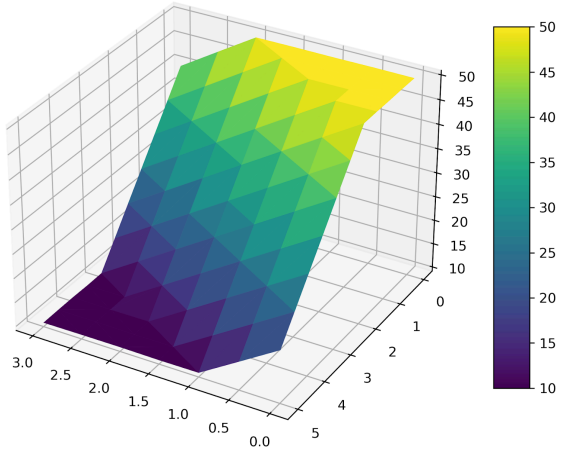


Surface plot



Heat map

We ran 100,000 epochs using a job script within our conda environment and achieved a minimum loss of 0.00037, which is well below the target value of 0.005. The resulting PINN temperature field shows excellent agreement with the Gauss–Seidel finite-difference solution, as illustrated in the figure below.

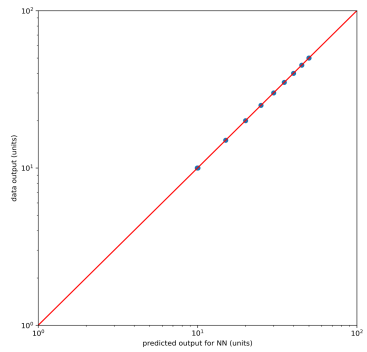
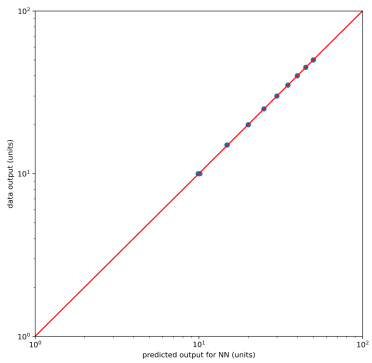
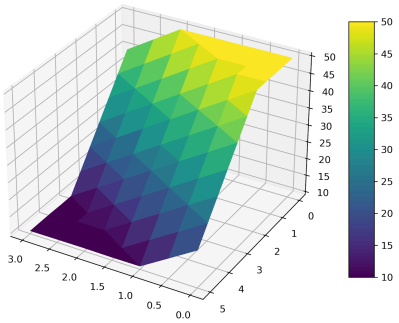
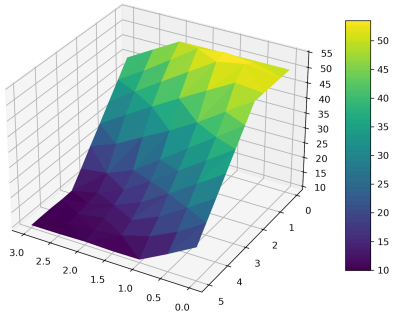
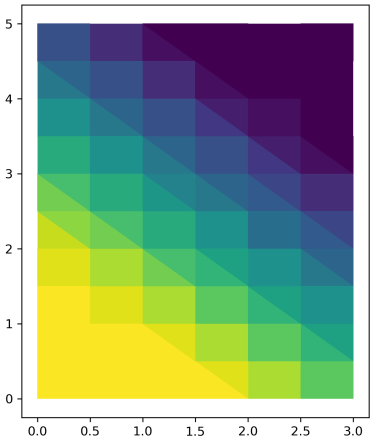
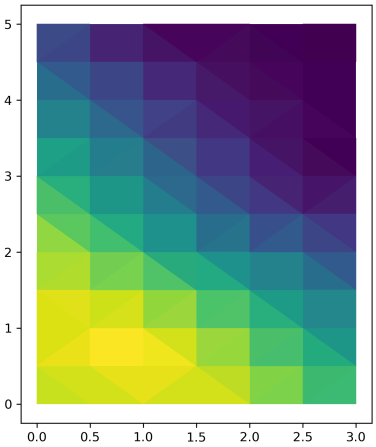
Gauss-Siedel finite-difference solution	PINN model solution
	

Task 3.2.3

KAPPA = 5.0

Last 6 epochs are shown here from the base model run for kappa = 5.0

Epoch 99995/100000
1/1 ——————————————————————— 0s 33ms/step - loss: 0.0030
Epoch 99996/100000
1/1 ——————————————————————— 0s 33ms/step - loss: 0.0030
Epoch 99997/100000
1/1 ——————————————————————— 0s 33ms/step - loss: 0.0030
Epoch 99998/100000
1/1 ——————————————————————— 0s 34ms/step - loss: 0.0031
Epoch 99999/100000
1/1 ——————————————————————— 0s 33ms/step - loss: 0.0032
Epoch 100000/100000
1/1 ——————————————————————— 0s 33ms/step - loss: 0.0033
best epoch = 99982
smallest loss = 0.0029064263217151165

	Kappa =1.3	Kappa =5.0
Log -log plot		
Surface plot		
Heat Map		

With κ = 5.0, the smallest loss we obtained was 0.00290, which is larger than the minimum loss achieved using κ = 1.3 (0.00037) under the same model configuration. This indicates that increasing κ slows the convergence of the solver. Accordingly, the heatmap and surface plot for κ = 1.3 show a more fully developed and accurate temperature field compared to those generated with κ = 5.0.

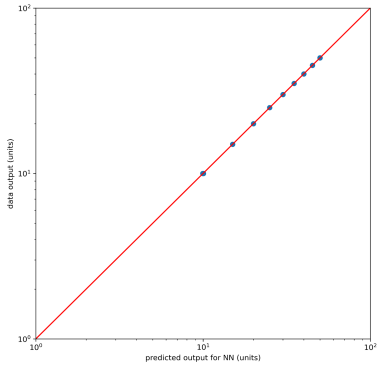
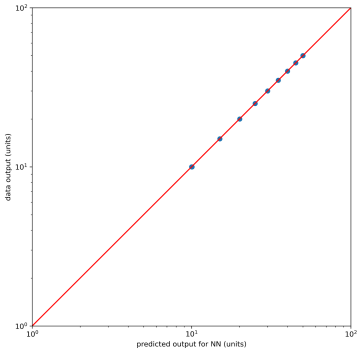
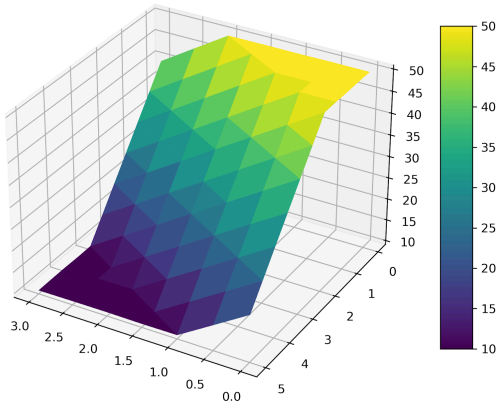
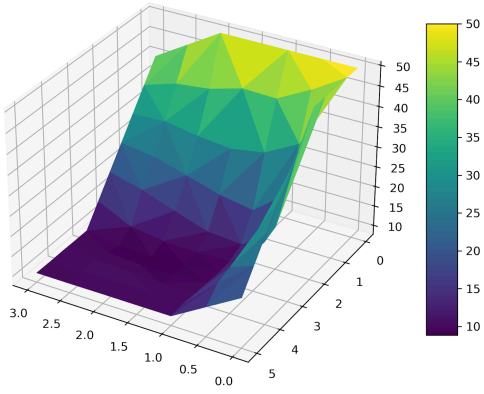
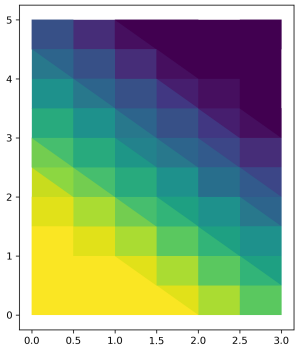
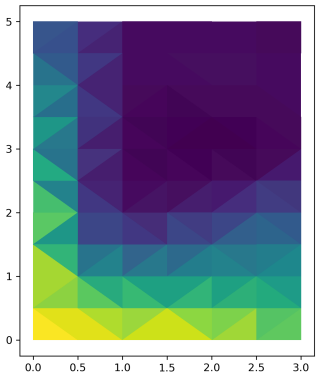
Task 3.2.4

Base model	Updated model
model = Sequential([Dense(5, input_dim=2, activation='elu'), Dense(10, activation='elu'), Dense(1)])	model = Sequential([Dense(8, input_dim=2, activation='elu'), Dense(16, activation='elu'), Dense(1) # Output temperature T

Last 6 epochs are shown here from the updated model run

```
Epoch 99995/100000
1/1 ——————————————————————— 0s 40ms/step - loss: 2.6422e-04
Epoch 99996/100000
1/1 ——————————————————————— 0s 39ms/step - loss: 3.5773e-04
Epoch 99997/100000
1/1 ——————————————————————— 0s 40ms/step - loss: 5.0253e-04
Epoch 99998/100000
1/1 ——————————————————————— 0s 41ms/step - loss: 7.2110e-04
Epoch 99999/100000
1/1 ——————————————————————— 0s 39ms/step - loss: 0.0010
Epoch 100000/100000
1/1 ——————————————————————— 0s 53ms/step - loss: 0.0014
best epoch =  99964
smallest loss = 0.00010163153638131917
```

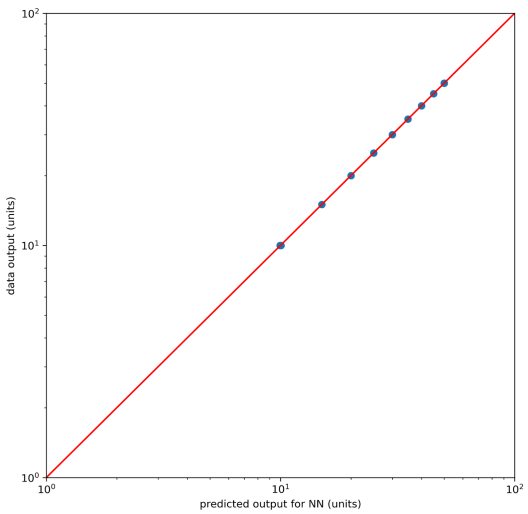
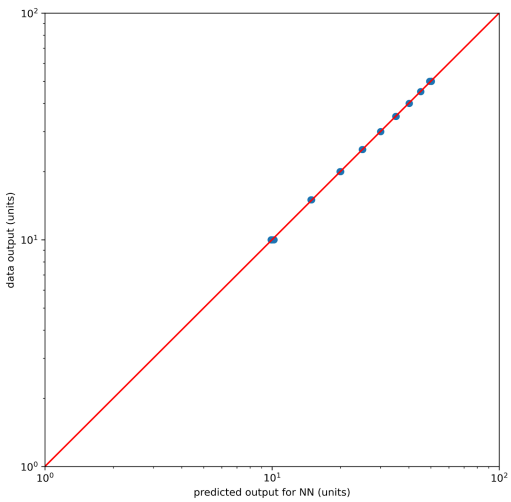
The updated model, which uses larger dense layers than the base configuration, achieved a minimum loss of 0.00010163, compared to 0.00037596 for the base model. Although the log–log plots for both models appear nearly identical, the differences become clear in the surface and heat-map visualizations. The updated model produces sharper gradients and more detailed structure across both the surface plot and the heat-map contours, reflecting its improved accuracy.

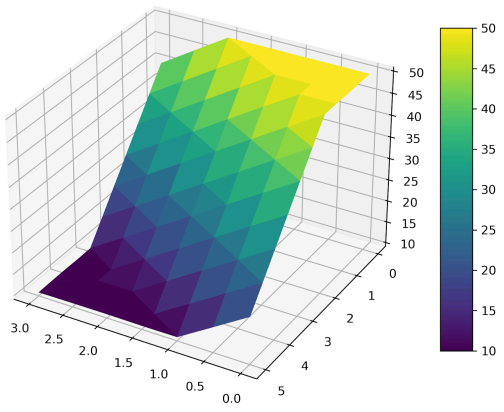
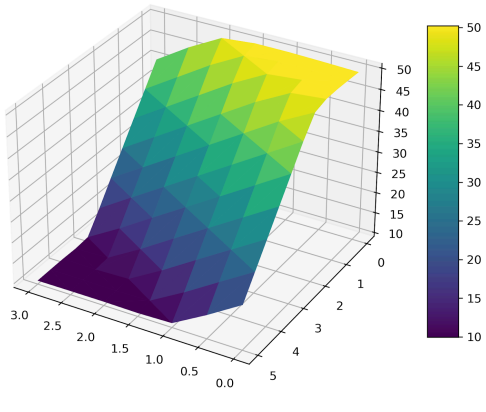
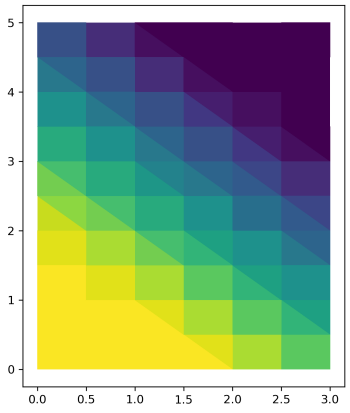
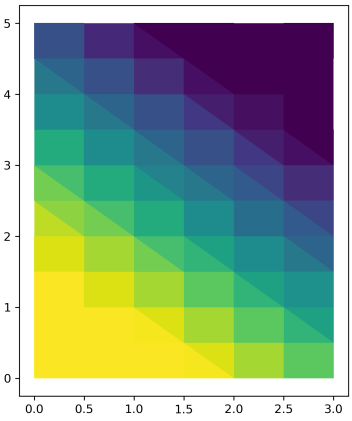
	Base model	Updated model
Log-log plot		
Surface Plot		
Heat Map		

Task 3.2.5

Last 6 epochs are shown here from the hexagonal model run

Epoch 99996/100000
2/2 ——————————————————————————— 0s 19ms/step - loss: 0.0327
Epoch 99997/100000
2/2 ——————————————————————————— 0s 19ms/step - loss: 0.0347
Epoch 99998/100000
2/2 ——————————————————————————— 0s 20ms/step - loss: 0.0337
Epoch 99999/100000
2/2 ——————————————————————————— 0s 19ms/step - loss: 0.0337
Epoch 100000/100000
2/2 ——————————————————————————— 0s 19ms/step - loss: 0.0340
best epoch = 99945
smallest loss = 0.03044164553284645

	Rectangular domain	Hexagonal domain
Log-log plot		

Surface plot		
Heat map		

For the hexagonal domain, the smallest loss we obtained was 0.03044, which is above the target threshold of 0.005, indicating that convergence is significantly slower compared to the rectangular domain since we used the same model run configuration. Despite the slower convergence, the resulting temperature fields for both domains are nearly identical.

In summary, we explored the construction and application of custom-loss Physics-Informed Neural Networks to model steady-state heat conduction. After developing familiarity with neural-network training and normalization through a preliminary PV power prediction task, we implemented a PINN framework that enforces both boundary conditions and the Laplacian residual through a tailored loss function. The improved network architecture achieved strong accuracy, with losses as low as 0.00010 and temperature fields that closely matched Gauss–Seidel finite-difference solutions. Although higher κ values and the hexagonal domain slowed convergence, the predicted temperature fields remained consistent with the benchmark solution. Overall, the project demonstrated the effectiveness of custom-loss PINNs in solving PDE-based heat transfer problems and highlighted how model structure and loss weighting influence convergence and solution quality.

Appendix:

Job script:

```
#!/bin/bash
#SBATCH --job-name=PINN
#SBATCH --partition=savio3
#SBATCH --qos=aiolos_savio3_normal
#SBATCH --account=co_aiolos
#SBATCH --nodes=2
#SBATCH --ntasks-per-node=40
#SBATCH --time=03:59:59
#SBATCH --mail-type=all
#SBATCH --mail-user=m_elius@berkeley.edu
#SBATCH --output=job_%j.out
#SBATCH --error=job_%j.err

# 1) Start clean and go to your project dir
module purge
cd /global/scratch/users/melius/Fall_2025/ME249_Machine_Learning/me249/prj3

# 2) Activate your conda env
source ~/.bashrc
conda activate my_kernel

# 3) Ensure Conda's libstdc++ is found before system one (fixes GLIBCXX errors)
export LD_LIBRARY_PATH="${CONDA_PREFIX}/lib:${LD_LIBRARY_PATH}"

# 4) Avoid accidental Python contamination + oversubscription
unset PYTHONHOME PYTHONPATH
export PYTHONNOUSERSITE=1
export OMP_NUM_THREADS=1
export OPENBLAS_NUM_THREADS=1
export MKL_NUM_THREADS=1
export NUMEXPR_NUM_THREADS=1

# 5) (Optional) Convert notebook to script
jupyter nbconvert --to script CodeP3.4F25_hexagonal_shape.ipynb

# 6) (Optional) quick diagnostics to the job log
```

```
python - <<'PY'
import sys, subprocess
print("Python:", sys.version)
try:
    import pandas._libs.window.aggregations as m
    print("Pandas ext:", m.__file__)
    subprocess.run(["bash","-lc", f"ldd {m.__file__} | grep libstdc++ || true"])
except Exception as e:
    print("Pandas import test failed:", e)
PY

# 7) Run your code
python CodeP3.4F25_hexagonal_shape.py
```

In [12]:
```
# CodeP3.4F25 – V.P. Carey, ME249 F25

#import useful packages
import keras
import pandas as pd
from keras.models import Sequential
import numpy as np
import keras.backend as kb
import tensorflow as tf
#the following 2 lines are only needed for Mac OS machines
import os
os.environ['KMP_DUPLICATE_LIB_OK']='True'

#create input data array

#convert to:
# BC's for PINN conduction model with HE enforcement in interior points

# x(cm), y(cm)   These are (x,y) coordinates of boundary location where temp
xdata = []
ND = 32
xdata =  [  [ 0.0, 0.0 ], [ 0.0, 0.5 ], [ 0.0, 1.0 ], [ 0.0, 1.5 ],[ 0.0, 2.
            [ 0.0, 3.5 ], [ 0.0, 4.0 ], [ 0.0, 4.5 ], [ 0.0, 5.0 ],
            [ 3.0, 0.0 ], [ 3.0, 0.5 ], [ 3.0, 1.0 ], [ 3.0, 1.5 ],[ 3.0, 2.
            [ 3.0, 3.5 ], [ 3.0, 4.0 ], [ 3.0, 4.5 ], [ 3.0, 5.0 ]]

xdata.append([  0.5, 0.0 ])
xdata.append([  1.0, 0.0 ])
xdata.append([  1.5, 0.0 ])
xdata.append([  2.0, 0.0 ])
xdata.append([  2.5, 0.0 ])

xdata.append([  0.5, 5.0 ])
xdata.append([  1.0, 5.0 ])
xdata.append([  1.5, 5.0 ])
xdata.append([  2.0, 5.0 ])
xdata.append([  2.5, 5.0 ])


# add more here if needed



xarray= np.array(xdata)
print (xdata)
print (xarray)



#convert to:
# meadian values of output variables

# T (deg C)   These are specified (measured) temperatures for each pair of (
```

```
ydata = []

ydata = [ [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 45. ], [ 40. ], [ 3
         [ 40. ], [ 35. ],  [ 30. ], [ 25. ], [ 20. ], [ 15. ], [ 10. ], [ 1

ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 45. ])

ydata.append([ 15. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])

# add more here if needed

yarray= np.array(ydata)
print (ydata)
print (yarray)

data_inputs = np.array(xdata)
data_outputs = np.array(ydata)

#boundary_x = ...  # x-coordinates for the boundary data
#boundary_y = ...  # y-coordinates for the boundary data
#boundary_T = ...  # Boundary temperature data (T_data)

# Copy the first column to a 1D numpy array
#vector = arr[:, 0].copy()
boundary_xnp = xarray[:, 0].copy()
# Copy the first column to a 1D numpy array
boundary_ynp = xarray[:, 1].copy()
boundary_Tnp = yarray[:, 0].copy()

# Convert the arrays to a TensorFlow tensors
boundary_x = tf.convert_to_tensor(boundary_xnp, dtype=tf.float32)
boundary_y = tf.convert_to_tensor(boundary_ynp, dtype=tf.float32)
boundary_T_ytrain = tf.convert_to_tensor(boundary_Tnp, dtype=tf.float32)

boundary_points_xtrain = tf.stack([boundary_x, boundary_y], axis=1)

print(boundary_points_xtrain)
print(boundary_T_ytrain)
```

```
[[0.0, 0.0], [0.0, 0.5], [0.0, 1.0], [0.0, 1.5], [0.0, 2.0], [0.0, 2.5], [0.
0, 3.0], [0.0, 3.5], [0.0, 4.0], [0.0, 4.5], [0.0, 5.0], [3.0, 0.0], [3.0,
0.5], [3.0, 1.0], [3.0, 1.5], [3.0, 2.0], [3.0, 2.5], [3.0, 3.0], [3.0, 3.
5], [3.0, 4.0], [3.0, 4.5], [3.0, 5.0], [0.5, 0.0], [1.0, 0.0], [1.5, 0.0],
[2.0, 0.0], [2.5, 0.0], [0.5, 5.0], [1.0, 5.0], [1.5, 5.0], [2.0, 5.0], [2.
5, 5.0]]
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]]
[[50.0], [50.0], [50.0], [50.0], [50.0], [45.0], [40.0], [35.0], [30.0], [2
5.0], [20.0], [40.0], [35.0], [30.0], [25.0], [20.0], [15.0], [10.0], [10.
0], [10.0], [10.0], [10.0], [50.0], [50.0], [50.0], [50.0], [45.0], [15.0],
[10.0], [10.0], [10.0], [10.0]]
[[50.]
 [50.]
 [50.]
 [50.]
 [50.]
 [45.]
 [40.]
 [35.]
 [30.]
 [25.]
 [20.]
 [40.]
 [35.]
 [30.]
```

```
 [25.]
 [20.]
 [15.]
 [10.]
 [10.]
 [10.]
 [10.]
 [10.]
 [50.]
 [50.]
 [50.]
 [50.]
 [45.]
 [15.]
 [10.]
 [10.]
 [10.]
 [10.]]
tf.Tensor(
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]], shape=(32, 2), dtype=float32)
tf.Tensor(
[50. 50. 50. 50. 50. 45. 40. 35. 30. 25. 20. 40. 35. 30. 25. 20. 15. 10.
 10. 10. 10. 10. 50. 50. 50. 50. 45. 15. 10. 10. 10. 10.], shape=(32,), dtyp
e=float32)
```

In [13]:
```
# define neural network model

    #As seen below, we have created four dense layers.
    #A dense layer is a layer in neural network that's fully connected.
    #In other words, all the neurons in one layer are connected to all other
    #In the first layer, we need to provide the input shape, which is 1 in c
    #The activation function we have chosen is elu, which stands for exponer

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Create your model (for example, a simple fully connected feedforward NN)
# Assume 2D input (x, y),  Output temperature T
model = Sequential([
        Dense(5, input_dim=2, activation='elu'),
        Dense(10, activation='elu'),
        Dense(1)
    ])

# Set the kappa value (this can be adjusted as per your needs)
kappa = 1.3

# Set the x_max and y_max values of your domain ( can adjust these according
x_max = 3.0
y_max = 5.0

#print summary of model features
model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Par
dense_3 (Dense)	(None, 5)	
dense_4 (Dense)	(None, 10)	
dense_5 (Dense)	(None, 1)	

Total params: 86 (344.00 B)

Trainable params: 86 (344.00 B)

Non-trainable params: 0 (0.00 B)

In [14]:
```
# ADD CODE HERE for custom_loss_function <<========
# PINN CUSTOM LOSS FUNCTION - V.P. Carey, ME249 F25
#import useful packages

import tensorflow as tf
from tensorflow.keras import backend as K


def custom_loss_function(kappa, x_max, y_max, epoch_tracker):
    """ Custom loss function for the neural network model.
        - kappa: weight for the Laplace residual term.
```

```
        - x_max, y_max: maximum x and y coordinates of the domain.
        - epoch_tracker: callback object to keep track of the current epoch.

    def loss(y_true, y_pred):
        # Define pointsBoundaryChk as a constant tensor
        # === COMPLETE 32 BOUNDARY POINTS FROM THE PROJECT FIGURE ===
        pointsBoundaryChk = tf.constant([
            # Bottom edge (y = 0)
            [0.0, 0.0],
            [0.5, 0.0],
            [1.0, 0.0],
            [1.5, 0.0],
            [2.0, 0.0],
            [2.5, 0.0],
            [3.0, 0.0],

            # Right edge (x = 3)
            [3.0, 0.5],
            [3.0, 1.0],
            [3.0, 1.5],
            [3.0, 2.0],
            [3.0, 2.5],
            [3.0, 3.0],
            [3.0, 3.5],
            [3.0, 4.0],
            [3.0, 4.5],
            [3.0, 5.0],

            # Top edge (y = 5)
            [0.5, 5.0],
            [1.0, 5.0],
            [1.5, 5.0],
            [2.0, 5.0],
            [2.5, 5.0],
            [3.0, 5.0],

            # Left edge (x = 0)
            [0.0, 0.5],
            [0.0, 1.0],
            [0.0, 1.5],
            [0.0, 2.0],
            [0.0, 2.5],
            [0.0, 3.0],
            [0.0, 3.5],
            [0.0, 4.0],
            [0.0, 4.5],
            [0.0, 5.0]

        ], dtype=tf.float32)


        # Boundary loss: Calculate squared error for the boundary condition
        boundary_loss = K.mean(K.square(y_pred - y_true))

        ESbdsum = 0.
        for _ in range(32):
```

```
            # Sample a random index and gather point
            idx = tf.random.uniform(shape=[], minval=0, maxval=32, dtype=tf.
            selected_point = tf.gather(pointsBoundaryChk, idx)

            # Stack selected_point as input for the model (shape becomes (1,
            input_tensor = tf.expand_dims(selected_point, axis=0)
            # Pass input_tensor as the input to the model
            T_pred_at_boundxy = model(input_tensor)  # Model prediction
            # add penalty if temperature at flux location is OUTSIDE RANGE 5
        #     if (T_pred_at_boundxy < 5.):
        #         ESbdsum += 20.*(5. - T_pred_at_boundxy)*(5. - T_pred_at_bo
        #     if (T_pred_at_boundxy > 55.):
        #         ESbdsum += 20.*(T_pred_at_boundxy - 55.)*(T_pred_at_boundx
        # boundary_loss += ESbdsum/32.
        # Compute lower and upper penalties without Python if-statements
            low_penalty  = tf.nn.relu(5.0  - T_pred_at_boundxy)
            high_penalty = tf.nn.relu(T_pred_at_boundxy - 55.0)

            # Square and scale (20×), then average over 32 samples
            ESbdsum += (20.0 * tf.square(low_penalty)  +
                        20.0 * tf.square(high_penalty)) / 32.0

        boundary_loss += ESbdsum


        # Check epoch from the callback
        epoch = epoch_tracker.epochs_trained

        # If current epoch is less than N_epInit, ignore the Laplace residua
        if epoch < epoch_tracker.N_epInit:
            total_loss = boundary_loss
        else:
            # 3. Randomly select an interior point from this list using Tens
            points = tf.constant([
                # x = 0.5
                [0.5, 0.5],
                [0.5, 1.0],
                [0.5, 1.5],
                [0.5, 2.0],
                [0.5, 2.5],
                [0.5, 3.0],
                [0.5, 3.5],
                [0.5, 4.0],
                [0.5, 4.5],

                # x = 1.0
                [1.0, 0.5],
                [1.0, 1.0],
                [1.0, 1.5],
                [1.0, 2.0],
                [1.0, 2.5],
                [1.0, 3.0],
                [1.0, 3.5],
                [1.0, 4.0],
                [1.0, 4.5],
```

```
            # x = 1.5
            [1.5, 0.5],
            [1.5, 1.0],
            [1.5, 1.5],
            [1.5, 2.0],
            [1.5, 2.5],
            [1.5, 3.0],
            [1.5, 3.5],
            [1.5, 4.0],
            [1.5, 4.5],

            # x = 2.0
            [2.0, 0.5],
            [2.0, 1.0],
            [2.0, 1.5],
            [2.0, 2.0],
            [2.0, 2.5],
            [2.0, 3.0],
            [2.0, 3.5],
            [2.0, 4.0],
            [2.0, 4.5],

            # x = 2.5
            [2.5, 0.5],
            [2.5, 1.0],
            [2.5, 1.5],
            [2.5, 2.0],
            [2.5, 2.5],
            [2.5, 3.0],
            [2.5, 3.5],
            [2.5, 4.0],
            [2.5, 4.5]
        ], dtype=tf.float32)


        meanLap_res = 0.
        meanTlimPen = 0.
        for _ in range(45):
            ##compute Laplacian residue here - need to do this once per
            # Use tf.random.uniform to sample an index for the interior
            idx = tf.random.uniform(shape=[], minval=0, maxval=45, dtype
            selected_point = tf.gather(points, idx)
            x_interior, y_interior = selected_point[0], selected_point[1

            # Use tf.GradientTape to compute second derivatives
            with tf.GradientTape(persistent=True) as tape:
                tape.watch(selected_point)
                # Pass the selected (x_interior, y_interior) to the mode
                T_pred_at_interior = model([x_interior, y_interior])  #

            # Compute first derivatives with respect to x and y
            dT_dx = tape.gradient(T_pred_at_interior, selected_point[0])
            dT_dy = tape.gradient(T_pred_at_interior, selected_point[1])

            # Compute second derivatives (Hessian)
            d2T_dx2 = tape.gradient(dT_dx, selected_point[0])
```

```
                d2T_dy2 = tape.gradient(dT_dy, selected_point[1])

                # Calculate the Laplacian (sum of second derivatives)
                laplacian = d2T_dx2 + d2T_dy2

                # Compute the Laplacian residual (sum of squared Laplacian)
                meanLap_res +=  K.square(laplacian)/45.

                # ADD PENALTY IF T below 5 or above 55

                # if (T_pred_at_interior < 5.):
                #     meanTlimPen += 20.*(5. - T_pred_at_interior)*(5. - T_p
                low_penalty = tf.nn.relu(5.0 - T_pred_at_interior)
                meanTlimPen += 20.0 * tf.square(low_penalty) / 45.0

                # if (T_pred_at_interior > 55.):
                #     meanTlimPen += 20.*(T_pred_at_interior - 55.)*(T_pred_
                high_penalty = tf.nn.relu(T_pred_at_interior - 55.0)
                meanTlimPen += 20.0 * tf.square(high_penalty) / 45.0


            #Combine three losses
            total_loss = boundary_loss + kappa * meanLap_res + meanTlimPen


        return total_loss
    return loss


from tensorflow.keras.callbacks import Callback

class EpochTracker(Callback):
    def __init__(self, N_epInit):
        super().__init__()
        self.N_epInit = N_epInit
        self.epochs_trained = 0

    def on_epoch_end(self, epoch, logs=None):
        self.epochs_trained = epoch

# Set the loss function with the required arguments
epoch_tracker = EpochTracker(N_epInit=2)
custom_loss = custom_loss_function(kappa=1.3, x_max=3.0, y_max=5.0, epoch_tr



# Compile the model with the custom loss
model.compile(optimizer='adam', loss=custom_loss)
```

In [15]:

```
#### After the compilation of the model, we'll try the fit method with 100 e


#The fit method takes three parameters; namely, x, y, and number of epochs.
#During model training, if all the batches of data are seen by the model onc
#we say that one epoch has been completed.
```

```
##### RUN FIT

# Assuming `boundary_points` and `boundary_T` are TensorFlow tensors

# Now, fit the model and pass the callback to `fit()`
historyData = model.fit(boundary_points_xtrain, boundary_T_ytrain,
                epochs=100000, batch_size=32,
                callbacks=[epoch_tracker])


loss_hist = historyData.history['loss']
#The above line will return a dictionary, access it's info like this:
best_epoch = np.argmin(historyData.history['loss']) + 1
print ('best epoch = ', best_epoch)
print('smallest loss =', np.min(loss_hist))

predictions = model.predict(data_inputs)
```

```
Epoch 1/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 4s 4s/step - loss: 1500.7461
Epoch 2/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 1499.0411
Epoch 3/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1499.8611
Epoch 4/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1496.0266
Epoch 5/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1487.0325
Epoch 6/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1484.7662
Epoch 7/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1480.0608
Epoch 8/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1479.1167
Epoch 9/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1473.4717
Epoch 10/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1454.7317
Epoch 11/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1469.1222
Epoch 12/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1476.0132
Epoch 13/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1461.0378
Epoch 14/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1459.0745
Epoch 15/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1475.5704
Epoch 16/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1463.8936
Epoch 17/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1463.6436
Epoch 18/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1439.9554
Epoch 19/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1464.2194
Epoch 20/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1428.5271
Epoch 21/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1451.6284
Epoch 22/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1454.3440
Epoch 23/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1432.4111
Epoch 24/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1426.4358
Epoch 25/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1432.4463
Epoch 26/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1437.8485
Epoch 27/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1442.1594
Epoch 28/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1418.2493
```

```
Epoch 29/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 1430.5325
Epoch 30/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 1388.9943
Epoch 31/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 1417.4482
Epoch 32/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 1397.8289
Epoch 33/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1411.0767
Epoch 34/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1409.6704
Epoch 35/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1414.3694
Epoch 36/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1372.2405
Epoch 37/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1380.0588
Epoch 38/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1404.5376
Epoch 39/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1385.7283
Epoch 40/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1383.5667
Epoch 41/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 1367.4462
Epoch 42/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1371.3159
Epoch 43/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1356.2915
Epoch 44/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 1358.1857
Epoch 45/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1364.7595
Epoch 46/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1374.4700
Epoch 47/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1372.9158
Epoch 48/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1382.7773
Epoch 49/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1367.8030
Epoch 50/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1324.8368
Epoch 51/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1375.6770
Epoch 52/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1352.9122
Epoch 53/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1358.2800
Epoch 54/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1330.8079
Epoch 55/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1323.2977
Epoch 56/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1336.7438
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Epoch 57/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1342.9265
Epoch 58/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1299.9315
Epoch 59/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1312.2039
Epoch 60/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1329.1584
Epoch 61/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1312.8452
Epoch 62/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1304.6062
Epoch 63/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 1308.8044
Epoch 64/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1304.8196
Epoch 65/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1303.3358
Epoch 66/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1263.6725
Epoch 67/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1300.6599
Epoch 68/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1308.2252
Epoch 69/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1302.0278
Epoch 70/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1288.2482
Epoch 71/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1272.7512
Epoch 72/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1288.7834
Epoch 73/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1253.8549
Epoch 74/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1261.3105
Epoch 75/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1283.0918
Epoch 76/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1264.8469
Epoch 77/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1291.8811
Epoch 78/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1237.5425
Epoch 79/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1257.1169
Epoch 80/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1251.4877
Epoch 81/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1242.5552
Epoch 82/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1249.0347
Epoch 83/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1256.2095
Epoch 84/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1232.2440
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Epoch 85/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1232.4360
Epoch 86/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1247.3909
Epoch 87/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1233.0892
Epoch 88/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1258.6860
Epoch 89/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1247.1313
Epoch 90/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1226.5863
Epoch 91/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1216.2134
Epoch 92/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1206.3787
Epoch 93/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1202.2118
Epoch 94/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1229.5632
Epoch 95/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1229.0355
Epoch 96/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1189.6389
Epoch 97/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1217.2141
Epoch 98/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1213.7512
Epoch 99/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1195.8973
Epoch 100/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1195.4742
Epoch 101/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1208.6819
Epoch 102/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1175.9980
Epoch 103/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1200.4410
Epoch 104/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1173.1116
Epoch 105/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1165.9292
Epoch 106/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1182.0848
Epoch 107/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1185.6436
Epoch 108/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1142.4716
Epoch 109/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1152.9205
Epoch 110/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1175.7183
Epoch 111/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1163.7826
Epoch 112/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1125.5383
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Epoch 113/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1164.0901
Epoch 114/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1160.6774
Epoch 115/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1167.1416
Epoch 116/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1144.2863
Epoch 117/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1134.4907
Epoch 118/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1146.0315
Epoch 119/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1132.0085
Epoch 120/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1139.7921
Epoch 121/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1126.1528
Epoch 122/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1128.5573
Epoch 123/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1139.5410
Epoch 124/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1118.6935
Epoch 125/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1141.4442
Epoch 126/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1116.9133
Epoch 127/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 1117.2922
Epoch 128/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1120.6200
Epoch 129/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1131.6602
Epoch 130/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1119.4896
Epoch 131/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1137.6118
Epoch 132/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1114.8381
Epoch 133/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1098.9033
Epoch 134/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1099.1873
Epoch 135/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1091.2781
Epoch 136/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1118.2223
Epoch 137/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1088.8545
Epoch 138/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1089.8395
Epoch 139/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1090.0481
Epoch 140/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1083.0170
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Epoch 141/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step - loss: 1079.7996
Epoch 142/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1102.4868
Epoch 143/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1083.8929
Epoch 144/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1078.7535
Epoch 145/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 1084.9183
Epoch 146/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1094.0333
Epoch 147/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1064.5291
Epoch 148/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1062.6932
Epoch 149/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1066.8252
Epoch 150/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1061.9390
Epoch 151/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1074.9540
Epoch 152/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1048.8093
Epoch 153/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1049.4036
Epoch 154/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1058.4480
Epoch 155/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1074.6144
Epoch 156/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1040.6067
Epoch 157/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1027.7787
Epoch 158/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1042.3403
Epoch 159/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1040.5945
Epoch 160/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1061.1158
Epoch 161/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1028.4922
Epoch 162/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 1031.9387
Epoch 163/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 1044.4210
Epoch 164/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1038.2565
Epoch 165/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1012.9929
Epoch 166/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1011.3938
Epoch 167/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1023.9899
Epoch 168/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1036.6367
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Epoch 169/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1030.8785
Epoch 170/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1026.5430
Epoch 171/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1022.9033
Epoch 172/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1015.6661
Epoch 173/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1043.0647
Epoch 174/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1024.4377
Epoch 175/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1024.4976
Epoch 176/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1025.6653
Epoch 177/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1024.6459
Epoch 178/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1017.9253
Epoch 179/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1004.0917
Epoch 180/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1004.5266
Epoch 181/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1002.0059
Epoch 182/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1002.7405
Epoch 183/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1023.9153
Epoch 184/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1004.3701
Epoch 185/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 986.8263
Epoch 186/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 1008.0667
Epoch 187/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 991.0884
Epoch 188/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1003.6035
Epoch 189/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1008.2393
Epoch 190/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1000.2707
Epoch 191/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 986.7160
Epoch 192/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 976.8127
Epoch 193/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1000.2024
Epoch 194/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 991.6771
Epoch 195/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 1001.5352
Epoch 196/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 975.2083
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Epoch 197/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 978.4572
Epoch 198/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 978.2898
Epoch 199/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 980.5406
Epoch 200/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 975.1281
Epoch 201/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 979.4527
Epoch 202/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 974.1371
Epoch 203/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 968.7694
Epoch 204/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 966.6357
Epoch 205/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 976.8928
Epoch 206/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 972.8111
Epoch 207/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 959.1162
Epoch 208/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 967.9368
Epoch 209/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 961.5198
Epoch 210/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 976.7350
Epoch 211/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 963.2534
Epoch 212/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 967.2407
Epoch 213/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 966.9256
Epoch 214/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 958.2469
Epoch 215/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 956.2304
Epoch 216/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 961.2422
Epoch 217/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 963.6641
Epoch 218/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 963.2695
Epoch 219/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 963.7019
Epoch 220/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 951.5724
Epoch 221/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 973.7245
Epoch 222/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 954.6342
Epoch 223/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 950.8190
Epoch 224/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 941.0992
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Epoch 225/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 943.0041
Epoch 226/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 950.1320
Epoch 227/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 946.6165
Epoch 228/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 937.4487
Epoch 229/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 945.4646
Epoch 230/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 948.8493
Epoch 231/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 936.3728
Epoch 232/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 936.7831
Epoch 233/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 936.6689
Epoch 234/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 937.0267
Epoch 235/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 933.1567
Epoch 236/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 928.8753
Epoch 237/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 933.4943
Epoch 238/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 930.0409
Epoch 239/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 938.8946
Epoch 240/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 944.1274
Epoch 241/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 933.0881
Epoch 242/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 938.1753
Epoch 243/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 926.8564
Epoch 244/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 923.9096
Epoch 245/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 925.2552
Epoch 246/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 928.3380
Epoch 247/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 919.2562
Epoch 248/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 929.6661
Epoch 249/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 925.5399
Epoch 250/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 924.6979
Epoch 251/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 916.5624
Epoch 252/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 927.8597
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Epoch 253/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 913.1513
Epoch 254/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 925.0112
Epoch 255/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 924.1938
Epoch 256/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 910.5848
Epoch 257/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 918.0220
Epoch 258/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 916.4753
Epoch 259/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 910.3014
Epoch 260/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 914.8324
Epoch 261/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 913.5532
Epoch 262/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 912.2914
Epoch 263/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 905.7420
Epoch 264/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 913.6880
Epoch 265/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 904.8145
Epoch 266/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 905.8950
Epoch 267/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 907.7665
Epoch 268/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 902.1814
Epoch 269/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 902.0769
Epoch 270/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 902.1316
Epoch 271/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 903.0184
Epoch 272/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 897.3724
Epoch 273/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 899.3828
Epoch 274/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 899.5548
Epoch 275/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 897.3447
Epoch 276/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 901.4561
Epoch 277/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 893.8257
Epoch 278/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 895.7557
Epoch 279/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 893.0919
Epoch 280/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 889.2241
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Epoch 281/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 892.9592
Epoch 282/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 887.5957
Epoch 283/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 886.8444
Epoch 284/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 893.5035
Epoch 285/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 890.4227
Epoch 286/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 884.6595
Epoch 287/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 881.6169
Epoch 288/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 891.8135
Epoch 289/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 883.6171
Epoch 290/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 880.5347
Epoch 291/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 881.4591
Epoch 292/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 881.3051
Epoch 293/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 885.8071
Epoch 294/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 876.2851
Epoch 295/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 879.4350
Epoch 296/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 876.9474
Epoch 297/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 876.1816
Epoch 298/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 871.8207
Epoch 299/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 871.0399
Epoch 300/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 874.1937
Epoch 301/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 872.5822
Epoch 302/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 875.4304
Epoch 303/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 870.9485
Epoch 304/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 867.2106
Epoch 305/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 873.5519
Epoch 306/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 866.1494
Epoch 307/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 866.0328
Epoch 308/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 869.4796
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Epoch 309/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 867.3774
Epoch 310/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 868.5096
Epoch 311/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 864.7597
Epoch 312/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 861.7514
Epoch 313/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 861.8940
Epoch 314/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 858.3054
Epoch 315/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 859.8442
Epoch 316/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 856.6643
Epoch 317/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 855.0517
Epoch 318/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 854.2534
Epoch 319/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 857.8513
Epoch 320/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 853.2035
Epoch 321/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 855.7781
Epoch 322/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 853.2173
Epoch 323/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 854.6541
Epoch 324/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 852.7678
Epoch 325/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 853.6296
Epoch 326/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 849.4871
Epoch 327/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 848.5956
Epoch 328/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 845.6080
Epoch 329/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 848.0325
Epoch 330/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 847.1084
Epoch 331/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 843.6666
Epoch 332/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 844.7864
Epoch 333/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 843.4575
Epoch 334/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 840.6692
Epoch 335/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 841.6121
Epoch 336/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 838.5792
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Epoch 337/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 839.4410
Epoch 338/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 838.2494
Epoch 339/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 838.7729
Epoch 340/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 839.7813
Epoch 341/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 836.9153
Epoch 342/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 834.8498
Epoch 343/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 837.0615
Epoch 344/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 834.1020
Epoch 345/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 830.5715
Epoch 346/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 830.9507
Epoch 347/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 831.2834
Epoch 348/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 830.3444
Epoch 349/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 829.5660
Epoch 350/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 827.4361
Epoch 351/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 825.2086
Epoch 352/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 825.5801
Epoch 353/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 824.5271
Epoch 354/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 823.6088
Epoch 355/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 823.8655
Epoch 356/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 822.8126
Epoch 357/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 827.0722
Epoch 358/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 820.1678
Epoch 359/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 818.9706
Epoch 360/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 818.0339
Epoch 361/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 819.1016
Epoch 362/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 817.0723
Epoch 363/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 815.2164
Epoch 364/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 813.4511
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Epoch 365/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 813.3802
Epoch 366/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 812.4746
Epoch 367/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 811.5590
Epoch 368/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 810.5765
Epoch 369/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 808.8539
Epoch 370/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 809.4755
Epoch 371/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 807.0115
Epoch 372/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 806.7662
Epoch 373/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 805.8090
Epoch 374/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 804.1863
Epoch 375/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 803.2690
Epoch 376/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 802.9874
Epoch 377/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 803.2955
Epoch 378/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 800.4857
Epoch 379/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 799.5575
Epoch 380/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 799.2048
Epoch 381/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 798.2585
Epoch 382/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 796.7710
Epoch 383/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 796.3660
Epoch 384/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 795.4187
Epoch 385/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 794.9632
Epoch 386/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 793.9964
Epoch 387/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 792.1039
Epoch 388/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 791.6125
Epoch 389/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 790.2269
Epoch 390/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 789.7047
Epoch 391/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 788.3477
Epoch 392/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 787.4072
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Epoch 393/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 787.2142
Epoch 394/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 785.5231
Epoch 395/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 785.2720
Epoch 396/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 784.2988
Epoch 397/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 783.3231
Epoch 398/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 782.3453
Epoch 399/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 781.3656
Epoch 400/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 779.8213
Epoch 401/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 779.6725
Epoch 402/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 778.1628
Epoch 403/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 776.9456
Epoch 404/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 775.9856
Epoch 405/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 775.2480
Epoch 406/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 774.0647
Epoch 407/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 773.1036
Epoch 408/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 772.5226
Epoch 409/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 771.1788
Epoch 410/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 770.3851
Epoch 411/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 769.4105
Epoch 412/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 769.0378
Epoch 413/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 767.4547
Epoch 414/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 766.4739
Epoch 415/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 765.3688
Epoch 416/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 764.3961
Epoch 417/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 763.5309
Epoch 418/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 762.6489
Epoch 419/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 761.6584
Epoch 420/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 760.4975
```

```
Epoch 421/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 759.6758
Epoch 422/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 758.6838
Epoch 423/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 757.6262
Epoch 424/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 756.8146
Epoch 425/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 755.6502
Epoch 426/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 754.6138
Epoch 427/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 753.6717
Epoch 428/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 752.7190
Epoch 429/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 751.6597
Epoch 430/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 750.7018
Epoch 431/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 749.7113
Epoch 432/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 748.7607
Epoch 433/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 747.7953
Epoch 434/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 746.7625
Epoch 435/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 745.7521
Epoch 436/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 744.7471
Epoch 437/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 743.7465
Epoch 438/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 742.7584
Epoch 439/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 741.7613
Epoch 440/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 740.7698
Epoch 441/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 739.7761
Epoch 442/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 738.7809
Epoch 443/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 737.7864
Epoch 444/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 736.7916
Epoch 445/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 735.7964
Epoch 446/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 734.8008
Epoch 447/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 733.8047
Epoch 448/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 732.8082
```

```
Epoch 449/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 731.8112
Epoch 450/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 730.8137
Epoch 451/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 729.8157
Epoch 452/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 728.8171
Epoch 453/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 727.8182
Epoch 454/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 726.8187
Epoch 455/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 725.8188
Epoch 456/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 724.8184
Epoch 457/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 723.8175
Epoch 458/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 722.8162
Epoch 459/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 721.8143
Epoch 460/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 720.8121
Epoch 461/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 719.8094
Epoch 462/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 718.8063
Epoch 463/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 717.8029
Epoch 464/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 716.7990
Epoch 465/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 715.7948
Epoch 466/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 714.7902
Epoch 467/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 713.7853
Epoch 468/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 712.7800
Epoch 469/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 711.7745
Epoch 470/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 710.7687
Epoch 471/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 709.7626
Epoch 472/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 708.7563
Epoch 473/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 707.7498
Epoch 474/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 706.7430
Epoch 475/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 705.7362
Epoch 476/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 704.7291
```

```
Epoch 477/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 703.7220
Epoch 478/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 702.7147
Epoch 479/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 701.7074
Epoch 480/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 700.6999
Epoch 481/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 699.6924
Epoch 482/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 698.6849
Epoch 483/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 697.6774
Epoch 484/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 696.6699
Epoch 485/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 695.6626
Epoch 486/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 694.6552
Epoch 487/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 693.6479
Epoch 488/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 692.6409
Epoch 489/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 691.6339
Epoch 490/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 690.6271
Epoch 491/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step - loss: 689.6205
Epoch 492/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 688.6141
Epoch 493/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 687.6080
Epoch 494/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 686.6021
Epoch 495/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 685.5966
Epoch 496/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 684.5913
Epoch 497/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 683.5864
Epoch 498/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 682.5820
Epoch 499/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 681.5779
Epoch 500/500
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 680.5742
best epoch =  500
smallest loss = 680.57421875
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 50ms/step
```

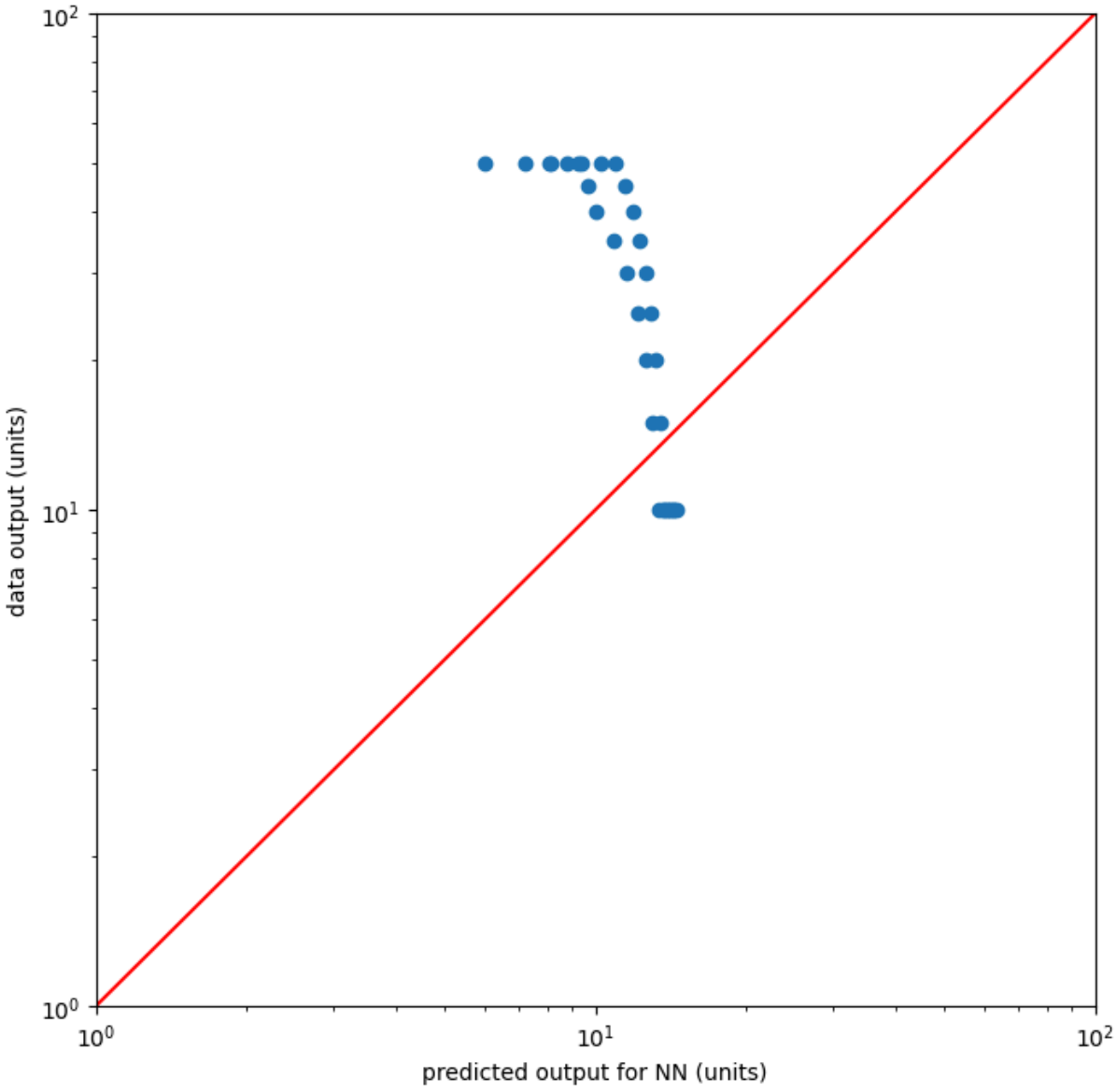
In [16]:
```
predictions = model.predict(data_inputs)
```

```
#SETTING UP PLOT
#matplotlib inline
# importing the required module
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [8, 8] # for square canvas
#========

'''CALCULATE PREDICTED VALUES AND RETRIEVE DATA VALUES TO PLOT'''

plt.scatter(predictions[:, 0], data_outputs[:, 0])
#plt.title('Genetic Algorithm training of Neural Network ==> output = alpha'
plt.xlabel('predicted output for NN (units)')
plt.ylabel('data output (units)')
plt.loglog()
plt.xlim(xmax = 100, xmin = 1.)
plt.ylim(ymax = 100, ymin = 1.)
# Generate red y=x line
x_data = np.linspace(1.0, 100.0, num=3)
y_data = x_data
plt.plot(x_data, y_data, color='red')
plt.savefig("log-log_plot_3.4.png", dpi=300, bbox_inches='tight')
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
```



In [17]:
```
test = []
outpt=[]
predictions = model.predict(data_inputs)

#first point (row [0])comparison of data and prediction
# put in a loop to print comparion for all data points

xa=[]
ya=[]
za=[]

xao = np.linspace(0, 3, 7)  # 1D array for x values
yao = np.linspace(0, 5, 11)    # 1D array for y values
xa = np.linspace(0, 10, 77)    # 1D array for x values
ya = np.linspace(0, 10, 77)    # 1D array for y values
za = np.linspace(0, 10, 77)    # 1D array for z values

x =-0.50
ia=-1
while (x < 2.51):
```

```
    x = x + 0.5
    y = -0.5
    while (y < 4.51):
        ia=ia+1
        y = y + 0.50
        xa[ia]=x
        ya[ia]=y

        test = [[ x , y ]]
        testarray = np.array(test)
        outpt = model.predict(testarray)
        za[ia] = outpt
        print ('x= ', testarray[0][0], ', y= ', testarray[0][1],'T= ', outpt



X, Y, Z = xa, ya, za

import sys
#import csv
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import axes3d

# Plot X,Y,Z

# Creating figure
#fig = plt.figure(figsize =(16, 9))
#ax = plt.axes(projection ='3d')

fig = plt.figure()
#ax = fig.add_subplot(111, projection='3d')
ax = plt.axes(projection ='3d')
#ax.plot_trisurf(X, Y, Z, color='white', edgecolors='grey', alpha=0.5)
ax.plot_trisurf(X, Y, Z,
                linewidth = 0.2,
                antialiased = True
                , cmap='viridis');


surf = ax.plot_trisurf(
    X, Y, Z,
    linewidth=0.2,
    antialiased=True,
    cmap='viridis'
)


#  ADD COLORBAR
fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)



# fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)

ax.invert_yaxis()
```

```
ax.invert_xaxis()

plt.savefig("trisurf_plot.png", dpi=300, bbox_inches='tight')
plt.show()



# Plot the data
plt.figure(figsize=(5, 6))
plt.tripcolor(xa, ya, za, shading='flat', cmap='viridis')
plt.savefig("Heat_map.png", dpi=300, bbox_inches='tight')
plt.colorbar(label='Z value')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Heat Map of Z values over X and Y domain')
plt.show()
```

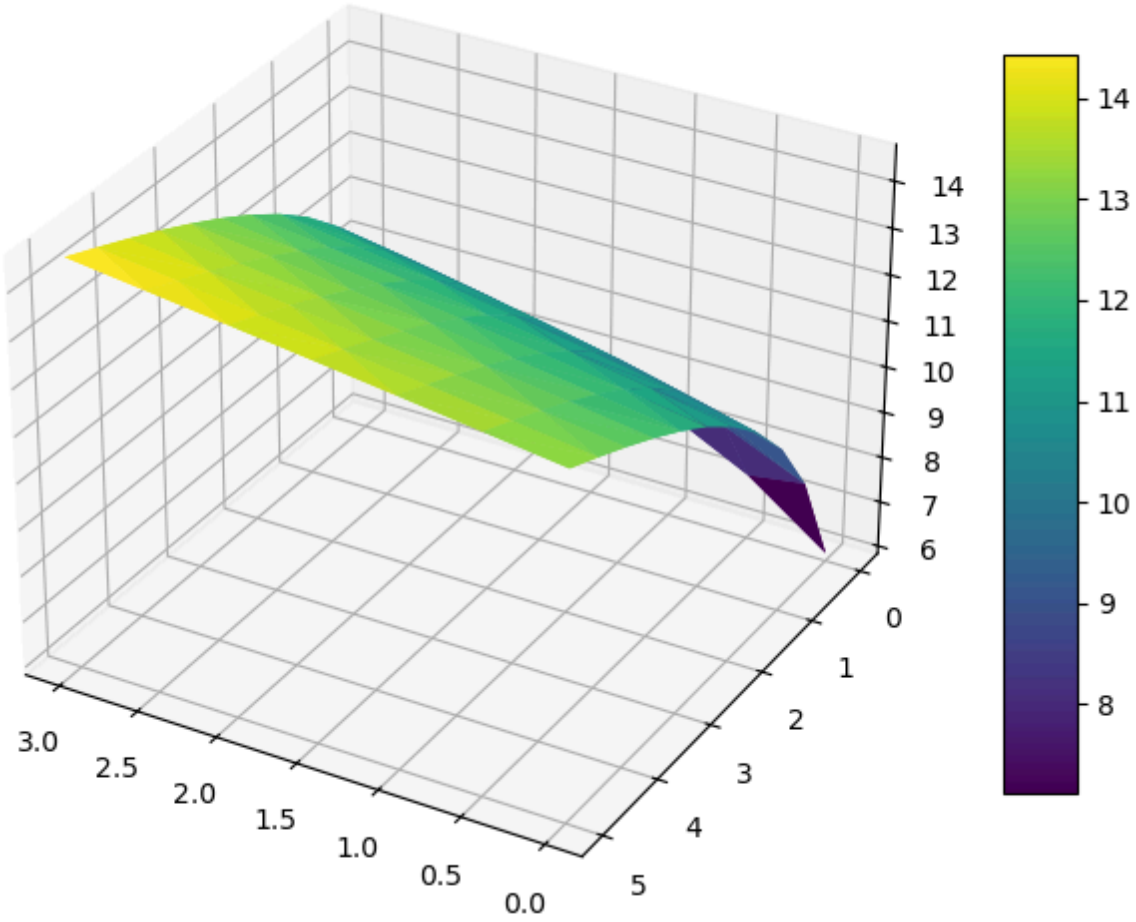
```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 52ms/step
x=  0.0 , y=  0.0 T=  6.0094495
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  0.5 T=  8.109011
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  1.0 T=  9.397686
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  1.5 T=  10.280795
```

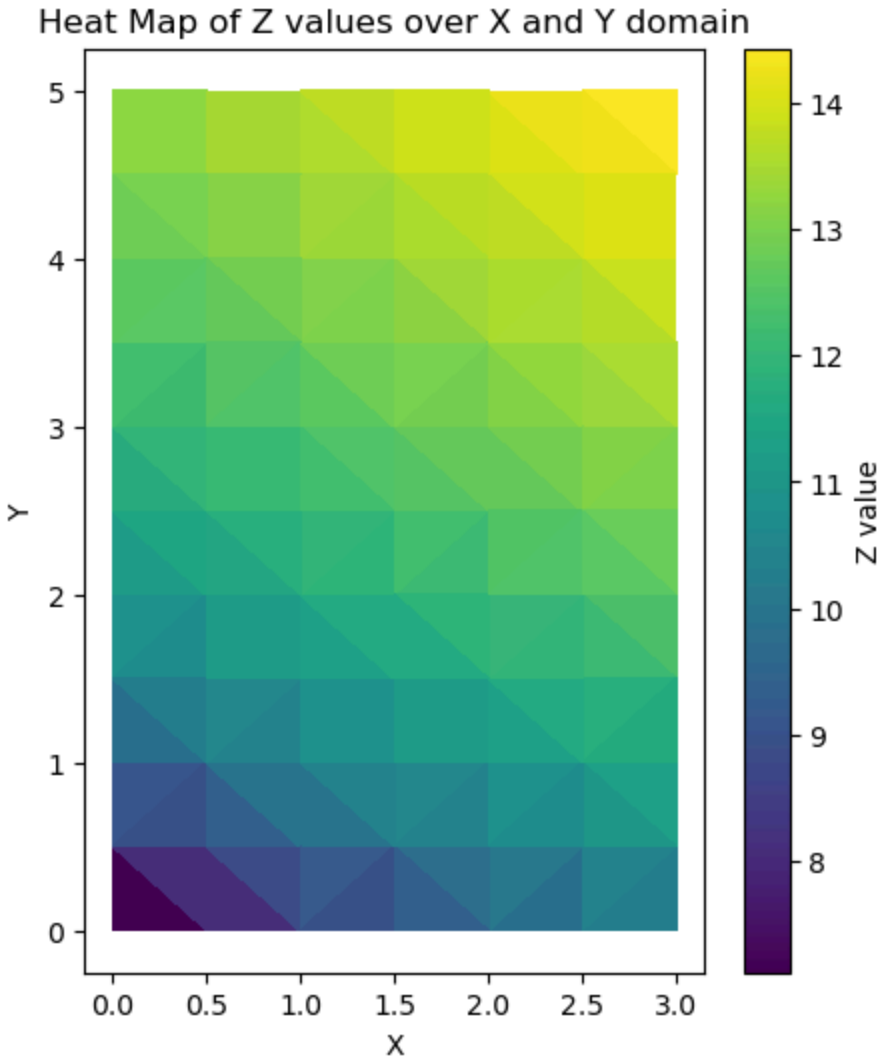
```
/tmp/ipykernel_4153360/3841726189.py:32: DeprecationWarning: Conversion of a
n array with ndim > 0 to a scalar is deprecated, and will error in future. E
nsure you extract a single element from your array before performing this op
eration. (Deprecated NumPy 1.25.)
  za[ia] = outpt
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  2.0 T=  10.937163
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  2.5 T=  11.458237
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  3.0 T=  11.89495
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  0.0 , y=  3.5 T=  12.277219
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  4.0 T=  12.623282
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.0 , y=  4.5 T=  12.944598
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  0.0 , y=  5.0 T=  13.248539
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  0.5 , y=  0.0 T=  7.2526593
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.5 , y=  0.5 T=  8.864651
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.5 , y=  1.0 T=  9.923031
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.5 , y=  1.5 T=  10.684241
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.5 , y=  2.0 T=  11.271755
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.5 , y=  2.5 T=  11.752211
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  0.5 , y=  3.0 T=  12.164108
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  0.5 , y=  3.5 T=  12.530715
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  4.0 T=  12.866551
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  0.5 , y=  4.5 T=  13.180935
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  0.5 , y=  5.0 T=  13.47998
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.0 , y=  0.0 T=  8.117424
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.0 , y=  0.5 T=  9.439673
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.0 , y=  1.0 T=  10.351308
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.0 , y=  1.5 T=  11.031774
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.0 , y=  2.0 T=  11.57256
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.0 , y=  2.5 T=  12.024953
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.0 , y=  3.0 T=  12.419438
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.0 , y=  3.5 T=  12.774877
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.0 , y=  4.0 T=  13.103286
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.0 , y=  4.5 T=  13.412522
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.0 , y=  5.0 T=  13.707836
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.5 , y=  0.0 T=  8.760659
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.5 , y=  0.5 T=  9.899927
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.5 , y=  1.0 T=  10.71481
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.5 , y=  1.5 T=  11.340707
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.5 , y=  2.0 T=  11.849384
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.5 , y=  2.5 T=  12.282241
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.5 , y=  3.0 T=  12.664445
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.5 , y=  3.5 T=  13.0118885
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.5 , y=  4.0 T=  13.334878
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  1.5 , y=  4.5 T=  13.64027
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  1.5 , y=  5.0 T=  13.9327135
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  2.0 , y=  0.0 T=  9.264804
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  0.5 T=  10.284008
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  1.0 T=  11.033725
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  1.5 T=  11.6223
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  2.0 T=  12.108789
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  2.5 T=  12.528017
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  3.0 T=  12.901566
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  3.5 T=  13.243292
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  4.0 T=  13.562331
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  4.5 T=  13.864845
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.0 , y=  5.0 T=  14.155062
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  2.5 , y=  0.0 T=  9.677266
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  2.5 , y=  0.5 T=  10.615811
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.5 , y=  1.0 T=  11.321117
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  2.5 , y=  1.5 T=  11.884065
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  2.5 , y=  2.0 T=  12.355246
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  2.5 , y=  2.5 T=  12.765023
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.5 , y=  3.0 T=  13.132524
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  2.5 , y=  3.5 T=  13.470198
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.5 , y=  4.0 T=  13.786373
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.5 , y=  4.5 T=  14.086725
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  2.5 , y=  5.0 T=  14.375208
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  3.0 , y=  0.0 T=  10.027026
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  3.0 , y=  0.5 T=  10.910701
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  3.0 , y=  1.0 T=  11.585661
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  3.0 , y=  1.5 T=  12.131104
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
x=  3.0 , y=  2.0 T=  12.591849
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  3.0 , y=  2.5 T=  12.995189
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  3.0 , y=  3.0 T=  13.358544
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  3.0 , y=  3.5 T=  13.693401
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  3.0 , y=  4.0 T=  14.007525
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  3.0 , y=  4.5 T=  14.30627
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
x=  3.0 , y=  5.0 T=  14.59339
```


14
13
12
11
10
9
8
7
6
0
1
2
3
4
5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
14
13
12
11
10
9
8


Heat Map of Z values over X and Y domain
Y
X
Z value
0
1
2
3
4
5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
8
9
10
11
12
13
14

In [ ]:

In [ ]:

In [2]:
```
'''>>>>> start CodeP3.2F25
    V.P. Carey ME249, Fall 2025

Intro to Neural Network Modeling
Keras model for PV power system'''

#import useful packages
import keras
import pandas as pd
from keras.models import Sequential
import numpy as np
import keras.backend as kb
import tensorflow as tf
#the follwoing 2 lines are only needed for Mac OS machines
import os
os.environ['KMP_DUPLICATE_LIB_OK']='True'

# # define meadian values of input variables - add your values here
# Tamed = 1.  #make sure Tamed does not = 0
# IDmed = 1.
# RLmed = 1.

# #create input data array
# xdata = []

# #Part 1 input data: Air temp (degC), ID (W/sqm), load resistance (ohms)
# xdata = [[-10.0, 350, 4.464]]
# xdata.append([-10.0, 650, 4.464])
# xdata.append([-10.0, 950, 4.464])
# xdata.append([-10.0, 1250, 4.464])
# ''' ADD THE REST OF THE INPUT DATA AND CONVERT TO:
# xdata = [[-10.0/Tamed, 350/IDmed, 4.464/RLmed]]
# xdata.append([-10.0/Tamed, 650/IDmed, 4.464/RLmed])
# xdata.append([-10.0/Tamed, 950/IDmed, 4.464/RLmed])
# xdata.append([-10.0/Tamed, 1251/IDmed, 4.464/RLmed])
# ... etc.'''

# xarray= np.array(xdata)
# print (xdata)
# print (xarray)

# # define meadian values of output variables - add your values here
# VLmed = 1.
# Wdmed = 1

# #create output data array
# ydata = []

# #Part 1 output data: VL (V) and Power out Wd (W)
# ydata = [[18.9, 80.3]]
# ydata.append([23.5, 124.6])
# ydata.append([24.8, 138.6])
# ydata.append([25.6, 146.9])
# ''' ADD THE REST OF THE OUPUT DATA AND CONVERT TO:
```

```
# ydata = [[18.9/VLmed, 80.3/Wdmed]]
# ydata.append([23.5/VLmed, 124.6/Wdmed])
# ydata.append([24.8/VLmed, 138.6/Wdmed])
# ydata.append([25.6/VLmed, 146.9/Wdmed])
# ... etc.'''

# yarray= np.array(ydata)
# print (ydata)
# print (yarray)




'''>>>>> CodeP3.2F25
    V.P. Carey ME249, Fall 2025
Intro to Neural Network Modeling
Keras model for PV power system'''

#import useful packages
import keras
import pandas as pd
from keras.models import Sequential
import numpy as np
import keras.backend as kb
import tensorflow as tf

#the following 2 lines are only needed for Mac OS machines
import os
os.environ['KMP_DUPLICATE_LIB_OK']='True'

# ===== PART 1: Define raw data first, convert temp to Kelvin, then calculat
# Raw input data: Air temp (degC), ID (W/sqm), load resistance (ohms)
xdata_raw_celsius = [[-10.0, 350, 4.464],
  [-10.0, 650, 4.464],
  [-10.0, 950, 4.464],
  [-10.0, 1250, 4.464],
  [10.0, 350, 4.464],
  [10.0, 650, 4.464],
  [10.0, 950, 4.464],
  [10.0, 1250, 4.464],
  [30.0, 350, 4.464],
  [30.0, 650, 4.464],
  [30.0, 950, 4.464],
  [30.0, 1250, 4.466],
  [-10.0, 350, 6.696],
  [-10.0, 650, 6.696],
  [-10.0, 950, 6.696],
  [-10.0, 1250, 6.696],
  [10.0, 350, 6.696],
  [10.0, 650, 6.696],
  [10.0, 950, 6.696],
  [10.0, 1250, 6.696],
  [30.0, 350, 6.696],
  [30.0, 650, 6.696],
  [30.0, 950, 6.696],
  [30.0, 1250, 6.696],
```

```
  [-10.0, 350, 8.928],
  [-10.0, 650, 8.928],
  [-10.0, 950, 8.928],
  [-10.0, 1250, 8.928],
  [10.0, 350, 8.928],
  [10.0, 650, 8.928],
  [10.0, 950, 8.928],
  [10.0, 1250, 8.928],
  [30.0, 350, 8.928],
  [30.0, 650, 8.928],
  [30.0, 950, 8.928],
  [30.0, 1250, 8.928]]

# Convert temperature from Celsius to Kelvin (K = degC + 273.15)
xdata_raw = [[row[0] + 273.15, row[1], row[2]] for row in xdata_raw_celsius]

print("===== Temperature Conversion =====")
print("First 3 samples (Celsius -> Kelvin):")
for i in range(3):
    print(f"  {xdata_raw_celsius[i][0]:6.2f} °C  ->  {xdata_raw[i][0]:6.2f}
print()

# Raw output data: VL (V) and Power out (W)
ydata_raw = [[18.9, 80.3],
 [23.5, 124.6],
 [24.8, 138.6],
 [25.6, 146.9],
 [19.2, 83.1],
 [25.0, 140.5],
 [26.5, 157.6],
 [27.3, 167.5],
 [19.4, 84.7],
 [26.4, 156.7],
 [28.1, 177.7],
 [29.0, 189.4],
 [22.4, 75.2],
 [24.8, 92.2],
 [25.8, 99.7],
 [26.4, 104.6],
 [23.6, 83.7],
 [26.5, 104.9],
 [27.6, 113.8],
 [28.3, 119.6],
 [24.8, 92.0],
 [28.1, 118.2],
 [29.3, 128.8],
 [30.1, 135.5],
 [23.5, 62.2],
 [25.4, 72.5],
 [26.3, 77.6],
 [26.9, 81.1],
 [25.0, 70.3],
 [27.1, 82.7],
 [28.1, 88.7],
 [28.7, 92.8],
 [26.5, 78.6],
```

```
 [28.8, 93.5],
 [29.9, 100.5],
 [30.6, 105.2]]

# Convert to numpy arrays and calculate medians
xarray_raw = np.array(xdata_raw)
yarray_raw = np.array(ydata_raw)

# Calculate input variable medians from data
Tamed = np.median(xarray_raw[:, 0])   # Air temperature median (K)
IDmed = np.median(xarray_raw[:, 1])   # Irradiance (ID) median (W/sqm)
RLmed = np.median(xarray_raw[:, 2])   # Load resistance median (ohms)

# Calculate output variable medians from data
VLmed = np.median(yarray_raw[:, 0])   # Load voltage (VL) median (V)
Wdmed = np.median(yarray_raw[:, 1])   # Power output (Wd) median (W)

print("===== Calculated Median Values =====")
print(f"Temperature median: {Tamed:.2f} K")
print(f"Irradiance median: {IDmed:.2f} W/sqm")
print(f"Load resistance median: {RLmed:.4f} ohms")
print(f"Voltage median: {VLmed:.2f} V")
print(f"Power median: {Wdmed:.2f} W\n")

# ===== PART 2: Create normalized input data array =====
# Normalize input data by dividing by medians
xdata = []
for row in xdata_raw:
    xdata.append([row[0]/Tamed, row[1]/IDmed, row[2]/RLmed])

xarray = np.array(xdata)
print("===== Normalized Input Data =====")
print("First 5 rows (normalized):")
print(xarray[:5])
print(f"Input shape: {xarray.shape}\n")

# ===== PART 3: Create normalized output data array =====
# Normalize output data by dividing by medians
ydata = []
for row in ydata_raw:
    ydata.append([row[0]/VLmed, row[1]/Wdmed])

yarray = np.array(ydata)
print("===== Normalized Output Data =====")
print("First 5 rows (normalized):")
print(yarray[:5])
print(f"Output shape: {yarray.shape}\n")

# ===== PART 4: Split data into training (2/3) and validation (1/3) sets ===
# Set random seed for reproducibility
np.random.seed(42)

# Generate random indices and shuffle them
total_samples = len(xarray)
indices = np.arange(total_samples)
np.random.shuffle(indices)
```

```
# Calculate split point (2/3 for training, 1/3 for validation)
split_point = int(2/3 * total_samples)

# Split indices
train_indices = indices[:split_point]
val_indices = indices[split_point:]

# Create training and validation sets
x_train = xarray[train_indices]
y_train = yarray[train_indices]
x_val = xarray[val_indices]
y_val = yarray[val_indices]

print("===== Data Split =====")
print(f"Total samples: {total_samples}")
print(f"Training set size: {len(x_train)} ({len(x_train)/total_samples*100:.
print(f"Validation set size: {len(x_val)} ({len(x_val)/total_samples*100:.1f
print(f"\nTraining input shape: {x_train.shape}")
print(f"Training output shape: {y_train.shape}")
print(f"Validation input shape: {x_val.shape}")
print(f"Validation output shape: {y_val.shape}\n")
```

```
===== Temperature Conversion =====
First 3 samples (Celsius -> Kelvin):
  -10.00 °C  ->  263.15 K
  -10.00 °C  ->  263.15 K
  -10.00 °C  ->  263.15 K

===== Calculated Median Values =====
Temperature median: 283.15 K
Irradiance median: 800.00 W/sqm
Load resistance median: 6.6960 ohms
Voltage median: 26.45 V
Power median: 100.10 W

===== Normalized Input Data =====
First 5 rows (normalized):
[[0.92936606 0.4375     0.66666667]
 [0.92936606 0.8125     0.66666667]
 [0.92936606 1.1875     0.66666667]
 [0.92936606 1.5625     0.66666667]
 [1.         0.4375     0.66666667]]
Input shape: (36, 3)

===== Normalized Output Data =====
First 5 rows (normalized):
[[0.71455577 0.8021978 ]
 [0.88846881 1.24475524]
 [0.93761815 1.38461538]
 [0.96786389 1.46753247]
 [0.72589792 0.83016983]]
Output shape: (36, 2)

===== Data Split =====
Total samples: 36
Training set size: 24 (66.7%)
Validation set size: 12 (33.3%)

Training input shape: (24, 3)
Training output shape: (24, 2)
Validation input shape: (12, 3)
Validation output shape: (12, 2)
```

In [3]:

```python
# define neural network model

#As seen below, we have created four dense layers.
#A dense layer is a layer in neural network that's fully connected.
#In other words, all the neurons in one layer are connected to all other neu
#In the first layer, we need to provide the input shape, which is 1 in our c
#The activation function we have chosen is elu, which stands for exponential

from keras import backend as K
#initialize weights with values between -0.2 and 0.5
# initializer = keras.initializers.RandomUniform(minval= -0.2, maxval=0.5)
initializer = keras.initializers.RandomUniform(minval= -0.2, maxval=0.5)

model = keras.Sequential([
```

```
    keras.layers.Dense(6, activation='elu', input_shape=[3],  kernel_initial
    keras.layers.Dense(12, activation='elu',  kernel_initializer=initializer
    keras.layers.Dense(6, activation='elu', kernel_initializer=initializer),
    keras.layers.Dense(2,  kernel_initializer=initializer)
  ])
```

```
/global/home/users/kwyang0331/.conda/envs/wrf/lib/python3.10/site-packages/k
eras/src/layers/core/dense.py:95: UserWarning: Do not pass an `input_shape`/
`input_dim` argument to a layer. When using Sequential models, prefer using
an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

In [22]:
```
#We're using RMSprop as our optimizer here. RMSprop stands for Root Mean Squ
#It's one of the most popular gradient descent optimization algorithms for d
#RMSprop is an optimizer that's reliable and fast.
#We're compiling the mode using the model.compile function. The loss functio
#is mean squared error. After the compilation of the model, we'll use the fi
#Number of epochs can be varied.

#from tf.keras import optimizers
# rms = keras.optimizers.RMSprop(0.020)
rms = keras.optimizers.RMSprop(0.005)
#model.compile(loss='mse',optimizer=rms)

from tensorflow.keras import backend as K
def mean_double_squared_error(y_true, y_pred):
    return K.mean(K.square(y_pred-y_true)*K.square(y_pred-y_true))

model.compile(optimizer="rmsprop",loss=mean_double_squared_error)
```

In [23]:
```
#After the compilation of the model, we'll use the fit method with 500 epoch
#I started with epochs value of 100 and then tested the model after training
#The prediction was not that good. Then I modified the number of epochs to 2
#Accuracy had improved slightly, but figured I'd give it one more try. Final
#I found acceptable prediction accuracy.

#The fit method takes three parameters; namely, x, y, and number of epochs.
#During model training, if all the batches of data are seen by the model onc
#we say that one epoch has been completed.

# Add an early stopping callback
es = keras.callbacks.EarlyStopping(
    monitor='loss',
    mode='min',
    patience = 80,
    restore_best_weights = True,
    verbose=0)
# Add a checkpoint where loss is minimum, and save that model
mc = keras.callbacks.ModelCheckpoint('best_model.keras', monitor='loss',
                     mode='min',  verbose=0, save_best_only=True)



# historyData = model.fit(x_train,y_train,epochs=800,callbacks=[es])
# This runs silently (verbose=0) and uses your callbacks (es, mc)
historyData = model.fit(xarray, yarray, epochs=6000, verbose=0)
```

```
loss_hist = historyData.history['loss']

best_loss = np.min(loss_hist)
best_epoch = np.argmin(loss_hist) + 1
print(f"Best MSE (Smallest Loss): {best_loss}")
print(f"Best Epoch: {best_epoch}")

# #The above line will return a dictionary, access it's info like this:
# best_epoch = np.argmin(historyData.history['loss']) + 1
# print ('best epoch = ', best_epoch)
# print('smallest loss =', np.min(loss_hist))

model.save('./best_model.keras')
```

```
Best MSE (Smallest Loss): 7.750137420714509e-09
Best Epoch: 5997
```

In [24]:
```
# Get the loss from the very last epoch (the end of training)
final_loss = loss_hist[-1]
print(f"Loss at end of training: {final_loss}")
```

```
Loss at end of training: 7.751622455032248e-09
```

In [27]:
```
# This line of code can be used to reconstruct the saved model.

recon_model = keras.models.load_model(best_model.keras)

# the name of the model is now "recon_model". You can then use this model to
# See the previous project for code to do the comparisons.
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Input In [27], in <cell line: 3>()
      1 # This line of code can be used to reconstruct the saved model.
----> 3 recon_model = keras.models.load_model(best_model.keras)
      5 # the name of the model is now "recon_model". You can then use this
model to do predictions for comparisons.
      6 # See the previous project for code to do the comparisons.

NameError: name 'best_model' is not defined
```

In [28]:
```
test = []
outpt=[]
predictions = model.predict(xarray)

#put in a loop to print comparison for all data points
for i in range(len(xarray)):
    test = [[xarray[i][0], xarray[i][1], xarray[i][2]]]
    testarray = np.array(test)
    outpt = model.predict(testarray, verbose=0)
    print('row [' + str(i) + '] data:  T1= ', xarray[i][0]*Tamed, ', ID= ',
        ', RL= ', xarray[i][2]*RLmed, ', power= ', yarray[i][1]*Wdmed, \
        ',  predicted power = ', outpt[0][1]*Wdmed)

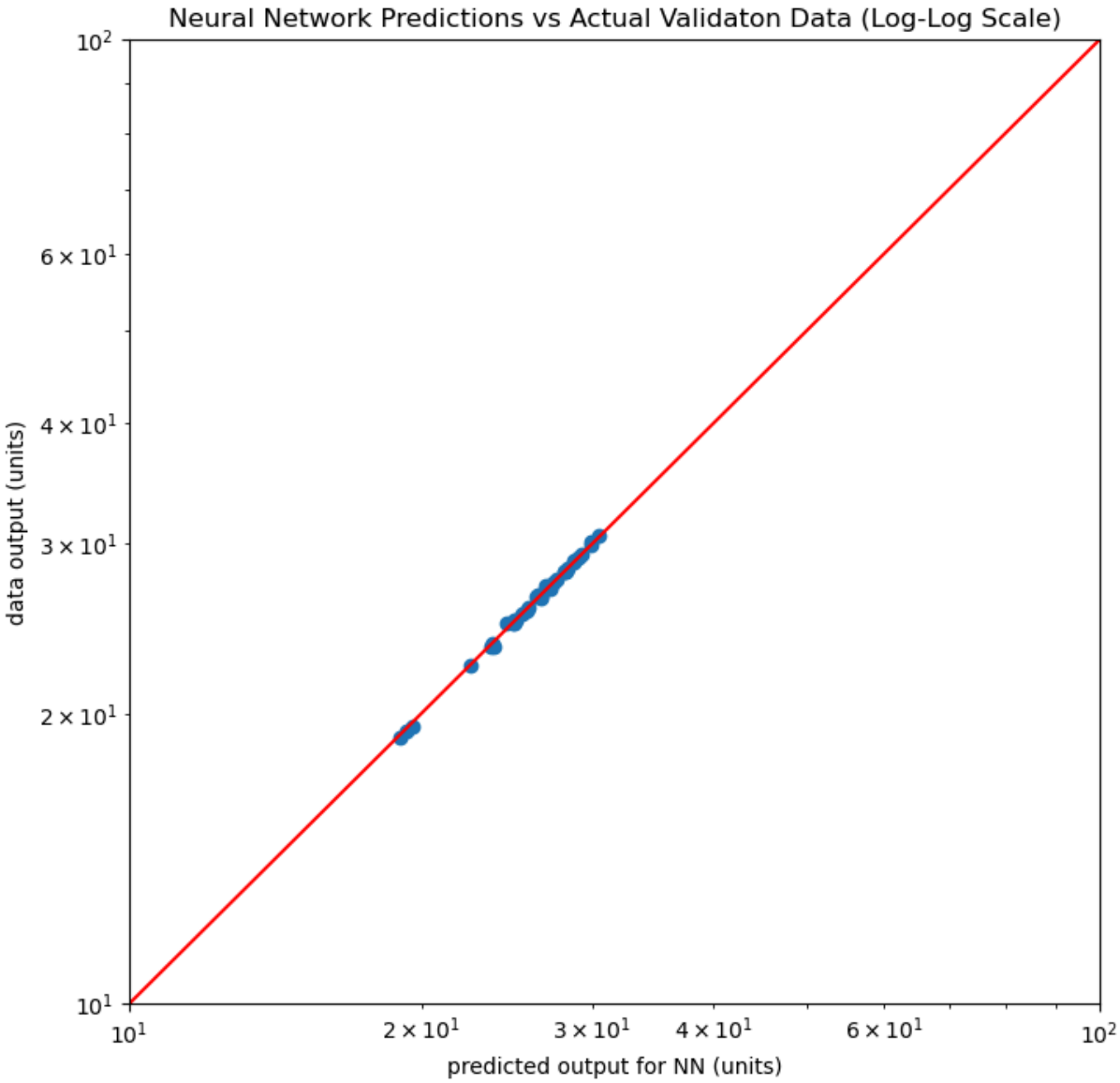
#SETTING UP PLOT
%matplotlib inline
```

```
# importing the required module
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [8, 8] # for square canvas
#========
'''CALCULATE PREDICTED VALUES AND RETRIEVE DATA VALUES TO PLOT'''
# Denormalize for plotting
predictions_denorm = predictions[:, 0] * VLmed
actual_denorm = yarray[:, 0] * VLmed

plt.scatter(predictions_denorm, actual_denorm)
plt.title('Neural Network Predictions vs Actual Validaton Data (Log-Log Scal
plt.xlabel('predicted output for NN (units)')
plt.ylabel('data output (units)')
plt.loglog()
plt.xlim(xmax=100, xmin=10)
plt.ylim(ymax=100, ymin=10)
# Generate red y=x line using logspace for proper log scale
x_data = np.logspace(np.log10(1.0), np.log10(100.0), num=100)
y_data = x_data
plt.plot(x_data, y_data, color='red')
plt.show()
```

```
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step
row [0] data:  T1=  263.15 , ID=  350.0 , RL=  4.464 , power=  80.3 ,  predi
cted power =  80.0817374229431
row [1] data:  T1=  263.15 , ID=  650.0 , RL=  4.464 , power=  124.6 ,  pred
icted power =  122.87491796016693
row [2] data:  T1=  263.15 , ID=  950.0 , RL=  4.464 , power=  138.6 ,  pred
icted power =  139.40726096630095
row [3] data:  T1=  263.15 , ID=  1250.0 , RL=  4.464 , power=  146.9 ,  pre
dicted power =  148.23577995300292
row [4] data:  T1=  283.15 , ID=  350.0 , RL=  4.464 , power=  83.1 ,  predi
cted power =  82.45065274834633
row [5] data:  T1=  283.15 , ID=  650.0 , RL=  4.464 , power=  140.5 ,  pred
icted power =  140.380372941494
row [6] data:  T1=  283.15 , ID=  950.0 , RL=  4.464 , power=  157.6 ,  pred
icted power =  159.4145811200142
row [7] data:  T1=  283.15 , ID=  1250.0 , RL=  4.464 , power=  167.5 ,  pre
dicted power =  168.0419361233711
row [8] data:  T1=  303.15 , ID=  350.0 , RL=  4.464 , power=  84.7 ,  predi
cted power =  85.86603127717972
row [9] data:  T1=  303.15 , ID=  650.0 , RL=  4.464 , power=  156.7 ,  pred
icted power =  156.36477141380308
row [10] data:  T1=  303.15 , ID=  950.0 , RL=  4.464 , power=  177.7 ,  pre
dicted power =  178.97795152664185
row [11] data:  T1=  303.15 , ID=  1250.0 , RL=  4.466 , power=  189.4 ,  pr
edicted power =  187.55983043909072
row [12] data:  T1=  263.15 , ID=  350.0 , RL=  6.696 , power=  75.2 ,  pred
icted power =  75.93972589969634
row [13] data:  T1=  263.15 , ID=  650.0 , RL=  6.696 , power=  92.2 ,  pred
icted power =  92.34159512519835
row [14] data:  T1=  263.15 , ID=  950.0 , RL=  6.696 , power=  99.7 ,  pred
icted power =  98.73750720024108
row [15] data:  T1=  263.15 , ID=  1250.0 , RL=  6.696 , power=  104.5999999
9999998 ,  predicted power =  104.09948556423187
row [16] data:  T1=  283.15 , ID=  350.0 , RL=  6.696 , power=  83.7 ,  pred
icted power =  84.5221537232399
row [17] data:  T1=  283.15 , ID=  650.0 , RL=  6.696 , power=  104.9 ,  pre
dicted power =  104.9388899564743
row [18] data:  T1=  283.15 , ID=  950.0 , RL=  6.696 , power=  113.80000000
000001 ,  predicted power =  113.10446753501891
row [19] data:  T1=  283.15 , ID=  1250.0 , RL=  6.696 , power=  119.6 ,  pr
edicted power =  119.27088704109191
row [20] data:  T1=  303.15 , ID=  350.0 , RL=  6.696 , power=  92.0 ,  pred
icted power =  92.26192545294761
row [21] data:  T1=  303.15 , ID=  650.0 , RL=  6.696 , power=  118.20000000
000002 ,  predicted power =  118.24275269508361
row [22] data:  T1=  303.15 , ID=  950.0 , RL=  6.696 , power=  128.8 ,  pre
dicted power =  128.45547033548354
row [23] data:  T1=  303.15 , ID=  1250.0 , RL=  6.696 , power=  135.5 ,  pr
edicted power =  135.77866030931472
row [24] data:  T1=  263.15 , ID=  350.0 , RL=  8.928 , power=  62.2 ,  pred
icted power =  62.45204357504844
row [25] data:  T1=  263.15 , ID=  650.0 , RL=  8.928 , power=  72.5 ,  pred
icted power =  72.82461367249489
row [26] data:  T1=  263.15 , ID=  950.0 , RL=  8.928 , power=  77.6 ,  pred
icted power =  78.83839412927627
row [27] data:  T1=  263.15 , ID=  1250.0 , RL=  8.928 , power=  81.1 ,  pre
```

```
dicted power =  82.33076340556144
row [28] data:  T1=  283.15 , ID=  350.0 , RL=  8.928 , power=  70.3 ,  pred
icted power =  70.30604251027107
row [29] data:  T1=  283.15 , ID=  650.0 , RL=  8.928 , power=  82.7 ,  pred
icted power =  81.66248204708099
row [30] data:  T1=  283.15 , ID=  950.0 , RL=  8.928 , power=  88.7 ,  pred
icted power =  88.35598746538162
row [31] data:  T1=  283.15 , ID=  1250.0 , RL=  8.928 , power=  92.8 ,  pre
dicted power =  92.49231895208358
row [32] data:  T1=  303.15 , ID=  350.0 , RL=  8.928 , power=  78.6 ,  pred
icted power =  79.32460406422615
row [33] data:  T1=  303.15 , ID=  650.0 , RL=  8.928 , power=  93.5 ,  pred
icted power =  92.27662075757979
row [34] data:  T1=  303.15 , ID=  950.0 , RL=  8.928 , power=  100.5 ,  pre
dicted power =  99.9954742014408
row [35] data:  T1=  303.15 , ID=  1250.0 , RL=  8.928 , power=  105.2000000
0000002 ,  predicted power =  104.82500283718109
```



In [ ]:
```
test = []
outpt=[]
predictions = model.predict(x_train)
```

```
#put in a loop to print comparison for all data points
for i in range(len(x_train)):
    test = [[x_train[i][0], x_train[i][1], x_train[i][2]]]
    testarray = np.array(test)
    outpt = model.predict(testarray, verbose=0)
    print('row [' + str(i) + '] data:  T1= ', xarray[i][0], ', ID= ', xarray
        ', RL= ', xarray[i][2], ', power= ', yarray[i][1]*Wdmed, \
        ',  predicted power = ', outpt[0][1]*Wdmed)

#SETTING UP PLOT
%matplotlib inline
# importing the required module
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [8, 8] # for square canvas
#========
'''CALCULATE PREDICTED VALUES AND RETRIEVE DATA VALUES TO PLOT'''
# Denormalize for plotting
predictions_denorm = predictions[:, 0]
actual_denorm = y_train[:, 0]

plt.scatter(predictions_denorm, actual_denorm)
plt.title('Neural Network Predictions vs Actual Training Data (Log-Log Scale
plt.xlabel('predicted output for NN (units)')
plt.ylabel('data output (units)')
plt.loglog()
plt.xlim(xmax=10, xmin=0.1)
plt.ylim(ymax=10, ymin=0.1)
# Generate red y=x line using logspace for proper log scale
x_data = np.logspace(np.log10(0.1), np.log10(10.0), num=100)
y_data = x_data
plt.plot(x_data, y_data, color='red')
plt.show()
```

In [ ]:

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# ===== FIXED PARAMETER =====
T_fixed_celsius = 20.0  # Fixed air temperature in Celsius
T_fixed_kelvin = T_fixed_celsius + 273.15  # Convert to Kelvin
T_normalized = T_fixed_kelvin / Tamed  # Normalize using median

print("===== Fixed Parameters =====")
print(f"Air Temperature: {T_fixed_celsius}°C = {T_fixed_kelvin:.2f} K")
print(f"Normalized Temperature: {T_normalized:.4f}\n")

# ===== CREATE GRID FOR PREDICTIONS =====
# Load resistance range: 4 to 8 Ohms
RL_min, RL_max = 4.0, 8.0
RL_points = 20  # Number of points in RL direction

# Irradiance range: 500 to 1200 W/m^2
ID_min, ID_max = 500.0, 1200.0
ID_points = 20  # Number of points in ID direction
```

```
# Create meshgrid
RL_array = np.linspace(RL_min, RL_max, RL_points)
ID_array = np.linspace(ID_min, ID_max, ID_points)
RL_mesh, ID_mesh = np.meshgrid(RL_array, ID_array)

print("===== Input Ranges =====")
print(f"Load Resistance (RL): {RL_min} - {RL_max} Ohms ({RL_points} points)"
print(f"Irradiance (ID): {ID_min} - {ID_max} W/m² ({ID_points} points)")
print(f"Total predictions: {RL_points * ID_points}\n")

# ===== NORMALIZE INPUT DATA FOR MODEL =====
# Flatten meshgrid for model prediction
RL_flat = RL_mesh.flatten()
ID_flat = ID_mesh.flatten()
T_flat = np.full_like(RL_flat, T_normalized)

# Normalize by medians
RL_norm = RL_flat / RLmed
ID_norm = ID_flat / IDmed
T_norm = T_flat / Tamed

# Create input array [T, ID, RL] for the model
input_data = np.column_stack([T_norm, ID_norm, RL_norm])

print("===== Normalized Input Statistics =====")
print(f"Temperature (normalized): {T_norm[0]:.4f}")
print(f"Irradiance (normalized): min={ID_norm.min():.4f}, max={ID_norm.max()
print(f"Load Resistance (normalized): min={RL_norm.min():.4f}, max={RL_norm.

# ===== GET PREDICTIONS FROM MODEL =====
predictions_normalized = model.predict(input_data, verbose=0)

# Extract power output (second column) and denormalize
power_normalized = predictions_normalized[:, 1]  # Column 1 is power output
power_denorm = power_normalized * Wdmed  # Denormalize using median

# Reshape back to grid
power_grid = power_denorm.reshape(RL_mesh.shape)

print("===== Power Output Statistics =====")
print(f"Minimum Power: {power_grid.min():.2f} W")
print(f"Maximum Power: {power_grid.max():.2f} W")
print(f"Mean Power: {power_grid.mean():.2f} W\n")

# ===== CREATE 3D SURFACE PLOT =====
fig = plt.figure(figsize=(12, 8))
ax = fig.add_subplot(111, projection='3d')

# Create surface plot
surf = ax.plot_surface(RL_mesh, ID_mesh, power_grid, cmap='viridis',
                       edgecolor='none', alpha=0.9, antialiased=True)

# Labels and title
ax.set_xlabel('Load Resistance RL (Ohms)', fontsize=11, fontweight='bold')
ax.set_ylabel('Irradiance ID (W/m²)', fontsize=11, fontweight='bold')
ax.set_zlabel('Power Output Wd (Watts)', fontsize=11, fontweight='bold')
```

```
ax.set_title(f'Solar Power Output vs RL and ID\n(Fixed Temperature: {T_fixed
             fontsize=13, fontweight='bold')

# Add colorbar
cbar = fig.colorbar(surf, ax=ax, pad=0.1, shrink=0.8)
cbar.set_label('Power (W)', fontsize=11, fontweight='bold')

# Adjust viewing angle for better visualization
ax.view_init(elev=25, azim=45)

plt.savefig("surface_plot.png", dpi=300, bbox_inches='tight')
plt.show()

# ===== CREATE CONTOUR PLOT =====
fig, ax = plt.subplots(figsize=(10, 7))

# Create filled contour plot
contourf = ax.contourf(RL_mesh, ID_mesh, power_grid, levels=20, cmap='viridi
contour = ax.contour(RL_mesh, ID_mesh, power_grid, levels=10, colors='black'
                     alpha=0.3, linewidths=0.5)

# Add contour labels
ax.clabel(contour, inline=True, fontsize=9, fmt='%.1f W')

# Labels and title
ax.set_xlabel('Load Resistance RL (Ohms)', fontsize=11, fontweight='bold')
ax.set_ylabel('Irradiance ID (W/m²)', fontsize=11, fontweight='bold')
ax.set_title(f'Power Output Contour Map\n(Fixed Temperature: {T_fixed_celsiu
             fontsize=13, fontweight='bold')

# Add colorbar
cbar = fig.colorbar(contourf, ax=ax)
cbar.set_label('Power (W)', fontsize=11, fontweight='bold')
plt.savefig("Contour_plot.png", dpi=300, bbox_inches='tight')

plt.show()

# ===== PRINT SAMPLE PREDICTIONS =====
print("===== Sample Predictions =====\n")
sample_indices = [0, RL_points-1, RL_points*ID_points//2, RL_points*ID_point
for idx in sample_indices:
    if idx < len(RL_flat):
        print(f"RL = {RL_flat[idx]:.2f} Ω, ID = {ID_flat[idx]:.0f} W/m², Pow

# ===== OPTIONAL: Find optimal RL for maximum power at each ID =====
print("\n===== Optimal Load Resistance for Maximum Power =====\n")
print("ID (W/m²)  |  Optimal RL (Ω)  |  Max Power (W)")
print("-" * 50)
for j in range(0, ID_points, ID_points//5):
    max_power_idx = np.argmax(power_grid[j, :])
    optimal_rl = RL_array[max_power_idx]
    max_power = power_grid[j, max_power_idx]
    print(f"{ID_array[j]:6.0f}     |     {optimal_rl:6.2f}      |     {max_p
```

In [ ]:

In [ ]:

In [1]:
```
# CodeP3.4F25 – V.P. Carey, ME249 F25

#import useful packages
import keras
import pandas as pd
from keras.models import Sequential
import numpy as np
import keras.backend as kb
import tensorflow as tf
#the following 2 lines are only needed for Mac OS machines
import os
os.environ['KMP_DUPLICATE_LIB_OK']='True'

#create input data array

#convert to:
# BC's for PINN conduction model with HE enforcement in interior points

# x(cm), y(cm)   These are (x,y) coordinates of boundary location where temp
xdata = []
ND = 32
xdata =  [  [ 0.0, 0.0 ], [ 0.0, 0.5 ], [ 0.0, 1.0 ], [ 0.0, 1.5 ],[ 0.0, 2.
            [ 0.0, 3.5 ], [ 0.0, 4.0 ], [ 0.0, 4.5 ], [ 0.0, 5.0 ],
            [ 3.0, 0.0 ], [ 3.0, 0.5 ], [ 3.0, 1.0 ], [ 3.0, 1.5 ],[ 3.0, 2.
            [ 3.0, 3.5 ], [ 3.0, 4.0 ], [ 3.0, 4.5 ], [ 3.0, 5.0 ]]

xdata.append([  0.5, 0.0 ])
xdata.append([  1.0, 0.0 ])
xdata.append([  1.5, 0.0 ])
xdata.append([  2.0, 0.0 ])
xdata.append([  2.5, 0.0 ])

xdata.append([  0.5, 5.0 ])
xdata.append([  1.0, 5.0 ])
xdata.append([  1.5, 5.0 ])
xdata.append([  2.0, 5.0 ])
xdata.append([  2.5, 5.0 ])


# add more here if needed



xarray= np.array(xdata)
print (xdata)
print (xarray)



#convert to:
# meadian values of output variables

# T (deg C)   These are specified (measured) temperatures for each pair of (
```

```
ydata = []

ydata = [ [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 45. ], [ 40. ], [ 3
         [ 40. ], [ 35. ],  [ 30. ], [ 25. ], [ 20. ], [ 15. ], [ 10. ], [ 1

ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 45. ])

ydata.append([ 15. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])

# add more here if needed

yarray= np.array(ydata)
print (ydata)
print (yarray)

data_inputs = np.array(xdata)
data_outputs = np.array(ydata)

#boundary_x = ...  # x-coordinates for the boundary data
#boundary_y = ...  # y-coordinates for the boundary data
#boundary_T = ...  # Boundary temperature data (T_data)

# Copy the first column to a 1D numpy array
#vector = arr[:, 0].copy()
boundary_xnp = xarray[:, 0].copy()
# Copy the first column to a 1D numpy array
boundary_ynp = xarray[:, 1].copy()
boundary_Tnp = yarray[:, 0].copy()

# Convert the arrays to a TensorFlow tensors
boundary_x = tf.convert_to_tensor(boundary_xnp, dtype=tf.float32)
boundary_y = tf.convert_to_tensor(boundary_ynp, dtype=tf.float32)
boundary_T_ytrain = tf.convert_to_tensor(boundary_Tnp, dtype=tf.float32)

boundary_points_xtrain = tf.stack([boundary_x, boundary_y], axis=1)

print(boundary_points_xtrain)
print(boundary_T_ytrain)
```

```
[[0.0, 0.0], [0.0, 0.5], [0.0, 1.0], [0.0, 1.5], [0.0, 2.0], [0.0, 2.5], [0.
0, 3.0], [0.0, 3.5], [0.0, 4.0], [0.0, 4.5], [0.0, 5.0], [3.0, 0.0], [3.0,
0.5], [3.0, 1.0], [3.0, 1.5], [3.0, 2.0], [3.0, 2.5], [3.0, 3.0], [3.0, 3.
5], [3.0, 4.0], [3.0, 4.5], [3.0, 5.0], [0.5, 0.0], [1.0, 0.0], [1.5, 0.0],
[2.0, 0.0], [2.5, 0.0], [0.5, 5.0], [1.0, 5.0], [1.5, 5.0], [2.0, 5.0], [2.
5, 5.0]]
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]]
[[50.0], [50.0], [50.0], [50.0], [50.0], [45.0], [40.0], [35.0], [30.0], [2
5.0], [20.0], [40.0], [35.0], [30.0], [25.0], [20.0], [15.0], [10.0], [10.
0], [10.0], [10.0], [10.0], [50.0], [50.0], [50.0], [50.0], [45.0], [15.0],
[10.0], [10.0], [10.0], [10.0]]
[[50.]
 [50.]
 [50.]
 [50.]
 [50.]
 [45.]
 [40.]
 [35.]
 [30.]
 [25.]
 [20.]
 [40.]
 [35.]
 [30.]
```

```
 [25.]
 [20.]
 [15.]
 [10.]
 [10.]
 [10.]
 [10.]
 [10.]
 [50.]
 [50.]
 [50.]
 [50.]
 [45.]
 [15.]
 [10.]
 [10.]
 [10.]
 [10.]]
tf.Tensor(
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]], shape=(32, 2), dtype=float32)
tf.Tensor(
[50. 50. 50. 50. 50. 45. 40. 35. 30. 25. 20. 40. 35. 30. 25. 20. 15. 10.
 10. 10. 10. 10. 50. 50. 50. 50. 45. 15. 10. 10. 10. 10.], shape=(32,), dtyp
e=float32)
```

```
2025-11-25 11:17:55.526500: E external/local_xla/xla/stream_executor/cuda/cu
da_driver.cc:266] failed call to cuInit: CUDA_ERROR_NO_DEVICE: no CUDA-capab
le device is detected
```

In [2]:
```
# define neural network model

    #As seen below, we have created four dense layers.
    #A dense layer is a layer in neural network that's fully connected.
    #In other words, all the neurons in one layer are connected to all other
    #In the first layer, we need to provide the input shape, which is 1 in c
    #The activation function we have chosen is elu, which stands for exponen

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Create your model (for example, a simple fully connected feedforward NN)
# Assume 2D input (x, y),  Output temperature T
model = Sequential([
        Dense(5, input_dim=2, activation='elu'),
        Dense(10, activation='elu'),
        Dense(1)
    ])

# Set the kappa value (this can be adjusted as per your needs)
kappa = 5.0

# Set the x_max and y_max values of your domain ( can adjust these according
x_max = 3.0
y_max = 5.0

#print summary of model features
model.summary()
```

```
/global/home/users/melius/.conda/envs/my_kernel/lib/python3.11/site-package
s/keras/src/layers/core/dense.py:92: UserWarning: Do not pass an `input_shap
e`/`input_dim` argument to a layer. When using Sequential models, prefer usi
ng an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Par
dense (Dense)	(None, 5)	
dense_1 (Dense)	(None, 10)	
dense_2 (Dense)	(None, 1)	

Total params: 86 (344.00 B)

Trainable params: 86 (344.00 B)

Non-trainable params: 0 (0.00 B)

In [3]:
```
# ADD CODE HERE for custom_loss_function <<========
# PINN CUSTOM LOSS FUNCTION - V.P. Carey, ME249 F25
#import useful packages
```

```
import tensorflow as tf
from tensorflow.keras import backend as K


def custom_loss_function(kappa, x_max, y_max, epoch_tracker):
    """ Custom loss function for the neural network model.
        - kappa: weight for the Laplace residual term.
        - x_max, y_max: maximum x and y coordinates of the domain.
        - epoch_tracker: callback object to keep track of the current epoch.

    def loss(y_true, y_pred):
        # Define pointsBoundaryChk as a constant tensor
        # === COMPLETE 32 BOUNDARY POINTS FROM THE PROJECT FIGURE ===
        pointsBoundaryChk = tf.constant([
            # Bottom edge (y = 0)
            [0.0, 0.0],
            [0.5, 0.0],
            [1.0, 0.0],
            [1.5, 0.0],
            [2.0, 0.0],
            [2.5, 0.0],
            [3.0, 0.0],

            # Right edge (x = 3)
            [3.0, 0.5],
            [3.0, 1.0],
            [3.0, 1.5],
            [3.0, 2.0],
            [3.0, 2.5],
            [3.0, 3.0],
            [3.0, 3.5],
            [3.0, 4.0],
            [3.0, 4.5],
            [3.0, 5.0],

            # Top edge (y = 5)
            [0.5, 5.0],
            [1.0, 5.0],
            [1.5, 5.0],
            [2.0, 5.0],
            [2.5, 5.0],
            [3.0, 5.0],

            # Left edge (x = 0)
            [0.0, 0.5],
            [0.0, 1.0],
            [0.0, 1.5],
            [0.0, 2.0],
            [0.0, 2.5],
            [0.0, 3.0],
            [0.0, 3.5],
            [0.0, 4.0],
            [0.0, 4.5],
            [0.0, 5.0]
```

```
        ], dtype=tf.float32)


        # Boundary loss: Calculate squared error for the boundary condition
        boundary_loss = K.mean(K.square(y_pred - y_true))

        ESbdsum = 0.
        for _ in range(32):
            # Sample a random index and gather point
            idx = tf.random.uniform(shape=[], minval=0, maxval=32, dtype=tf.
            selected_point = tf.gather(pointsBoundaryChk, idx)

            # Stack selected_point as input for the model (shape becomes (1,
            input_tensor = tf.expand_dims(selected_point, axis=0)
            # Pass input_tensor as the input to the model
            T_pred_at_boundxy = model(input_tensor)  # Model prediction
            # add penalty if temperature at flux location is OUTSIDE RANGE 5
        #     if (T_pred_at_boundxy < 5.):
        #         ESbdsum += 20.*(5. - T_pred_at_boundxy)*(5. - T_pred_at_bo
        #     if (T_pred_at_boundxy > 55.):
        #         ESbdsum += 20.*(T_pred_at_boundxy - 55.)*(T_pred_at_boundx
        # boundary_loss += ESbdsum/32.
        # Compute lower and upper penalties without Python if-statements
            low_penalty  = tf.nn.relu(5.0  - T_pred_at_boundxy)
            high_penalty = tf.nn.relu(T_pred_at_boundxy - 55.0)

            # Square and scale (20×), then average over 32 samples
            ESbdsum += (20.0 * tf.square(low_penalty)  +
                        20.0 * tf.square(high_penalty)) / 32.0

        boundary_loss += ESbdsum


        # Check epoch from the callback
        epoch = epoch_tracker.epochs_trained

        # If current epoch is less than N_epInit, ignore the Laplace residua
        if epoch < epoch_tracker.N_epInit:
            total_loss = boundary_loss
        else:
            # 3. Randomly select an interior point from this list using Tens
            points = tf.constant([
                # x = 0.5
                [0.5, 0.5],
                [0.5, 1.0],
                [0.5, 1.5],
                [0.5, 2.0],
                [0.5, 2.5],
                [0.5, 3.0],
                [0.5, 3.5],
                [0.5, 4.0],
                [0.5, 4.5],

                # x = 1.0
                [1.0, 0.5],
                [1.0, 1.0],
```

```
            [1.0, 1.5],
            [1.0, 2.0],
            [1.0, 2.5],
            [1.0, 3.0],
            [1.0, 3.5],
            [1.0, 4.0],
            [1.0, 4.5],

            # x = 1.5
            [1.5, 0.5],
            [1.5, 1.0],
            [1.5, 1.5],
            [1.5, 2.0],
            [1.5, 2.5],
            [1.5, 3.0],
            [1.5, 3.5],
            [1.5, 4.0],
            [1.5, 4.5],

            # x = 2.0
            [2.0, 0.5],
            [2.0, 1.0],
            [2.0, 1.5],
            [2.0, 2.0],
            [2.0, 2.5],
            [2.0, 3.0],
            [2.0, 3.5],
            [2.0, 4.0],
            [2.0, 4.5],

            # x = 2.5
            [2.5, 0.5],
            [2.5, 1.0],
            [2.5, 1.5],
            [2.5, 2.0],
            [2.5, 2.5],
            [2.5, 3.0],
            [2.5, 3.5],
            [2.5, 4.0],
            [2.5, 4.5]
        ], dtype=tf.float32)


        meanLap_res = 0.
        meanTlimPen = 0.
        for _ in range(45):
            ##compute Laplacian residue here - need to do this once per
            # Use tf.random.uniform to sample an index for the interior
            idx = tf.random.uniform(shape=[], minval=0, maxval=45, dtype
            selected_point = tf.gather(points, idx)
            x_interior, y_interior = selected_point[0], selected_point[1

            # Use tf.GradientTape to compute second derivatives
            with tf.GradientTape(persistent=True) as tape:
                tape.watch(selected_point)
                # Pass the selected (x_interior, y_interior) to the mode
```

```
                    T_pred_at_interior = model([x_interior, y_interior])  #

                # Compute first derivatives with respect to x and y
                dT_dx = tape.gradient(T_pred_at_interior, selected_point[0])
                dT_dy = tape.gradient(T_pred_at_interior, selected_point[1])

                # Compute second derivatives (Hessian)
                d2T_dx2 = tape.gradient(dT_dx, selected_point[0])
                d2T_dy2 = tape.gradient(dT_dy, selected_point[1])

                # Calculate the Laplacian (sum of second derivatives)
                laplacian = d2T_dx2 + d2T_dy2

                # Compute the Laplacian residual (sum of squared Laplacian)
                meanLap_res +=  K.square(laplacian)/45.

                # ADD PENALTY IF T below 5 or above 55

                # if (T_pred_at_interior < 5.):
                #     meanTlimPen += 20.*(5. - T_pred_at_interior)*(5. - T_p
                low_penalty = tf.nn.relu(5.0 - T_pred_at_interior)
                meanTlimPen += 20.0 * tf.square(low_penalty) / 45.0

                # if (T_pred_at_interior > 55.):
                #     meanTlimPen += 20.*(T_pred_at_interior - 55.)*(T_pred_
                high_penalty = tf.nn.relu(T_pred_at_interior - 55.0)
                meanTlimPen += 20.0 * tf.square(high_penalty) / 45.0


            #Combine three losses
            total_loss = boundary_loss + kappa * meanLap_res + meanTlimPen


        return total_loss
    return loss


from tensorflow.keras.callbacks import Callback

class EpochTracker(Callback):
    def __init__(self, N_epInit):
        super().__init__()
        self.N_epInit = N_epInit
        self.epochs_trained = 0

    def on_epoch_end(self, epoch, logs=None):
        self.epochs_trained = epoch

# Set the loss function with the required arguments
epoch_tracker = EpochTracker(N_epInit=2)
custom_loss = custom_loss_function(kappa=5.0, x_max=3.0, y_max=5.0, epoch_tr



# Compile the model with the custom loss
model.compile(optimizer='adam', loss=custom_loss)
```

In [8]:
```
#### After the compilation of the model, we'll try the fit method with 100 e


#The fit method takes three parameters; namely, x, y, and number of epochs.
#During model training, if all the batches of data are seen by the model onc
#we say that one epoch has been completed.


##### RUN FIT

# Assuming `boundary_points` and `boundary_T` are TensorFlow tensors

# Now, fit the model and pass the callback to `fit()`
historyData = model.fit(boundary_points_xtrain, boundary_T_ytrain,
                epochs=100000, batch_size=32,
                callbacks=[epoch_tracker])


loss_hist = historyData.history['loss']
#The above line will return a dictionary, access it's info like this:
best_epoch = np.argmin(historyData.history['loss']) + 1
print ('best epoch = ', best_epoch)
print('smallest loss =', np.min(loss_hist))

predictions = model.predict(data_inputs)
```

```
Epoch 1/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1580
Epoch 2/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1580
Epoch 3/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1580
Epoch 4/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1580
Epoch 5/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 6/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1579
Epoch 7/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1579
Epoch 8/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1579
Epoch 9/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 10/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1579
Epoch 11/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1579
Epoch 12/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 13/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 14/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 15/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 16/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 17/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 18/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 19/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 20/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 21/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 22/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1578
Epoch 23/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 24/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1578
Epoch 25/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 26/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1578
Epoch 27/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 28/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1578
```

```
Epoch 29/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 30/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 31/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 32/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 33/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 34/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 35/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1578
Epoch 36/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 37/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 38/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 39/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 40/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 41/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1577
Epoch 42/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 43/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 44/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 45/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 46/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 47/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1577
Epoch 48/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 49/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 50/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 51/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 52/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1576
Epoch 53/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1576
Epoch 54/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1576
Epoch 55/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1576
Epoch 56/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1576
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Epoch 57/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 58/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1577
Epoch 59/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 60/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 61/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 62/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1578
Epoch 63/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 64/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1580
Epoch 65/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1580
Epoch 66/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1581
Epoch 67/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1580
Epoch 68/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1579
Epoch 69/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1578
Epoch 70/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1576
Epoch 71/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1575
Epoch 72/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1575
Epoch 73/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1576
Epoch 74/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1576
Epoch 75/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1577
Epoch 76/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1577
Epoch 77/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1576
Epoch 78/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
Epoch 79/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
Epoch 80/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
Epoch 81/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
Epoch 82/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
Epoch 83/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
Epoch 84/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
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Epoch 85/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1575
Epoch 86/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1574
Epoch 87/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1574
Epoch 88/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1574
Epoch 89/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1574
Epoch 90/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1574
Epoch 91/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1574
Epoch 92/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1574
Epoch 93/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1574
Epoch 94/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1574
Epoch 95/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1574
Epoch 96/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1574
Epoch 97/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1574
Epoch 98/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1574
Epoch 99/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1574
Epoch 100/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1573
Epoch 101/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1573
Epoch 102/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1573
Epoch 103/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 104/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 105/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 106/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 107/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 108/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 109/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 110/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 111/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 112/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1573
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Epoch 113/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1573
Epoch 114/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1573
Epoch 115/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 116/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1572
Epoch 117/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 118/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1572
Epoch 119/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 120/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1572
Epoch 121/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 122/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1572
Epoch 123/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 124/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 125/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 126/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1572
Epoch 127/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 128/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 129/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1572
Epoch 130/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1571
Epoch 131/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 132/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 133/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1571
Epoch 134/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1571
Epoch 135/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 136/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1571
Epoch 137/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 138/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 139/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 140/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
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Epoch 141/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 142/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 143/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 144/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 145/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1571
Epoch 146/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 147/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 148/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 149/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 150/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 151/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1570
Epoch 152/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1570
Epoch 153/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1570
Epoch 154/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1570
Epoch 155/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 156/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 157/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 158/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1570
Epoch 159/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 160/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1570
Epoch 161/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 162/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 163/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 164/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 165/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 166/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1569
Epoch 167/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1569
Epoch 168/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
```

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Epoch 169/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1569
Epoch 170/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 171/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 172/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1569
Epoch 173/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 174/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 175/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1569
Epoch 176/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 177/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 178/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1568
Epoch 179/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 180/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 181/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 182/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 183/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 184/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1568
Epoch 185/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 186/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 187/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 188/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 189/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 190/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1568
Epoch 191/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1568
Epoch 192/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 193/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 194/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 195/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 196/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1567
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Epoch 197/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 198/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 199/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 200/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 201/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 202/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1567
Epoch 203/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 204/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 205/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 206/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 207/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1567
Epoch 208/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 209/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 210/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 211/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
Epoch 212/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1566
Epoch 213/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
Epoch 214/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
Epoch 215/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 216/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
Epoch 217/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
Epoch 218/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 219/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1566
Epoch 220/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 221/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 222/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1566
Epoch 223/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 224/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
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Epoch 225/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 226/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 227/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1567
Epoch 228/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1567
Epoch 229/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1568
Epoch 230/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1569
Epoch 231/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 232/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1570
Epoch 233/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1570
Epoch 234/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1569
Epoch 235/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1567
Epoch 236/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
Epoch 237/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1565
Epoch 238/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1565
Epoch 239/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1565
Epoch 240/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1566
Epoch 241/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1566
Epoch 242/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1566
Epoch 243/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1565
Epoch 244/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1565
Epoch 245/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1564
Epoch 246/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1564
Epoch 247/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 248/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1565
Epoch 249/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1565
Epoch 250/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 251/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 252/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1564
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Epoch 253/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 254/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 255/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 256/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 257/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 258/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1564
Epoch 259/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 260/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 261/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1563
Epoch 262/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1563
Epoch 263/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 264/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 265/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 266/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1563
Epoch 267/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 268/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 269/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 270/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1563
Epoch 271/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 272/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 273/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 274/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 275/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1562
Epoch 276/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 277/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 278/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 279/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 280/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
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Epoch 281/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 282/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 283/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 284/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 285/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1562
Epoch 286/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 287/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1561
Epoch 288/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 289/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 290/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 291/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1561
Epoch 292/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 293/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1561
Epoch 294/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 295/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 296/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 297/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1561
Epoch 298/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 299/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 300/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1561
Epoch 301/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1561
Epoch 302/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1560
Epoch 303/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1560
Epoch 304/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1560
Epoch 305/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1560
Epoch 306/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1560
Epoch 307/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1560
Epoch 308/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1560
```

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Epoch 309/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1560
Epoch 310/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1560
Epoch 311/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1560
Epoch 312/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1560
Epoch 313/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1560
Epoch 314/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1560
Epoch 315/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1560
Epoch 316/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1560
Epoch 317/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 318/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 319/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 320/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 321/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 322/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 323/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1559
Epoch 324/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 60ms/step - loss: 0.1559
Epoch 325/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 326/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1559
Epoch 327/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 328/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 329/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1559
Epoch 330/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1559
Epoch 331/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 332/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 333/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 334/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1558
Epoch 335/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 336/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
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Epoch 337/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1558
Epoch 338/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 339/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 340/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 341/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 342/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 343/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 344/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 345/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 346/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 347/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 348/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1557
Epoch 349/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 350/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 351/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1557
Epoch 352/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 353/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 354/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 355/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 356/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 357/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 358/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1557
Epoch 359/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 360/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 361/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 362/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1557
Epoch 363/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 364/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1556
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Epoch 365/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 366/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 367/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1556
Epoch 368/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 369/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 370/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1556
Epoch 371/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 372/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 373/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 374/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 375/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 376/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 377/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 378/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 379/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1555
Epoch 380/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 381/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 382/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1555
Epoch 383/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 384/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 385/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 386/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 387/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 388/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 389/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1555
Epoch 390/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 391/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 392/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
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Epoch 393/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 394/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1555
Epoch 395/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 396/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 397/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 398/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1557
Epoch 399/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1558
Epoch 400/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 401/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 402/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1559
Epoch 403/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1558
Epoch 404/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1556
Epoch 405/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1555
Epoch 406/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 60ms/step - loss: 0.1554
Epoch 407/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1554
Epoch 408/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1554
Epoch 409/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1555
Epoch 410/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1555
Epoch 411/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1555
Epoch 412/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1554
Epoch 413/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1554
Epoch 414/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1553
Epoch 415/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 416/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 417/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1554
Epoch 418/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1554
Epoch 419/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1554
Epoch 420/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
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Epoch 421/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 422/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 423/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 424/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 425/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 426/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 427/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 428/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 429/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1552
Epoch 430/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 431/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 432/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 433/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 434/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1552
Epoch 435/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 436/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1552
Epoch 437/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1552
Epoch 438/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 439/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1552
Epoch 440/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1552
Epoch 441/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
Epoch 442/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
Epoch 443/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
Epoch 444/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1551
Epoch 445/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1551
Epoch 446/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
Epoch 447/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1551
Epoch 448/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
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Epoch 449/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
Epoch 450/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1551
Epoch 451/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1551
Epoch 452/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1551
Epoch 453/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1551
Epoch 454/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
Epoch 455/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1551
Epoch 456/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1550
Epoch 457/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1550
Epoch 458/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1550
Epoch 459/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1550
Epoch 460/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1550
Epoch 461/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1550
Epoch 462/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1550
Epoch 463/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 464/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 465/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 466/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 467/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 468/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 469/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 470/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1550
Epoch 471/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1549
Epoch 472/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1549
Epoch 473/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1549
Epoch 474/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 475/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 476/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
```

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Epoch 477/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1549
Epoch 478/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 479/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1549
Epoch 480/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 481/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 482/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 483/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 484/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1549
Epoch 485/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1549
Epoch 486/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 487/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 488/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 489/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 490/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1548
Epoch 491/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 492/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 493/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 494/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 495/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 496/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1548
Epoch 497/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1548
Epoch 498/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1548
Epoch 499/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1548
Epoch 500/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1548
Epoch 501/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1547
Epoch 502/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1547
Epoch 503/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1547
Epoch 504/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1547
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Epoch 505/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 506/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1547
Epoch 507/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1547
Epoch 508/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1547
Epoch 509/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 510/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 511/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 512/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1547
Epoch 513/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 514/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1547
Epoch 515/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 516/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1546
Epoch 517/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1546
Epoch 518/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1546
Epoch 519/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1546
Epoch 520/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1546
Epoch 521/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1546
Epoch 522/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1546
Epoch 523/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1546
Epoch 524/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1546
Epoch 525/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1546
Epoch 526/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1546
Epoch 527/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1546
Epoch 528/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1546
Epoch 529/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1546
Epoch 530/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1546
Epoch 531/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 532/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1545
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Epoch 533/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 534/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 535/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 536/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 537/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 538/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1545
Epoch 539/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 540/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 541/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 542/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 543/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 544/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1545
Epoch 545/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1545
Epoch 546/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 547/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 548/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 549/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 550/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 551/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 552/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 553/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 554/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1544
Epoch 555/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1544
Epoch 556/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 557/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1544
Epoch 558/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 559/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 560/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
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Epoch 561/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 562/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1543
Epoch 563/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 564/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1543
Epoch 565/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 566/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 567/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 568/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 569/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1543
Epoch 570/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1543
Epoch 571/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 572/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 573/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 574/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 575/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 576/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 577/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 578/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 579/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 580/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 581/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 582/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1544
Epoch 583/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 584/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 585/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1549
Epoch 586/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1552
Epoch 587/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1553
Epoch 588/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1551
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Epoch 589/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1547
Epoch 590/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 591/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1541
Epoch 592/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 593/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1545
Epoch 594/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1546
Epoch 595/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1544
Epoch 596/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1542
Epoch 597/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1541
Epoch 598/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1542
Epoch 599/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 600/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1543
Epoch 601/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1542
Epoch 602/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1541
Epoch 603/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1541
Epoch 604/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1542
Epoch 605/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1542
Epoch 606/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1541
Epoch 607/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1541
Epoch 608/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1540
Epoch 609/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1541
Epoch 610/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1541
Epoch 611/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1541
Epoch 612/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 613/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 614/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 615/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 616/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
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Epoch 617/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 618/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1540
Epoch 619/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 620/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 621/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 622/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1540
Epoch 623/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 624/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 625/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 626/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 627/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1539
Epoch 628/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 629/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1539
Epoch 630/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 631/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 632/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 633/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1539
Epoch 634/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 635/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1539
Epoch 636/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 637/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 638/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 639/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 640/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 641/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 642/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 643/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 644/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
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Epoch 645/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 646/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 647/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1538
Epoch 648/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 649/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 650/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1538
Epoch 651/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1537
Epoch 652/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1537
Epoch 653/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1537
Epoch 654/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1537
Epoch 655/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1537
Epoch 656/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1537
Epoch 657/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1537
Epoch 658/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1537
Epoch 659/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1537
Epoch 660/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1537
Epoch 661/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1537
Epoch 662/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1537
Epoch 663/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1537
Epoch 664/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1537
Epoch 665/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1537
Epoch 666/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1537
Epoch 667/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 668/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 669/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1536
Epoch 670/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 671/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 672/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
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Epoch 673/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 674/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 675/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1536
Epoch 676/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 677/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1536
Epoch 678/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 679/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 680/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 681/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1536
Epoch 682/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1535
Epoch 683/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1535
Epoch 684/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1535
Epoch 685/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1535
Epoch 686/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1535
Epoch 687/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1535
Epoch 688/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1535
Epoch 689/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1535
Epoch 690/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1535
Epoch 691/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1535
Epoch 692/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1535
Epoch 693/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1535
Epoch 694/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1535
Epoch 695/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1535
Epoch 696/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1534
Epoch 697/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1534
Epoch 698/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1534
Epoch 699/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1534
Epoch 700/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1534
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Epoch 701/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1534
Epoch 702/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1534
Epoch 703/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1534
Epoch 704/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1534
Epoch 705/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1534
Epoch 706/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1534
Epoch 707/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1534
Epoch 708/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1534
Epoch 709/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1534
Epoch 710/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1534
Epoch 711/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 712/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1533
Epoch 713/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1533
Epoch 714/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 715/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 716/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1533
Epoch 717/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1533
Epoch 718/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1533
Epoch 719/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 720/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 721/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 722/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 723/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 724/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1533
Epoch 725/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1533
Epoch 726/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 727/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 728/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
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Epoch 729/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 730/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 731/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 732/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 733/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 734/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 735/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1532
Epoch 736/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 737/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1532
Epoch 738/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 739/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1532
Epoch 740/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 741/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 742/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 743/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 744/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 745/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 746/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 747/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1531
Epoch 748/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1531
Epoch 749/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 750/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 751/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 752/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1531
Epoch 753/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1531
Epoch 754/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1531
Epoch 755/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 756/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1530
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Epoch 757/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 758/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 759/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 760/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1530
Epoch 761/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 762/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 763/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 764/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 765/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1530
Epoch 766/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1530
Epoch 767/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1530
Epoch 768/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1530
Epoch 769/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1529
Epoch 770/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 771/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1529
Epoch 772/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 773/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 774/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 775/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 776/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 777/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 778/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 779/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1529
Epoch 780/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 781/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 782/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1529
Epoch 783/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 784/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1528
```

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Epoch 785/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 786/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 787/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 788/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 789/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 790/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1528
Epoch 791/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 792/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 793/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 794/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 795/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 796/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1528
Epoch 797/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 798/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 799/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 800/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1527
Epoch 801/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1527
Epoch 802/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1527
Epoch 803/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 804/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 805/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 806/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 807/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 808/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1527
Epoch 809/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1527
Epoch 810/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1527
Epoch 811/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1526
Epoch 812/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
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Epoch 813/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 814/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 815/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 816/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 817/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 818/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 819/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1526
Epoch 820/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1526
Epoch 821/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1526
Epoch 822/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1526
Epoch 823/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 824/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 825/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1525
Epoch 826/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 827/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1525
Epoch 828/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 829/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 830/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 831/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 832/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1525
Epoch 833/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 834/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 835/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 59ms/step - loss: 0.1525
Epoch 836/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 837/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1525
Epoch 838/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 839/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 840/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
```

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Epoch 841/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 842/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1524
Epoch 843/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 844/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 845/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 846/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 847/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 848/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 849/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1524
Epoch 850/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1524
Epoch 851/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1524
Epoch 852/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1523
Epoch 853/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1523
Epoch 854/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1523
Epoch 855/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 856/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 857/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 858/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 859/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 860/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 861/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 862/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 863/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 864/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 865/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1525
Epoch 866/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 867/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1527
Epoch 868/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1528
```

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Epoch 869/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 870/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1528
Epoch 871/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1526
Epoch 872/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1524
Epoch 873/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1522
Epoch 874/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1522
Epoch 875/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1522
Epoch 876/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1523
Epoch 877/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1524
Epoch 878/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1524
Epoch 879/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1523
Epoch 880/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1522
Epoch 881/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 882/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 883/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1522
Epoch 884/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - loss: 0.1522
Epoch 885/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1522
Epoch 886/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1522
Epoch 887/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 888/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 889/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 890/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 891/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 892/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 893/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1521
Epoch 894/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1520
Epoch 895/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1520
Epoch 896/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
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Epoch 897/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1520
Epoch 898/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1520
Epoch 899/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
Epoch 900/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
Epoch 901/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
Epoch 902/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
Epoch 903/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
Epoch 904/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1520
Epoch 905/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
Epoch 906/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1520
Epoch 907/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 908/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 909/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 910/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 911/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1519
Epoch 912/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 913/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 914/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 915/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1519
Epoch 916/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 917/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 918/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 919/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 920/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1519
Epoch 921/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 922/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 923/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 924/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
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Epoch 925/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 926/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 927/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 60ms/step - loss: 0.1518
Epoch 928/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 929/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 930/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 931/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 932/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 933/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1518
Epoch 934/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1517
Epoch 935/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1517
Epoch 936/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1517
Epoch 937/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1517
Epoch 938/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1517
Epoch 939/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1517
Epoch 940/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1517
Epoch 941/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1517
Epoch 942/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1517
Epoch 943/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1517
Epoch 944/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1517
Epoch 945/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1517
Epoch 946/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1517
Epoch 947/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 948/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 949/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 950/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 951/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 952/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
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Epoch 953/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 954/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 955/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 956/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 957/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1516
Epoch 958/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 959/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 960/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1516
Epoch 961/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1515
Epoch 962/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1515
Epoch 963/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1515
Epoch 964/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1515
Epoch 965/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1515
Epoch 966/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1515
Epoch 967/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1515
Epoch 968/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1515
Epoch 969/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1515
Epoch 970/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1515
Epoch 971/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1515
Epoch 972/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1515
Epoch 973/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1515
Epoch 974/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1514
Epoch 975/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1514
Epoch 976/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1514
Epoch 977/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1514
Epoch 978/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1514
Epoch 979/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1514
Epoch 980/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1514
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Epoch 981/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1514
Epoch 982/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1514
Epoch 983/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1514
Epoch 984/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1514
Epoch 985/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1514
Epoch 986/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1514
Epoch 987/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1513
Epoch 988/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 59ms/step - loss: 0.1513
Epoch 989/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1513
Epoch 990/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1513
Epoch 991/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1513
Epoch 992/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1513
Epoch 993/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1513
Epoch 994/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1513
Epoch 995/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1513
Epoch 996/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1513
Epoch 997/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1513
Epoch 998/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1513
Epoch 999/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1513
Epoch 1000/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1001/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1002/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1003/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1004/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1005/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1006/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1512
Epoch 1007/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1008/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
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Epoch 1009/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1512
Epoch 1010/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1512
Epoch 1011/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1512
Epoch 1012/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1512
Epoch 1013/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1511
Epoch 1014/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1511
Epoch 1015/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1016/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1017/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1018/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1019/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1020/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1021/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1022/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1023/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1024/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1511
Epoch 1025/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1026/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1027/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1028/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1029/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1030/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1510
Epoch 1031/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1032/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1033/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1034/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1035/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1036/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1510
```

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Epoch 1037/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
Epoch 1038/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1039/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1040/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1509
Epoch 1041/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1509
Epoch 1042/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1509
Epoch 1043/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1509
Epoch 1044/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1045/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1509
Epoch 1046/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1047/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1048/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1509
Epoch 1049/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1050/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1051/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1052/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1508
Epoch 1053/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1054/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1055/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1508
Epoch 1056/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1057/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1058/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1059/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1508
Epoch 1060/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1061/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1508
Epoch 1062/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1063/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1064/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1510
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Epoch 1065/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1511
Epoch 1066/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1512
Epoch 1067/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1513
Epoch 1068/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1513
Epoch 1069/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1512
Epoch 1070/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1511
Epoch 1071/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1509
Epoch 1072/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1507
Epoch 1073/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1507
Epoch 1074/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1507
Epoch 1075/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1508
Epoch 1076/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1509
Epoch 1077/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1508
Epoch 1078/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1508
Epoch 1079/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1507
Epoch 1080/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1506
Epoch 1081/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1506
Epoch 1082/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1506
Epoch 1083/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1507
Epoch 1084/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1507
Epoch 1085/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1506
Epoch 1086/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1506
Epoch 1087/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1506
Epoch 1088/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1506
Epoch 1089/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1506
Epoch 1090/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1506
Epoch 1091/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1506
Epoch 1092/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1506
```

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Epoch 1093/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1505
Epoch 1094/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1505
Epoch 1095/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1505
Epoch 1096/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1505
Epoch 1097/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1505
Epoch 1098/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1505
Epoch 1099/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1505
Epoch 1100/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1505
Epoch 1101/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1504
Epoch 1102/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1504
Epoch 1103/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1104/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1504
Epoch 1105/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1106/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1107/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1108/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1504
Epoch 1109/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1110/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1111/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1112/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1504
Epoch 1113/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1504
Epoch 1114/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1503
Epoch 1115/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1116/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1117/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1118/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1119/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1503
Epoch 1120/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
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Epoch 1121/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1503
Epoch 1122/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1123/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1124/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1125/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1503
Epoch 1126/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1127/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1502
Epoch 1128/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1129/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1130/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1131/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1132/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1133/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1134/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1135/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1136/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1502
Epoch 1137/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1502
Epoch 1138/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1139/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1140/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1141/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1142/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1143/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1144/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1501
Epoch 1145/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1146/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1147/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1148/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
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Epoch 1149/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1501
Epoch 1150/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1501
Epoch 1151/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1152/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1153/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1154/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1155/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1156/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1157/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1158/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1159/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1160/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1161/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1162/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1500
Epoch 1163/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1164/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1165/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1166/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1167/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1168/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1169/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1170/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1171/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1172/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1173/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1174/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1499
Epoch 1175/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1176/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
```

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Epoch 1177/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1178/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1179/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1180/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1498
Epoch 1181/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1182/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1183/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1184/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1185/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1186/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1498
Epoch 1187/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1188/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1189/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1190/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1191/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1497
Epoch 1192/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1497
Epoch 1193/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1194/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1195/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1497
Epoch 1196/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1497
Epoch 1197/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1198/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1496
Epoch 1199/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1200/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1201/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1496
Epoch 1202/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1203/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1204/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
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Epoch 1205/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1206/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1207/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1208/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1209/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1496
Epoch 1210/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1211/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1212/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1213/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1214/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1215/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1216/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1217/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1218/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1219/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1220/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1221/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
Epoch 1222/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1494
Epoch 1223/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1494
Epoch 1224/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1494
Epoch 1225/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1494
Epoch 1226/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1494
Epoch 1227/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1494
Epoch 1228/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1494
Epoch 1229/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1494
Epoch 1230/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1494
Epoch 1231/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1494
Epoch 1232/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1494
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Epoch 1233/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1234/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1235/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1236/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1237/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1238/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1239/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1493
Epoch 1240/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1241/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1493
Epoch 1242/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1243/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1244/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1245/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1246/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1492
Epoch 1247/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1248/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1249/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1250/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1251/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1492
Epoch 1252/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1253/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1254/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1255/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1256/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1257/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1493
Epoch 1258/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1493
Epoch 1259/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1494
Epoch 1260/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1495
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Epoch 1261/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1496
Epoch 1262/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1263/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1497
Epoch 1264/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1496
Epoch 1265/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1494
Epoch 1266/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1267/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1491
Epoch 1268/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1491
Epoch 1269/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1491
Epoch 1270/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1271/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1493
Epoch 1272/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1492
Epoch 1273/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1491
Epoch 1274/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1490
Epoch 1275/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1490
Epoch 1276/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1490
Epoch 1277/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1491
Epoch 1278/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1491
Epoch 1279/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1490
Epoch 1280/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1490
Epoch 1281/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1282/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1283/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1489
Epoch 1284/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1490
Epoch 1285/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1286/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1287/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1288/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
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Epoch 1289/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1290/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1291/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1292/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1489
Epoch 1293/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1488
Epoch 1294/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1488
Epoch 1295/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1488
Epoch 1296/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1488
Epoch 1297/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1488
Epoch 1298/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1488
Epoch 1299/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1488
Epoch 1300/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1488
Epoch 1301/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1488
Epoch 1302/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1303/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1304/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1305/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1306/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1487
Epoch 1307/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1308/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1309/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1310/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1311/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1487
Epoch 1312/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1487
Epoch 1313/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1486
Epoch 1314/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
Epoch 1315/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
Epoch 1316/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
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Epoch 1317/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
Epoch 1318/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1486
Epoch 1319/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1486
Epoch 1320/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
Epoch 1321/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
Epoch 1322/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
Epoch 1323/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1486
Epoch 1324/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1325/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1326/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1327/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1328/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1329/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1330/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1485
Epoch 1331/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1485
Epoch 1332/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1333/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1334/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1335/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1485
Epoch 1336/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1337/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1338/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1339/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1340/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1341/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1342/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1484
Epoch 1343/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1344/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
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Epoch 1345/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1346/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1484
Epoch 1347/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1483
Epoch 1348/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1483
Epoch 1349/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1350/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1351/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1352/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1353/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1483
Epoch 1354/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1355/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1356/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1357/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1483
Epoch 1358/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1359/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1360/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1482
Epoch 1361/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1362/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1363/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1364/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1365/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1366/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1482
Epoch 1367/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1482
Epoch 1368/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1482
Epoch 1369/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1481
Epoch 1370/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1481
Epoch 1371/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1481
Epoch 1372/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1481
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Epoch 1373/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1481
Epoch 1374/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1481
Epoch 1375/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1481
Epoch 1376/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1481
Epoch 1377/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1481
Epoch 1378/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1481
Epoch 1379/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1380/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1381/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1382/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1383/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1480
Epoch 1384/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1480
Epoch 1385/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1480
Epoch 1386/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1387/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1388/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1389/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1480
Epoch 1390/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1391/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1392/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1393/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1394/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1395/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1396/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1397/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1398/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1399/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1400/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
```

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Epoch 1401/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1402/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1403/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1404/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1405/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1406/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1407/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1408/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1409/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1410/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1411/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1477
Epoch 1412/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1477
Epoch 1413/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1477
Epoch 1414/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1477
Epoch 1415/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1477
Epoch 1416/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1477
Epoch 1417/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1477
Epoch 1418/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1477
Epoch 1419/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1477
Epoch 1420/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1477
Epoch 1421/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1422/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1423/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1424/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1425/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1476
Epoch 1426/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1427/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1476
Epoch 1428/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
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Epoch 1429/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1430/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1431/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1476
Epoch 1432/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1433/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1434/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1435/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1436/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1437/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1438/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1439/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1440/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1441/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1442/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1443/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1444/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1445/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1446/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1447/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1448/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1449/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1450/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1474
Epoch 1451/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1452/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1453/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1454/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1455/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1473
Epoch 1456/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
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Epoch 1457/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1473
Epoch 1458/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1459/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1460/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1461/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1473
Epoch 1462/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1463/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1472
Epoch 1464/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1472
Epoch 1465/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1472
Epoch 1466/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1467/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1473
Epoch 1468/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1469/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1474
Epoch 1470/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1471/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1476
Epoch 1472/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1473/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1479
Epoch 1474/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1479
Epoch 1475/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1478
Epoch 1476/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1475
Epoch 1477/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1472
Epoch 1478/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1471
Epoch 1479/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1472
Epoch 1480/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1473
Epoch 1481/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1474
Epoch 1482/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1473
Epoch 1483/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1472
Epoch 1484/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
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Epoch 1485/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
Epoch 1486/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1471
Epoch 1487/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1472
Epoch 1488/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1471
Epoch 1489/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1471
Epoch 1490/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
Epoch 1491/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
Epoch 1492/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
Epoch 1493/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
Epoch 1494/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
Epoch 1495/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1470
Epoch 1496/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1469
Epoch 1497/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1469
Epoch 1498/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1469
Epoch 1499/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1469
Epoch 1500/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1469
Epoch 1501/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1469
Epoch 1502/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1469
Epoch 1503/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1468
Epoch 1504/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1469
Epoch 1505/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1469
Epoch 1506/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
Epoch 1507/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
Epoch 1508/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
Epoch 1509/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
Epoch 1510/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
Epoch 1511/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
Epoch 1512/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
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Epoch 1513/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1468
Epoch 1514/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1515/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1516/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1517/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1518/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1519/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1520/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1521/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1467
Epoch 1522/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1523/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1467
Epoch 1524/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1525/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1526/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1527/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1528/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1529/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1530/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1531/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1532/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1466
Epoch 1533/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1466
Epoch 1534/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1465
Epoch 1535/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1465
Epoch 1536/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1465
Epoch 1537/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1465
Epoch 1538/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1465
Epoch 1539/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1465
Epoch 1540/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1465
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Epoch 1541/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1465
Epoch 1542/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1465
Epoch 1543/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1465
Epoch 1544/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1464
Epoch 1545/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1464
Epoch 1546/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1464
Epoch 1547/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1464
Epoch 1548/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1464
Epoch 1549/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1464
Epoch 1550/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1464
Epoch 1551/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1464
Epoch 1552/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1464
Epoch 1553/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1464
Epoch 1554/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1463
Epoch 1555/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1463
Epoch 1556/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1463
Epoch 1557/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1463
Epoch 1558/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1463
Epoch 1559/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1463
Epoch 1560/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1463
Epoch 1561/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1463
Epoch 1562/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1463
Epoch 1563/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1463
Epoch 1564/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
Epoch 1565/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1462
Epoch 1566/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
Epoch 1567/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
Epoch 1568/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
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Epoch 1569/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
Epoch 1570/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
Epoch 1571/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1462
Epoch 1572/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
Epoch 1573/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1462
Epoch 1574/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1575/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1461
Epoch 1576/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1577/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1578/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1579/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1580/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1581/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1582/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1461
Epoch 1583/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1461
Epoch 1584/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1585/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1586/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1587/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1588/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1460
Epoch 1589/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1590/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1460
Epoch 1591/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1592/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1593/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1460
Epoch 1594/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1459
Epoch 1595/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1459
Epoch 1596/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1459
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Epoch 1597/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1459
Epoch 1598/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1459
Epoch 1599/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1459
Epoch 1600/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1459
Epoch 1601/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1459
Epoch 1602/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1459
Epoch 1603/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1459
Epoch 1604/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1458
Epoch 1605/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1458
Epoch 1606/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1458
Epoch 1607/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1458
Epoch 1608/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1458
Epoch 1609/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1458
Epoch 1610/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1458
Epoch 1611/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1458
Epoch 1612/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1458
Epoch 1613/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1614/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1615/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1616/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1617/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1618/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1457
Epoch 1619/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1620/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1621/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1622/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1457
Epoch 1623/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1456
Epoch 1624/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1456
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Epoch 1625/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1456
Epoch 1626/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1456
Epoch 1627/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1456
Epoch 1628/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1456
Epoch 1629/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1456
Epoch 1630/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1456
Epoch 1631/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1456
Epoch 1632/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1455
Epoch 1633/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1455
Epoch 1634/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1455
Epoch 1635/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1455
Epoch 1636/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1455
Epoch 1637/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1455
Epoch 1638/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1455
Epoch 1639/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1455
Epoch 1640/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1455
Epoch 1641/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1455
Epoch 1642/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1643/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1644/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1645/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1646/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1647/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1648/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1649/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1650/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1454
Epoch 1651/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1652/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
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Epoch 1653/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1654/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1655/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1656/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1657/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1658/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1659/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1660/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1452
Epoch 1661/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1662/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1663/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1664/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1665/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1666/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1667/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1668/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1669/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1670/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1671/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1672/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1673/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1674/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1675/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1676/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1677/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1678/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1679/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1450
Epoch 1680/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1450
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Epoch 1681/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1450
Epoch 1682/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1450
Epoch 1683/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1450
Epoch 1684/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1450
Epoch 1685/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1450
Epoch 1686/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1450
Epoch 1687/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1450
Epoch 1688/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1689/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1690/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1691/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1692/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1693/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1694/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1695/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1696/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1697/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1698/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1699/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1700/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1701/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1702/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1703/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1704/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1705/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1706/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1707/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1448
Epoch 1708/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
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Epoch 1709/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1448
Epoch 1710/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1711/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1712/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1713/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1449
Epoch 1714/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1450
Epoch 1715/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1451
Epoch 1716/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1717/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1453
Epoch 1718/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1452
Epoch 1719/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1450
Epoch 1720/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1721/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1446
Epoch 1722/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1446
Epoch 1723/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1446
Epoch 1724/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1447
Epoch 1725/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1448
Epoch 1726/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1447
Epoch 1727/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1446
Epoch 1728/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1445
Epoch 1729/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1445
Epoch 1730/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1445
Epoch 1731/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1446
Epoch 1732/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1446
Epoch 1733/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1445
Epoch 1734/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1445
Epoch 1735/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1736/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
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Epoch 1737/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1738/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1739/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1740/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1741/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1742/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1743/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1744/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1444
Epoch 1745/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1746/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1747/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1748/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1749/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1750/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1751/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1752/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1443
Epoch 1753/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1442
Epoch 1754/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1442
Epoch 1755/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1442
Epoch 1756/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1442
Epoch 1757/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1442
Epoch 1758/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1442
Epoch 1759/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1442
Epoch 1760/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1442
Epoch 1761/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1441
Epoch 1762/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1441
Epoch 1763/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1441
Epoch 1764/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1441
```

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Epoch 1765/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1441
Epoch 1766/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1441
Epoch 1767/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1441
Epoch 1768/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1441
Epoch 1769/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1441
Epoch 1770/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1771/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1772/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1773/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1774/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1775/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1440
Epoch 1776/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1777/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1778/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1440
Epoch 1779/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1439
Epoch 1780/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1439
Epoch 1781/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1439
Epoch 1782/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1439
Epoch 1783/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1439
Epoch 1784/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1439
Epoch 1785/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1439
Epoch 1786/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1439
Epoch 1787/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1439
Epoch 1788/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1789/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1790/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1791/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1792/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
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Epoch 1793/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1794/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1795/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1796/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1438
Epoch 1797/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1437
Epoch 1798/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1437
Epoch 1799/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1437
Epoch 1800/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1437
Epoch 1801/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1437
Epoch 1802/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1437
Epoch 1803/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1437
Epoch 1804/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1437
Epoch 1805/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1437
Epoch 1806/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1436
Epoch 1807/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1436
Epoch 1808/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1436
Epoch 1809/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1436
Epoch 1810/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1436
Epoch 1811/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1436
Epoch 1812/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1436
Epoch 1813/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1436
Epoch 1814/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1436
Epoch 1815/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1435
Epoch 1816/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
Epoch 1817/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
Epoch 1818/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
Epoch 1819/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
Epoch 1820/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
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Epoch 1821/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
Epoch 1822/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
Epoch 1823/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1435
Epoch 1824/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1825/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1826/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1827/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1828/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1829/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1830/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1831/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1434
Epoch 1832/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1433
Epoch 1833/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1433
Epoch 1834/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1433
Epoch 1835/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1433
Epoch 1836/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1433
Epoch 1837/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1433
Epoch 1838/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1433
Epoch 1839/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1433
Epoch 1840/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1433
Epoch 1841/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1842/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1843/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1844/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1845/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1846/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1847/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1848/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
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Epoch 1849/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1432
Epoch 1850/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1431
Epoch 1851/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1431
Epoch 1852/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1431
Epoch 1853/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1431
Epoch 1854/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1431
Epoch 1855/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1431
Epoch 1856/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1431
Epoch 1857/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1431
Epoch 1858/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1430
Epoch 1859/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1430
Epoch 1860/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1430
Epoch 1861/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1430
Epoch 1862/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1430
Epoch 1863/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1430
Epoch 1864/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1430
Epoch 1865/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1430
Epoch 1866/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1430
Epoch 1867/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1429
Epoch 1868/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1429
Epoch 1869/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1429
Epoch 1870/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1429
Epoch 1871/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1429
Epoch 1872/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1429
Epoch 1873/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1429
Epoch 1874/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1429
Epoch 1875/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1428
Epoch 1876/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1428
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Epoch 1877/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1428
Epoch 1878/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1428
Epoch 1879/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1428
Epoch 1880/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1428
Epoch 1881/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1428
Epoch 1882/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1428
Epoch 1883/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1427
Epoch 1884/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1427
Epoch 1885/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1427
Epoch 1886/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1427
Epoch 1887/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1427
Epoch 1888/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1427
Epoch 1889/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1427
Epoch 1890/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1427
Epoch 1891/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1427
Epoch 1892/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1426
Epoch 1893/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1426
Epoch 1894/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1426
Epoch 1895/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1426
Epoch 1896/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1426
Epoch 1897/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1426
Epoch 1898/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1426
Epoch 1899/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1426
Epoch 1900/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1425
Epoch 1901/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1425
Epoch 1902/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1425
Epoch 1903/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1425
Epoch 1904/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 59ms/step - loss: 0.1425
```

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Epoch 1905/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1425
Epoch 1906/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1425
Epoch 1907/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1425
Epoch 1908/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1425
Epoch 1909/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1910/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1911/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1912/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1424
Epoch 1913/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1914/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1915/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1916/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1917/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1425
Epoch 1918/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1425
Epoch 1919/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1426
Epoch 1920/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1427
Epoch 1921/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1428
Epoch 1922/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1428
Epoch 1923/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1428
Epoch 1924/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1427
Epoch 1925/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1425
Epoch 1926/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1423
Epoch 1927/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1422
Epoch 1928/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1422
Epoch 1929/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1423
Epoch 1930/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1424
Epoch 1931/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1424
Epoch 1932/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1423
```

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Epoch 1933/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1422
Epoch 1934/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1422
Epoch 1935/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1421
Epoch 1936/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1421
Epoch 1937/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1422
Epoch 1938/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1422
Epoch 1939/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1422
Epoch 1940/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1421
Epoch 1941/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1421
Epoch 1942/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1420
Epoch 1943/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1420
Epoch 1944/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1420
Epoch 1945/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1421
Epoch 1946/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1420
Epoch 1947/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1420
Epoch 1948/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1420
Epoch 1949/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1420
Epoch 1950/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1420
Epoch 1951/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1420
Epoch 1952/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1953/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1954/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1955/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1956/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1957/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1958/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1959/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1419
Epoch 1960/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1418
```

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Epoch 1961/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1418
Epoch 1962/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1418
Epoch 1963/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1418
Epoch 1964/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1418
Epoch 1965/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1418
Epoch 1966/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1418
Epoch 1967/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1418
Epoch 1968/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1417
Epoch 1969/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1417
Epoch 1970/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1417
Epoch 1971/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1417
Epoch 1972/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1417
Epoch 1973/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1417
Epoch 1974/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1417
Epoch 1975/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1417
Epoch 1976/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1416
Epoch 1977/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1416
Epoch 1978/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1416
Epoch 1979/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1416
Epoch 1980/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1416
Epoch 1981/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1416
Epoch 1982/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1416
Epoch 1983/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1416
Epoch 1984/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1415
Epoch 1985/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1415
Epoch 1986/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1415
Epoch 1987/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1415
Epoch 1988/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1415
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Epoch 1989/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1415
Epoch 1990/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1415
Epoch 1991/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1415
Epoch 1992/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1414
Epoch 1993/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1414
Epoch 1994/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1414
Epoch 1995/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1414
Epoch 1996/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1414
Epoch 1997/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1414
Epoch 1998/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1414
Epoch 1999/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1414
Epoch 2000/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1414
Epoch 2001/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1413
Epoch 2002/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1413
Epoch 2003/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1413
Epoch 2004/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1413
Epoch 2005/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1413
Epoch 2006/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1413
Epoch 2007/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1413
Epoch 2008/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1413
Epoch 2009/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1412
Epoch 2010/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1412
Epoch 2011/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1412
Epoch 2012/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1412
Epoch 2013/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1412
Epoch 2014/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1412
Epoch 2015/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1412
Epoch 2016/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1412
```

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Epoch 2017/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1411
Epoch 2018/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1411
Epoch 2019/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1411
Epoch 2020/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1411
Epoch 2021/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1411
Epoch 2022/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1411
Epoch 2023/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1411
Epoch 2024/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1411
Epoch 2025/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1410
Epoch 2026/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1410
Epoch 2027/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1410
Epoch 2028/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1410
Epoch 2029/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1410
Epoch 2030/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1410
Epoch 2031/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1410
Epoch 2032/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1410
Epoch 2033/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1409
Epoch 2034/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1409
Epoch 2035/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1409
Epoch 2036/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1409
Epoch 2037/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1409
Epoch 2038/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1409
Epoch 2039/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1409
Epoch 2040/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1409
Epoch 2041/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1408
Epoch 2042/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1408
Epoch 2043/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1408
Epoch 2044/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1408
```

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Epoch 2045/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1408
Epoch 2046/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1408
Epoch 2047/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1408
Epoch 2048/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1408
Epoch 2049/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1407
Epoch 2050/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1407
Epoch 2051/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1407
Epoch 2052/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1407
Epoch 2053/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1407
Epoch 2054/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1407
Epoch 2055/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1407
Epoch 2056/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1407
Epoch 2057/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2058/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2059/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2060/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2061/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2062/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2063/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2064/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1406
Epoch 2065/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1405
Epoch 2066/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1405
Epoch 2067/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1405
Epoch 2068/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1405
Epoch 2069/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1405
Epoch 2070/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1405
Epoch 2071/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1405
Epoch 2072/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1405
```

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Epoch 2073/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1404
Epoch 2074/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1404
Epoch 2075/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1404
Epoch 2076/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1404
Epoch 2077/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1404
Epoch 2078/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1404
Epoch 2079/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1404
Epoch 2080/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1404
Epoch 2081/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2082/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2083/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2084/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2085/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2086/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2087/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2088/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1403
Epoch 2089/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1402
Epoch 2090/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1402
Epoch 2091/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1402
Epoch 2092/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1402
Epoch 2093/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1402
Epoch 2094/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1402
Epoch 2095/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1402
Epoch 2096/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
Epoch 2097/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1401
Epoch 2098/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
Epoch 2099/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
Epoch 2100/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
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Epoch 2101/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
Epoch 2102/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
Epoch 2103/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
Epoch 2104/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2105/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1400
Epoch 2106/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2107/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2108/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2109/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2110/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2111/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2112/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2113/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2114/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2115/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2116/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1401
Epoch 2117/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1402
Epoch 2118/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1403
Epoch 2119/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1405
Epoch 2120/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2121/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1407
Epoch 2122/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1406
Epoch 2123/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1402
Epoch 2124/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1399
Epoch 2125/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1398
Epoch 2126/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1399
Epoch 2127/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2128/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1401
```

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Epoch 2129/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1400
Epoch 2130/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1398
Epoch 2131/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1397
Epoch 2132/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1397
Epoch 2133/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1398
Epoch 2134/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1399
Epoch 2135/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1398
Epoch 2136/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1397
Epoch 2137/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1396
Epoch 2138/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1396
Epoch 2139/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1397
Epoch 2140/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1397
Epoch 2141/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1396
Epoch 2142/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1396
Epoch 2143/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1396
Epoch 2144/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1396
Epoch 2145/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1396
Epoch 2146/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1396
Epoch 2147/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1395
Epoch 2148/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1395
Epoch 2149/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1395
Epoch 2150/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1395
Epoch 2151/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1395
Epoch 2152/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1395
Epoch 2153/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1394
Epoch 2154/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1394
Epoch 2155/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1394
Epoch 2156/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1394
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Epoch 2157/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1394
Epoch 2158/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1394
Epoch 2159/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1394
Epoch 2160/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1393
Epoch 2161/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1393
Epoch 2162/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1393
Epoch 2163/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1393
Epoch 2164/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1393
Epoch 2165/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1393
Epoch 2166/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1393
Epoch 2167/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1393
Epoch 2168/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1393
Epoch 2169/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1392
Epoch 2170/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1392
Epoch 2171/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1392
Epoch 2172/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1392
Epoch 2173/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1392
Epoch 2174/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1392
Epoch 2175/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1392
Epoch 2176/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1392
Epoch 2177/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1391
Epoch 2178/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1391
Epoch 2179/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1391
Epoch 2180/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1391
Epoch 2181/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1391
Epoch 2182/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1391
Epoch 2183/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1391
Epoch 2184/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1391
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Epoch 2185/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1390
Epoch 2186/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1390
Epoch 2187/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1390
Epoch 2188/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1390
Epoch 2189/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1390
Epoch 2190/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 59ms/step - loss: 0.1390
Epoch 2191/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1390
Epoch 2192/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1390
Epoch 2193/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1389
Epoch 2194/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1389
Epoch 2195/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1389
Epoch 2196/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1389
Epoch 2197/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1389
Epoch 2198/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1389
Epoch 2199/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1389
Epoch 2200/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1389
Epoch 2201/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1388
Epoch 2202/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1388
Epoch 2203/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1388
Epoch 2204/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1388
Epoch 2205/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1388
Epoch 2206/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1388
Epoch 2207/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1388
Epoch 2208/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1388
Epoch 2209/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1387
Epoch 2210/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1387
Epoch 2211/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1387
Epoch 2212/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1387
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Epoch 2213/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1387
Epoch 2214/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1387
Epoch 2215/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1387
Epoch 2216/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1387
Epoch 2217/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1386
Epoch 2218/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1386
Epoch 2219/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1386
Epoch 2220/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1386
Epoch 2221/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1386
Epoch 2222/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1386
Epoch 2223/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1386
Epoch 2224/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1386
Epoch 2225/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1385
Epoch 2226/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1385
Epoch 2227/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1385
Epoch 2228/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1385
Epoch 2229/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1385
Epoch 2230/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1385
Epoch 2231/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1385
Epoch 2232/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1385
Epoch 2233/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1384
Epoch 2234/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1384
Epoch 2235/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1384
Epoch 2236/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1384
Epoch 2237/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1384
Epoch 2238/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1384
Epoch 2239/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1384
Epoch 2240/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1384
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Epoch 2241/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1383
Epoch 2242/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1383
Epoch 2243/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1383
Epoch 2244/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1383
Epoch 2245/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1383
Epoch 2246/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1383
Epoch 2247/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1383
Epoch 2248/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1382
Epoch 2249/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1382
Epoch 2250/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1382
Epoch 2251/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1382
Epoch 2252/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1382
Epoch 2253/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1382
Epoch 2254/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1382
Epoch 2255/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1382
Epoch 2256/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1381
Epoch 2257/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1381
Epoch 2258/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1381
Epoch 2259/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1381
Epoch 2260/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1381
Epoch 2261/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1381
Epoch 2262/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1381
Epoch 2263/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1381
Epoch 2264/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1380
Epoch 2265/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1380
Epoch 2266/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1380
Epoch 2267/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1380
Epoch 2268/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1380
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Epoch 2269/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1380
Epoch 2270/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1380
Epoch 2271/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1380
Epoch 2272/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1379
Epoch 2273/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1379
Epoch 2274/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1379
Epoch 2275/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1379
Epoch 2276/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1379
Epoch 2277/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1379
Epoch 2278/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1379
Epoch 2279/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1379
Epoch 2280/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1378
Epoch 2281/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1378
Epoch 2282/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1378
Epoch 2283/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1378
Epoch 2284/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1378
Epoch 2285/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1378
Epoch 2286/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1378
Epoch 2287/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1377
Epoch 2288/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1377
Epoch 2289/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1377
Epoch 2290/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1377
Epoch 2291/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1377
Epoch 2292/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1377
Epoch 2293/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1377
Epoch 2294/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1377
Epoch 2295/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1376
Epoch 2296/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1376
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Epoch 2297/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1376
Epoch 2298/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1376
Epoch 2299/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1376
Epoch 2300/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1376
Epoch 2301/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1376
Epoch 2302/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1376
Epoch 2303/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1375
Epoch 2304/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1375
Epoch 2305/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1375
Epoch 2306/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1375
Epoch 2307/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1375
Epoch 2308/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1375
Epoch 2309/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1375
Epoch 2310/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1374
Epoch 2311/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1374
Epoch 2312/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1374
Epoch 2313/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1374
Epoch 2314/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1374
Epoch 2315/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1374
Epoch 2316/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1374
Epoch 2317/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1374
Epoch 2318/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1373
Epoch 2319/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1373
Epoch 2320/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1373
Epoch 2321/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1373
Epoch 2322/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1373
Epoch 2323/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1373
Epoch 2324/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1373
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Epoch 2325/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1373
Epoch 2326/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1372
Epoch 2327/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1372
Epoch 2328/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1372
Epoch 2329/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1372
Epoch 2330/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1372
Epoch 2331/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1372
Epoch 2332/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1372
Epoch 2333/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1372
Epoch 2334/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1371
Epoch 2335/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1371
Epoch 2336/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1371
Epoch 2337/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1371
Epoch 2338/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1371
Epoch 2339/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 59ms/step - loss: 0.1371
Epoch 2340/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1371
Epoch 2341/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2342/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2343/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2344/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2345/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2346/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2347/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2348/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1370
Epoch 2349/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1369
Epoch 2350/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
Epoch 2351/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
Epoch 2352/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
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Epoch 2353/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
Epoch 2354/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
Epoch 2355/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
Epoch 2356/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
Epoch 2357/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2358/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2359/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2360/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2361/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1368
Epoch 2362/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2363/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2364/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2365/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2366/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2367/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2368/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2369/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1368
Epoch 2370/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2371/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1369
Epoch 2372/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1370
Epoch 2373/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1371
Epoch 2374/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1372
Epoch 2375/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1373
Epoch 2376/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1372
Epoch 2377/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1370
Epoch 2378/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1368
Epoch 2379/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1366
Epoch 2380/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1366
```

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Epoch 2381/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1366
Epoch 2382/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1367
Epoch 2383/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1368
Epoch 2384/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1367
Epoch 2385/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1366
Epoch 2386/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1365
Epoch 2387/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1365
Epoch 2388/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1365
Epoch 2389/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1366
Epoch 2390/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1365
Epoch 2391/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1365
Epoch 2392/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1364
Epoch 2393/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1364
Epoch 2394/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1364
Epoch 2395/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1364
Epoch 2396/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1364
Epoch 2397/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1364
Epoch 2398/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1364
Epoch 2399/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1363
Epoch 2400/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1363
Epoch 2401/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1363
Epoch 2402/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1363
Epoch 2403/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1363
Epoch 2404/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1363
Epoch 2405/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1362
Epoch 2406/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1362
Epoch 2407/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1362
Epoch 2408/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1362
```

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Epoch 2409/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1362
Epoch 2410/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1362
Epoch 2411/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1362
Epoch 2412/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1362
Epoch 2413/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1361
Epoch 2414/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1361
Epoch 2415/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1361
Epoch 2416/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1361
Epoch 2417/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1361
Epoch 2418/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1361
Epoch 2419/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1361
Epoch 2420/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1361
Epoch 2421/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1361
Epoch 2422/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1360
Epoch 2423/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1360
Epoch 2424/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1360
Epoch 2425/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1360
Epoch 2426/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1360
Epoch 2427/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1360
Epoch 2428/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1360
Epoch 2429/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1360
Epoch 2430/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1359
Epoch 2431/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1359
Epoch 2432/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1359
Epoch 2433/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1359
Epoch 2434/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1359
Epoch 2435/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1359
Epoch 2436/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1359
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Epoch 2437/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1359
Epoch 2438/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1358
Epoch 2439/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1358
Epoch 2440/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1358
Epoch 2441/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1358
Epoch 2442/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1358
Epoch 2443/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1358
Epoch 2444/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1358
Epoch 2445/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1358
Epoch 2446/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1357
Epoch 2447/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1357
Epoch 2448/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1357
Epoch 2449/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1357
Epoch 2450/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1357
Epoch 2451/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1357
Epoch 2452/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1357
Epoch 2453/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1357
Epoch 2454/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1356
Epoch 2455/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1356
Epoch 2456/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1356
Epoch 2457/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1356
Epoch 2458/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1356
Epoch 2459/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1356
Epoch 2460/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1356
Epoch 2461/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1356
Epoch 2462/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1355
Epoch 2463/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1355
Epoch 2464/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1355
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Epoch 2465/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1355
Epoch 2466/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1355
Epoch 2467/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1355
Epoch 2468/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1355
Epoch 2469/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1355
Epoch 2470/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1354
Epoch 2471/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1354
Epoch 2472/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1354
Epoch 2473/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1354
Epoch 2474/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1354
Epoch 2475/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1354
Epoch 2476/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1354
Epoch 2477/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1354
Epoch 2478/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1354
Epoch 2479/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1353
Epoch 2480/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1353
Epoch 2481/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1353
Epoch 2482/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1353
Epoch 2483/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1353
Epoch 2484/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1353
Epoch 2485/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1353
Epoch 2486/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1353
Epoch 2487/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1352
Epoch 2488/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1352
Epoch 2489/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1352
Epoch 2490/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1352
Epoch 2491/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1352
Epoch 2492/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1352
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Epoch 2493/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1352
Epoch 2494/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1352
Epoch 2495/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1351
Epoch 2496/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1351
Epoch 2497/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1351
Epoch 2498/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1351
Epoch 2499/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1351
Epoch 2500/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1351
Epoch 2501/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1351
Epoch 2502/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1351
Epoch 2503/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1350
Epoch 2504/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1350
Epoch 2505/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1350
Epoch 2506/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1350
Epoch 2507/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1350
Epoch 2508/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1350
Epoch 2509/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1350
Epoch 2510/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1350
Epoch 2511/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1349
Epoch 2512/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1349
Epoch 2513/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1349
Epoch 2514/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1349
Epoch 2515/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1349
Epoch 2516/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1349
Epoch 2517/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1349
Epoch 2518/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1349
Epoch 2519/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1348
Epoch 2520/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1348
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Epoch 2521/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1348
Epoch 2522/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1348
Epoch 2523/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1348
Epoch 2524/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1348
Epoch 2525/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1348
Epoch 2526/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1348
Epoch 2527/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1347
Epoch 2528/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1347
Epoch 2529/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1347
Epoch 2530/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1347
Epoch 2531/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1347
Epoch 2532/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1347
Epoch 2533/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1347
Epoch 2534/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1347
Epoch 2535/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1346
Epoch 2536/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1346
Epoch 2537/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1346
Epoch 2538/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1346
Epoch 2539/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1346
Epoch 2540/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1346
Epoch 2541/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1346
Epoch 2542/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1346
Epoch 2543/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1346
Epoch 2544/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
Epoch 2545/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 59ms/step - loss: 0.1345
Epoch 2546/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
Epoch 2547/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
Epoch 2548/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
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Epoch 2549/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1345
Epoch 2550/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
Epoch 2551/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
Epoch 2552/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2553/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1344
Epoch 2554/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2555/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1344
Epoch 2556/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2557/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2558/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2559/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2560/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1343
Epoch 2561/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1343
Epoch 2562/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1343
Epoch 2563/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1343
Epoch 2564/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1343
Epoch 2565/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1343
Epoch 2566/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1343
Epoch 2567/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1343
Epoch 2568/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2569/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2570/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2571/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1342
Epoch 2572/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1342
Epoch 2573/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2574/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2575/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2576/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
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Epoch 2577/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2578/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1343
Epoch 2579/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1343
Epoch 2580/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2581/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
Epoch 2582/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1345
Epoch 2583/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1346
Epoch 2584/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1345
Epoch 2585/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1344
Epoch 2586/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1342
Epoch 2587/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1341
Epoch 2588/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1340
Epoch 2589/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1340
Epoch 2590/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1341
Epoch 2591/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1341
Epoch 2592/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1341
Epoch 2593/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1341
Epoch 2594/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1340
Epoch 2595/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1339
Epoch 2596/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1339
Epoch 2597/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1339
Epoch 2598/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1339
Epoch 2599/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1339
Epoch 2600/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1339
Epoch 2601/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1339
Epoch 2602/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1338
Epoch 2603/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1338
Epoch 2604/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1338
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Epoch 2605/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1338
Epoch 2606/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1338
Epoch 2607/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1338
Epoch 2608/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1338
Epoch 2609/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1338
Epoch 2610/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1337
Epoch 2611/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1337
Epoch 2612/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1337
Epoch 2613/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1337
Epoch 2614/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1337
Epoch 2615/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1337
Epoch 2616/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1337
Epoch 2617/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1337
Epoch 2618/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1336
Epoch 2619/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1336
Epoch 2620/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1336
Epoch 2621/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1336
Epoch 2622/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1336
Epoch 2623/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1336
Epoch 2624/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1336
Epoch 2625/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1336
Epoch 2626/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1335
Epoch 2627/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
Epoch 2628/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
Epoch 2629/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
Epoch 2630/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
Epoch 2631/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
Epoch 2632/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
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Epoch 2633/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
Epoch 2634/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1335
Epoch 2635/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2636/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2637/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2638/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2639/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2640/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2641/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2642/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1334
Epoch 2643/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2644/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2645/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1333
Epoch 2646/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2647/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2648/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2649/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2650/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2651/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1333
Epoch 2652/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1332
Epoch 2653/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1332
Epoch 2654/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1332
Epoch 2655/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1332
Epoch 2656/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1332
Epoch 2657/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1332
Epoch 2658/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1332
Epoch 2659/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1332
Epoch 2660/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1331
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Epoch 2661/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1331
Epoch 2662/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1331
Epoch 2663/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1331
Epoch 2664/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1331
Epoch 2665/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1331
Epoch 2666/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1331
Epoch 2667/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1331
Epoch 2668/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1331
Epoch 2669/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1330
Epoch 2670/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1330
Epoch 2671/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1330
Epoch 2672/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1330
Epoch 2673/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1330
Epoch 2674/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1330
Epoch 2675/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1330
Epoch 2676/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1330
Epoch 2677/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1329
Epoch 2678/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1329
Epoch 2679/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1329
Epoch 2680/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1329
Epoch 2681/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1329
Epoch 2682/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1329
Epoch 2683/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1329
Epoch 2684/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1329
Epoch 2685/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1328
Epoch 2686/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1328
Epoch 2687/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1328
Epoch 2688/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1328
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Epoch 2689/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1328
Epoch 2690/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1328
Epoch 2691/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1328
Epoch 2692/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1328
Epoch 2693/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1328
Epoch 2694/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2695/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2696/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2697/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2698/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2699/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2700/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2701/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1327
Epoch 2702/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1326
Epoch 2703/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1326
Epoch 2704/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1326
Epoch 2705/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1326
Epoch 2706/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1326
Epoch 2707/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1326
Epoch 2708/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1326
Epoch 2709/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1326
Epoch 2710/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1326
Epoch 2711/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2712/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2713/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2714/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2715/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2716/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1325
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Epoch 2717/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2718/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1325
Epoch 2719/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1324
Epoch 2720/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1324
Epoch 2721/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1324
Epoch 2722/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1324
Epoch 2723/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1324
Epoch 2724/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1324
Epoch 2725/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1324
Epoch 2726/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1324
Epoch 2727/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2728/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1323
Epoch 2729/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2730/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2731/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2732/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1323
Epoch 2733/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2734/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1323
Epoch 2735/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2736/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1322
Epoch 2737/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1322
Epoch 2738/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1322
Epoch 2739/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1322
Epoch 2740/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1322
Epoch 2741/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1322
Epoch 2742/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1322
Epoch 2743/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1322
Epoch 2744/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1322
```

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Epoch 2745/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1321
Epoch 2746/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2747/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2748/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2749/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2750/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2751/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2752/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2753/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2754/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2755/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2756/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1322
Epoch 2757/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1322
Epoch 2758/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2759/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1324
Epoch 2760/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2761/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1325
Epoch 2762/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1324
Epoch 2763/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1323
Epoch 2764/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1321
Epoch 2765/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1319
Epoch 2766/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1319
Epoch 2767/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1320
Epoch 2768/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1320
Epoch 2769/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1321
Epoch 2770/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1320
Epoch 2771/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1319
Epoch 2772/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1318
```

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Epoch 2773/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1318
Epoch 2774/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1318
Epoch 2775/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1319
Epoch 2776/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1319
Epoch 2777/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1319
Epoch 2778/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1318
Epoch 2779/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1317
Epoch 2780/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1317
Epoch 2781/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1317
Epoch 2782/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1318
Epoch 2783/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1318
Epoch 2784/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1317
Epoch 2785/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1317
Epoch 2786/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1317
Epoch 2787/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1317
Epoch 2788/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1317
Epoch 2789/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1317
Epoch 2790/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1316
Epoch 2791/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 60ms/step - loss: 0.1316
Epoch 2792/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1316
Epoch 2793/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1316
Epoch 2794/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1316
Epoch 2795/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1316
Epoch 2796/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1316
Epoch 2797/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1316
Epoch 2798/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1315
Epoch 2799/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1315
Epoch 2800/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1315
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Epoch 2801/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1315
Epoch 2802/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1315
Epoch 2803/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1315
Epoch 2804/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1315
Epoch 2805/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1315
Epoch 2806/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1314
Epoch 2807/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1314
Epoch 2808/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1314
Epoch 2809/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1314
Epoch 2810/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1314
Epoch 2811/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1314
Epoch 2812/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1314
Epoch 2813/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1314
Epoch 2814/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2815/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2816/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2817/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2818/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1313
Epoch 2819/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2820/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2821/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2822/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1313
Epoch 2823/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1312
Epoch 2824/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1312
Epoch 2825/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1312
Epoch 2826/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1312
Epoch 2827/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1312
Epoch 2828/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1312
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Epoch 2829/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1312
Epoch 2830/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1312
Epoch 2831/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1312
Epoch 2832/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1311
Epoch 2833/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2834/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2835/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2836/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2837/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2838/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2839/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2840/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1311
Epoch 2841/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1310
Epoch 2842/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 43ms/step - loss: 0.1310
Epoch 2843/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1310
Epoch 2844/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1310
Epoch 2845/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1310
Epoch 2846/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1310
Epoch 2847/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1310
Epoch 2848/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1310
Epoch 2849/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1310
Epoch 2850/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
Epoch 2851/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
Epoch 2852/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
Epoch 2853/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
Epoch 2854/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1309
Epoch 2855/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
Epoch 2856/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
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Epoch 2857/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
Epoch 2858/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1309
Epoch 2859/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1308
Epoch 2860/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2861/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2862/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2863/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2864/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2865/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2866/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2867/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1308
Epoch 2868/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1307
Epoch 2869/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1307
Epoch 2870/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1307
Epoch 2871/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1307
Epoch 2872/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1307
Epoch 2873/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1307
Epoch 2874/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1307
Epoch 2875/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1307
Epoch 2876/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1306
Epoch 2877/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1306
Epoch 2878/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1306
Epoch 2879/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1306
Epoch 2880/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1306
Epoch 2881/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1306
Epoch 2882/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1306
Epoch 2883/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1306
Epoch 2884/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1306
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Epoch 2885/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2886/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2887/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2888/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2889/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2890/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2891/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2892/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2893/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1305
Epoch 2894/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1304
Epoch 2895/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1304
Epoch 2896/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1304
Epoch 2897/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1304
Epoch 2898/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 42ms/step - loss: 0.1304
Epoch 2899/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1304
Epoch 2900/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1304
Epoch 2901/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1304
Epoch 2902/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1304
Epoch 2903/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1303
Epoch 2904/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1303
Epoch 2905/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1303
Epoch 2906/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1303
Epoch 2907/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1303
Epoch 2908/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1303
Epoch 2909/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1303
Epoch 2910/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1303
Epoch 2911/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1303
Epoch 2912/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1302
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Epoch 2913/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1302
Epoch 2914/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 44ms/step - loss: 0.1302
Epoch 2915/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1302
Epoch 2916/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1302
Epoch 2917/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1302
Epoch 2918/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1302
Epoch 2919/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1302
Epoch 2920/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1302
Epoch 2921/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1301
Epoch 2922/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1301
Epoch 2923/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1301
Epoch 2924/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1301
Epoch 2925/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1301
Epoch 2926/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1301
Epoch 2927/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1301
Epoch 2928/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1301
Epoch 2929/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1300
Epoch 2930/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1300
Epoch 2931/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1300
Epoch 2932/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1300
Epoch 2933/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1300
Epoch 2934/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1300
Epoch 2935/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1300
Epoch 2936/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1300
Epoch 2937/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1300
Epoch 2938/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - loss: 0.1300
Epoch 2939/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1300
Epoch 2940/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1300
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Epoch 2941/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1300
Epoch 2942/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1300
Epoch 2943/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1300
Epoch 2944/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1300
Epoch 2945/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1300
Epoch 2946/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1301
Epoch 2947/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1302
Epoch 2948/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1302
Epoch 2949/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1303
Epoch 2950/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1302
Epoch 2951/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1302
Epoch 2952/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1300
Epoch 2953/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1299
Epoch 2954/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1298
Epoch 2955/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1298
Epoch 2956/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1298
Epoch 2957/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1298
Epoch 2958/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1299
Epoch 2959/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1299
Epoch 2960/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1298
Epoch 2961/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1297
Epoch 2962/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1297
Epoch 2963/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1297
Epoch 2964/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1297
Epoch 2965/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1297
Epoch 2966/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1297
Epoch 2967/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1297
Epoch 2968/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1296
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Epoch 2969/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1296
Epoch 2970/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1296
Epoch 2971/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1296
Epoch 2972/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1296
Epoch 2973/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1296
Epoch 2974/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1296
Epoch 2975/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 45ms/step - loss: 0.1295
Epoch 2976/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 41ms/step - loss: 0.1295
Epoch 2977/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.1295
Epoch 2978/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1295
Epoch 2979/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1295
Epoch 2980/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1295
Epoch 2981/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1295
Epoch 2982/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1295
Epoch 2983/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1294
Epoch 2984/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1294
Epoch 2985/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1294
Epoch 2986/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1294
Epoch 2987/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.1294
Epoch 2988/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.1294
Epoch 2989/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 39ms/step - loss: 0.1294
Epoch 2990/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.1294
Epoch 2991/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1293
Epoch 2992/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1293
Epoch 2993/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 34ms/step - loss: 0.1293
Epoch 2994/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1293
Epoch 2995/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1293
Epoch 2996/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1293
```

```
Epoch 2997/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1293
Epoch 2998/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1293
Epoch 2999/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1292
Epoch 3000/3000
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step - loss: 0.1292
best epoch =  3000
smallest loss = 0.12922649085521698
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step
```

In [9]:

```
predictions = model.predict(data_inputs)

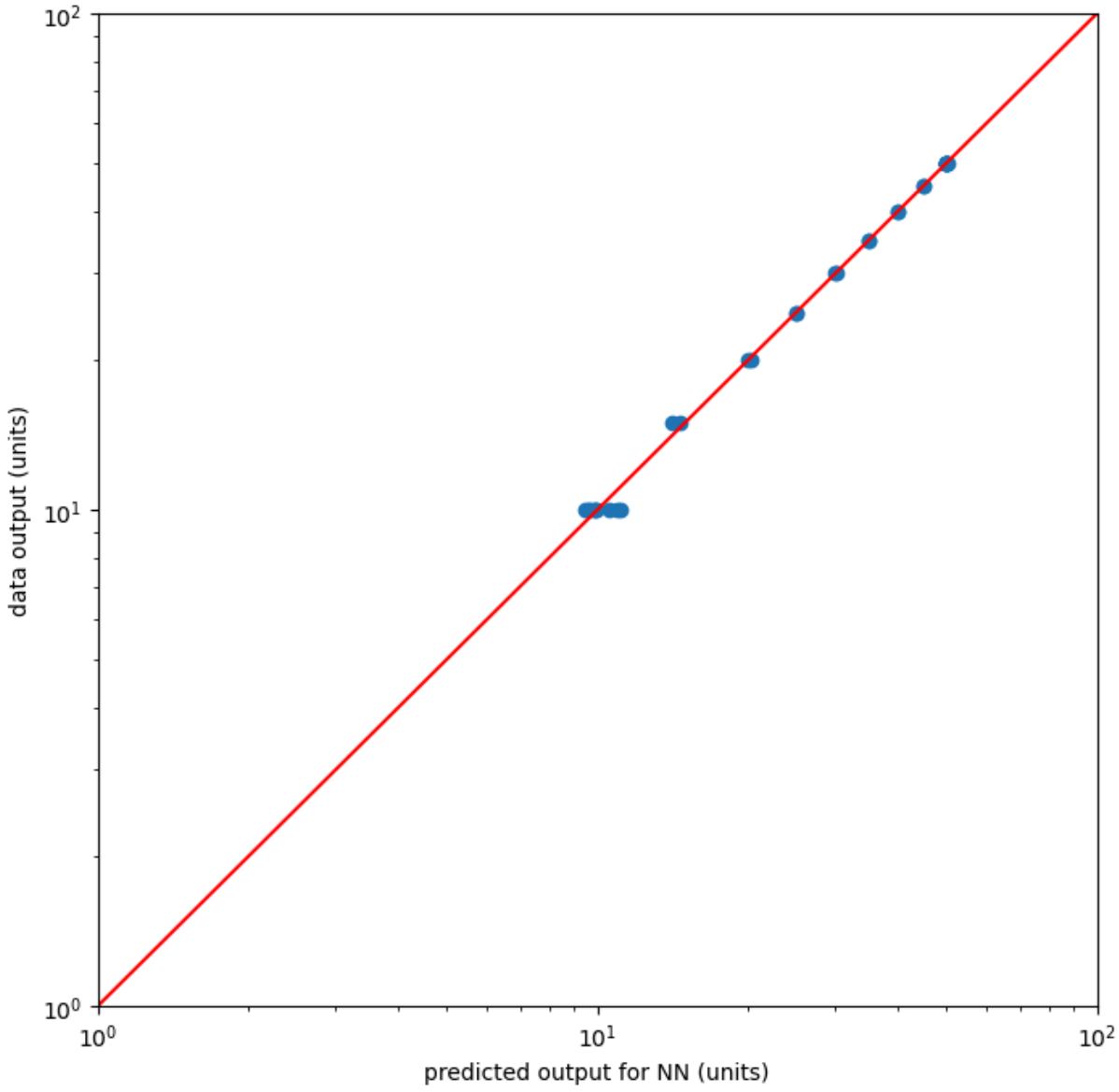


#SETTING UP PLOT
# matplotlib inline
# importing the required module
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [8, 8] # for square canvas
#========

'''CALCULATE PREDICTED VALUES AND RETRIEVE DATA VALUES TO PLOT'''

plt.scatter(predictions[:, 0], data_outputs[:, 0])
#plt.title('Genetic Algorithm training of Neural Network ==> output = alpha'
plt.xlabel('predicted output for NN (units)')
plt.ylabel('data output (units)')
plt.loglog()
plt.xlim(xmax = 100, xmin = 1.)
plt.ylim(ymax = 100, ymin = 1.)
# Generate red y=x line
x_data = np.linspace(1.0, 100.0, num=3)
y_data = x_data
plt.plot(x_data, y_data, color='red')
plt.savefig("log-log_plot_3.4_kappa_5.png", dpi=300, bbox_inches='tight')
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
```



In [10]:
```
test = []
outpt=[]
predictions = model.predict(data_inputs)

#first point (row [0])comparison of data and prediction
# put in a loop to print comparion for all data points

xa=[]
ya=[]
za=[]

xao = np.linspace(0, 3, 7)  # 1D array for x values
yao = np.linspace(0, 5, 11)    # 1D array for y values
xa = np.linspace(0, 10, 77)    # 1D array for x values
ya = np.linspace(0, 10, 77)    # 1D array for y values
za = np.linspace(0, 10, 77)    # 1D array for z values

x =-0.50
ia=-1
while (x < 2.51):
```

```
    x = x + 0.5
    y = -0.5
    while (y < 4.51):
        ia=ia+1
        y = y + 0.50
        xa[ia]=x
        ya[ia]=y

        test = [[ x , y ]]
        testarray = np.array(test)
        outpt = model.predict(testarray)
        za[ia] = outpt
        print ('x= ', testarray[0][0], ', y= ', testarray[0][1],'T= ', outpt



X, Y, Z = xa, ya, za

import sys
#import csv
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import axes3d

# Plot X,Y,Z

# Creating figure
#fig = plt.figure(figsize =(16, 9))
#ax = plt.axes(projection ='3d')

fig = plt.figure()
#ax = fig.add_subplot(111, projection='3d')
ax = plt.axes(projection ='3d')
#ax.plot_trisurf(X, Y, Z, color='white', edgecolors='grey', alpha=0.5)
ax.plot_trisurf(X, Y, Z,
                linewidth = 0.2,
                antialiased = True
                , cmap='viridis');


surf = ax.plot_trisurf(
    X, Y, Z,
    linewidth=0.2,
    antialiased=True,
    cmap='viridis'
)


#  ADD COLORBAR
fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)



# fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)

ax.invert_yaxis()
```

```
ax.invert_xaxis()

plt.savefig("trisurf_plot_kappa_5.png", dpi=300, bbox_inches='tight')
plt.show()



# Plot the data
plt.figure(figsize=(5, 6))
plt.tripcolor(xa, ya, za, shading='flat', cmap='viridis')
plt.savefig("Heat_map_kappa_5.png", dpi=300, bbox_inches='tight')
plt.colorbar(label='Z value')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Heat Map of Z values over X and Y domain')
plt.show()
```

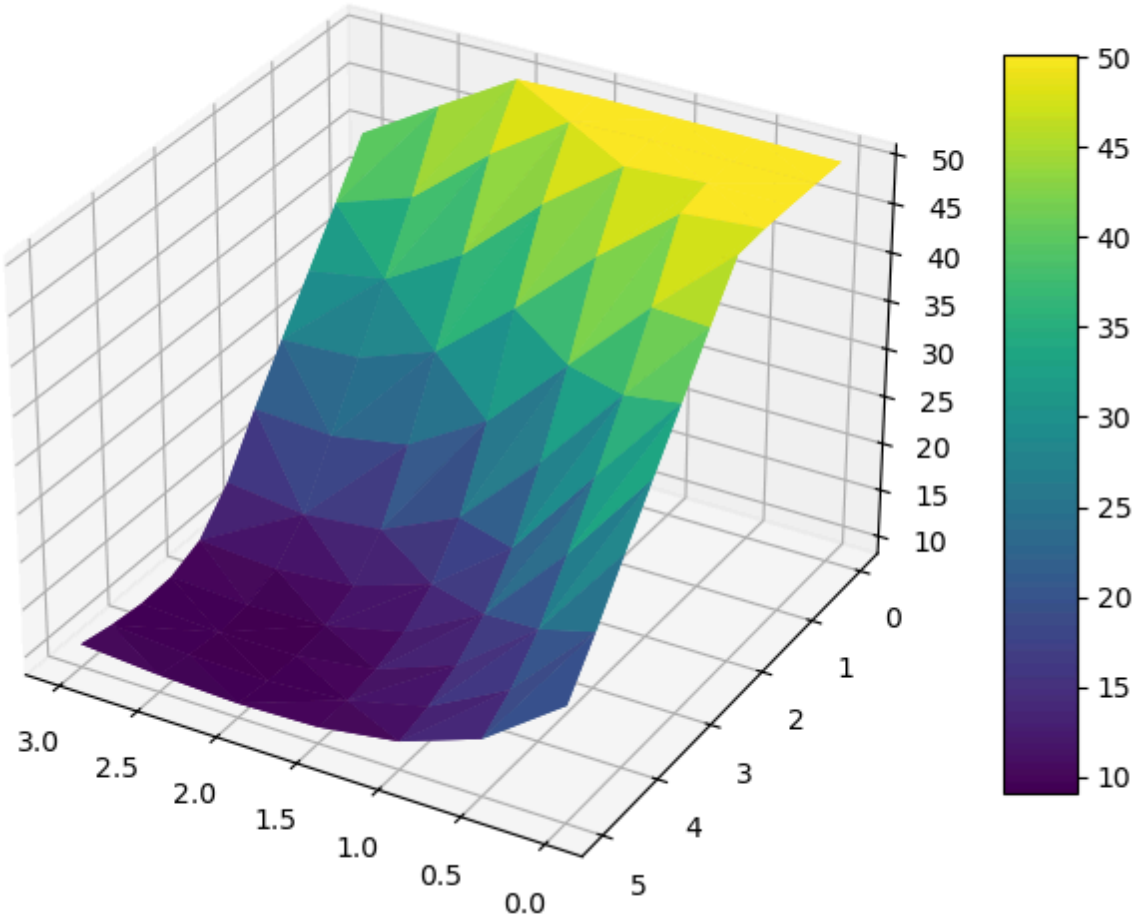
```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step
x=  0.0 , y=  0.0 T=  49.911327
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.0 , y=  0.5 T=  49.991688
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  1.0 T=  50.062546
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
```

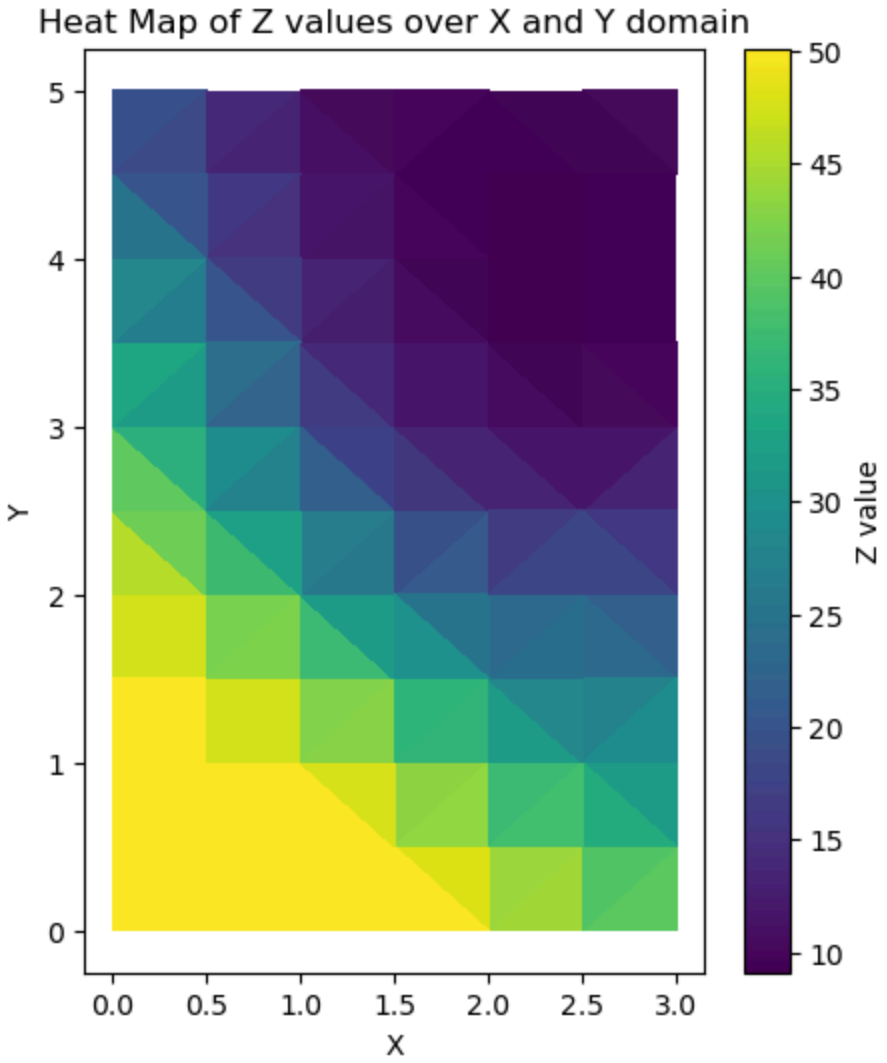
```
/tmp/ipykernel_1722156/3841726189.py:32: DeprecationWarning: Conversion of a
n array with ndim > 0 to a scalar is deprecated, and will error in future. E
nsure you extract a single element from your array before performing this op
eration. (Deprecated NumPy 1.25.)
  za[ia] = outpt
```

```
x=  0.0 , y=  1.5 T=  50.064568
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  2.0 T=  49.855915
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  2.5 T=  45.057415
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 33ms/step
x=  0.0 , y=  3.0 T=  39.98315
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.0 , y=  3.5 T=  35.03014
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  4.0 T=  30.051355
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  4.5 T=  24.923214
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  5.0 T=  20.244259
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  0.0 T=  49.99623
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  0.5 T=  50.07469
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  1.0 T=  50.125664
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  1.5 T=  49.376244
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  2.0 T=  42.34096
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  2.5 T=  35.931557
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  3.0 T=  30.21799
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  3.5 T=  24.833813
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  4.0 T=  19.847677
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  4.5 T=  16.35715
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  5.0 T=  14.10982
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.0 , y=  0.0 T=  50.067005
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.0 , y=  0.5 T=  50.12828
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  1.0 T=  49.643562
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.0 , y=  1.5 T=  42.81439
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  2.0 T=  34.14808
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.0 , y=  2.5 T=  26.5697
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  3.0 T=  20.299742
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  3.5 T=  16.061882
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  4.0 T=  13.4335375
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
```

```
x=  1.0 , y=  4.5 T=  11.888957
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  5.0 T=  11.093398
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.5 , y=  0.0 T=  50.110683
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  0.5 T=  49.78884
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.5 , y=  1.0 T=  43.628643
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.5 , y=  1.5 T=  35.26397
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.5 , y=  2.0 T=  25.962168
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  2.5 T=  18.35254
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.5 , y=  3.0 T=  13.937863
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
x=  1.5 , y=  3.5 T=  11.531447
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.5 , y=  4.0 T=  10.325313
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.5 , y=  4.5 T=  9.858531
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  5.0 T=  9.861281
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.0 , y=  0.0 T=  49.866642
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.0 , y=  0.5 T=  44.32533
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  1.0 T=  36.216614
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 35ms/step
x=  2.0 , y=  1.5 T=  28.414589
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.0 , y=  2.0 T=  21.05527
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  2.5 T=  14.545415
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  3.0 T=  11.129424
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  3.5 T=  9.610538
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  4.0 T=  9.120009
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  4.5 T=  9.204864
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  5.0 T=  9.623904
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  0.0 T=  45.005524
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  0.5 T=  37.683525
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  1.0 T=  31.411146
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  1.5 T=  25.438282
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
```

```
x=  2.5 , y=  2.0 T=  19.57347
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  2.5 T=  13.672854
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.5 , y=  3.0 T=  10.421851
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.5 , y=  3.5 T=  9.188373
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.5 , y=  4.0 T=  8.986114
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  4.5 T=  9.320186
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  5.0 T=  9.939872
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  0.0 T=  40.05996
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  0.5 T=  34.930573
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  1.0 T=  29.996775
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  1.5 T=  25.033537
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  2.0 T=  20.03112
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  2.5 T=  14.623942
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  3.0 T=  10.96321
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  3.5 T=  9.634059
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  4.0 T=  9.462626
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  4.5 T=  9.868518
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  5.0 T=  10.564704
```


50
45
40
35
30
25
20
15
10
0
1
2
3
4
5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
50
45
40
35
30
25
20
15
10


Heat Map of Z values over X and Y domain
Y
X
Z value
0
1
2
3
4
5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
10
15
20
25
30
35
40
45
50

In [ ]:

In [ ]:

In [2]:
```
# CodeP3.4F25 – V.P. Carey, ME249 F25

#import useful packages
import keras
import pandas as pd
from keras.models import Sequential
import numpy as np
import keras.backend as kb
import tensorflow as tf
#the following 2 lines are only needed for Mac OS machines
import os
os.environ['KMP_DUPLICATE_LIB_OK']='True'

#create input data array

#convert to:
# BC's for PINN conduction model with HE enforcement in interior points

# x(cm), y(cm)   These are (x,y) coordinates of boundary location where temp
xdata = []
ND = 32
xdata =  [  [ 0.0, 0.0 ], [ 0.0, 0.5 ], [ 0.0, 1.0 ], [ 0.0, 1.5 ],[ 0.0, 2.
            [ 0.0, 3.5 ], [ 0.0, 4.0 ], [ 0.0, 4.5 ], [ 0.0, 5.0 ],
            [ 3.0, 0.0 ], [ 3.0, 0.5 ], [ 3.0, 1.0 ], [ 3.0, 1.5 ],[ 3.0, 2.
            [ 3.0, 3.5 ], [ 3.0, 4.0 ], [ 3.0, 4.5 ], [ 3.0, 5.0 ]]

xdata.append([  0.5, 0.0 ])
xdata.append([  1.0, 0.0 ])
xdata.append([  1.5, 0.0 ])
xdata.append([  2.0, 0.0 ])
xdata.append([  2.5, 0.0 ])

xdata.append([  0.5, 5.0 ])
xdata.append([  1.0, 5.0 ])
xdata.append([  1.5, 5.0 ])
xdata.append([  2.0, 5.0 ])
xdata.append([  2.5, 5.0 ])


# add more here if needed



xarray= np.array(xdata)
print (xdata)
print (xarray)



#convert to:
# meadian values of output variables

# T (deg C)   These are specified (measured) temperatures for each pair of (
```

```
ydata = []

ydata = [ [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 45. ], [ 40. ], [ 3
         [ 40. ], [ 35. ],  [ 30. ], [ 25. ], [ 20. ], [ 15. ], [ 10. ], [ 1

ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 45. ])

ydata.append([ 15. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])

# add more here if needed

yarray= np.array(ydata)
print (ydata)
print (yarray)

data_inputs = np.array(xdata)
data_outputs = np.array(ydata)

#boundary_x = ...  # x-coordinates for the boundary data
#boundary_y = ...  # y-coordinates for the boundary data
#boundary_T = ...  # Boundary temperature data (T_data)

# Copy the first column to a 1D numpy array
#vector = arr[:, 0].copy()
boundary_xnp = xarray[:, 0].copy()
# Copy the first column to a 1D numpy array
boundary_ynp = xarray[:, 1].copy()
boundary_Tnp = yarray[:, 0].copy()

# Convert the arrays to a TensorFlow tensors
boundary_x = tf.convert_to_tensor(boundary_xnp, dtype=tf.float32)
boundary_y = tf.convert_to_tensor(boundary_ynp, dtype=tf.float32)
boundary_T_ytrain = tf.convert_to_tensor(boundary_Tnp, dtype=tf.float32)

boundary_points_xtrain = tf.stack([boundary_x, boundary_y], axis=1)

print(boundary_points_xtrain)
print(boundary_T_ytrain)
```

```
[[0.0, 0.0], [0.0, 0.5], [0.0, 1.0], [0.0, 1.5], [0.0, 2.0], [0.0, 2.5], [0.
0, 3.0], [0.0, 3.5], [0.0, 4.0], [0.0, 4.5], [0.0, 5.0], [3.0, 0.0], [3.0,
0.5], [3.0, 1.0], [3.0, 1.5], [3.0, 2.0], [3.0, 2.5], [3.0, 3.0], [3.0, 3.
5], [3.0, 4.0], [3.0, 4.5], [3.0, 5.0], [0.5, 0.0], [1.0, 0.0], [1.5, 0.0],
[2.0, 0.0], [2.5, 0.0], [0.5, 5.0], [1.0, 5.0], [1.5, 5.0], [2.0, 5.0], [2.
5, 5.0]]
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]]
[[50.0], [50.0], [50.0], [50.0], [50.0], [45.0], [40.0], [35.0], [30.0], [2
5.0], [20.0], [40.0], [35.0], [30.0], [25.0], [20.0], [15.0], [10.0], [10.
0], [10.0], [10.0], [10.0], [50.0], [50.0], [50.0], [50.0], [45.0], [15.0],
[10.0], [10.0], [10.0], [10.0]]
[[50.]
 [50.]
 [50.]
 [50.]
 [50.]
 [45.]
 [40.]
 [35.]
 [30.]
 [25.]
 [20.]
 [40.]
 [35.]
 [30.]
```

```
 [25.]
 [20.]
 [15.]
 [10.]
 [10.]
 [10.]
 [10.]
 [10.]
 [50.]
 [50.]
 [50.]
 [50.]
 [45.]
 [15.]
 [10.]
 [10.]
 [10.]
 [10.]]
tf.Tensor(
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]], shape=(32, 2), dtype=float32)
tf.Tensor(
[50. 50. 50. 50. 50. 45. 40. 35. 30. 25. 20. 40. 35. 30. 25. 20. 15. 10.
 10. 10. 10. 10. 50. 50. 50. 50. 45. 15. 10. 10. 10. 10.], shape=(32,), dtyp
e=float32)
```

In [3]:
```
# define neural network model

    #As seen below, we have created four dense layers.
    #A dense layer is a layer in neural network that's fully connected.
    #In other words, all the neurons in one layer are connected to all other
    #In the first layer, we need to provide the input shape, which is 1 in c
    #The activation function we have chosen is elu, which stands for exponer

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Create your model (for example, a simple fully connected feedforward NN)
# Assume 2D input (x, y),  Output temperature T
model = Sequential([
        Dense(8, input_dim=2, activation='elu'),
        Dense(16, activation='elu'),
        Dense(1)
    ])

# Set the kappa value (this can be adjusted as per your needs)
kappa = 1.3

# Set the x_max and y_max values of your domain ( can adjust these according
x_max = 3.0
y_max = 5.0

#print summary of model features
model.summary()
```

```
/global/home/users/melius/.conda/envs/my_kernel/lib/python3.11/site-package
s/keras/src/layers/core/dense.py:92: UserWarning: Do not pass an `input_shap
e`/`input_dim` argument to a layer. When using Sequential models, prefer usi
ng an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Par
dense (Dense)	(None, 8)	
dense_1 (Dense)	(None, 16)	
dense_2 (Dense)	(None, 1)	

Total params: 185 (740.00 B)

Trainable params: 185 (740.00 B)

Non-trainable params: 0 (0.00 B)

In [4]:
```
# ADD CODE HERE for custom_loss_function <<========
# PINN CUSTOM LOSS FUNCTION - V.P. Carey, ME249 F25
#import useful packages

import tensorflow as tf
from tensorflow.keras import backend as K
```

```python
def custom_loss_function(kappa, x_max, y_max, epoch_tracker):
    """ Custom loss function for the neural network model.
        - kappa: weight for the Laplace residual term.
        - x_max, y_max: maximum x and y coordinates of the domain.
        - epoch_tracker: callback object to keep track of the current epoch.

    def loss(y_true, y_pred):
        # Define pointsBoundaryChk as a constant tensor
        # === COMPLETE 32 BOUNDARY POINTS FROM THE PROJECT FIGURE ===
        pointsBoundaryChk = tf.constant([
            # Bottom edge (y = 0)
            [0.0, 0.0],
            [0.5, 0.0],
            [1.0, 0.0],
            [1.5, 0.0],
            [2.0, 0.0],
            [2.5, 0.0],
            [3.0, 0.0],

            # Right edge (x = 3)
            [3.0, 0.5],
            [3.0, 1.0],
            [3.0, 1.5],
            [3.0, 2.0],
            [3.0, 2.5],
            [3.0, 3.0],
            [3.0, 3.5],
            [3.0, 4.0],
            [3.0, 4.5],
            [3.0, 5.0],

            # Top edge (y = 5)
            [0.5, 5.0],
            [1.0, 5.0],
            [1.5, 5.0],
            [2.0, 5.0],
            [2.5, 5.0],
            [3.0, 5.0],

            # Left edge (x = 0)
            [0.0, 0.5],
            [0.0, 1.0],
            [0.0, 1.5],
            [0.0, 2.0],
            [0.0, 2.5],
            [0.0, 3.0],
            [0.0, 3.5],
            [0.0, 4.0],
            [0.0, 4.5],
            [0.0, 5.0]

        ], dtype=tf.float32)
```

```
        # Boundary loss: Calculate squared error for the boundary condition
        boundary_loss = K.mean(K.square(y_pred - y_true))

        ESbdsum = 0.
        for _ in range(32):
            # Sample a random index and gather point
            idx = tf.random.uniform(shape=[], minval=0, maxval=32, dtype=tf.
            selected_point = tf.gather(pointsBoundaryChk, idx)

            # Stack selected_point as input for the model (shape becomes (1,
            input_tensor = tf.expand_dims(selected_point, axis=0)
            # Pass input_tensor as the input to the model
            T_pred_at_boundxy = model(input_tensor)  # Model prediction
            # add penalty if temperature at flux location is OUTSIDE RANGE 5
        #     if (T_pred_at_boundxy < 5.):
        #         ESbdsum += 20.*(5. - T_pred_at_boundxy)*(5. - T_pred_at_bo
        #     if (T_pred_at_boundxy > 55.):
        #         ESbdsum += 20.*(T_pred_at_boundxy - 55.)*(T_pred_at_boundx
        # boundary_loss += ESbdsum/32.
        # Compute lower and upper penalties without Python if-statements
            low_penalty  = tf.nn.relu(5.0  - T_pred_at_boundxy)
            high_penalty = tf.nn.relu(T_pred_at_boundxy - 55.0)

            # Square and scale (20×), then average over 32 samples
            ESbdsum += (20.0 * tf.square(low_penalty)  +
                        20.0 * tf.square(high_penalty)) / 32.0

        boundary_loss += ESbdsum


        # Check epoch from the callback
        epoch = epoch_tracker.epochs_trained

        # If current epoch is less than N_epInit, ignore the Laplace residua
        if epoch < epoch_tracker.N_epInit:
            total_loss = boundary_loss
        else:
            # 3. Randomly select an interior point from this list using Tens
            points = tf.constant([
                # x = 0.5
                [0.5, 0.5],
                [0.5, 1.0],
                [0.5, 1.5],
                [0.5, 2.0],
                [0.5, 2.5],
                [0.5, 3.0],
                [0.5, 3.5],
                [0.5, 4.0],
                [0.5, 4.5],

                # x = 1.0
                [1.0, 0.5],
                [1.0, 1.0],
                [1.0, 1.5],
                [1.0, 2.0],
                [1.0, 2.5],
```

```
            [1.0, 3.0],
            [1.0, 3.5],
            [1.0, 4.0],
            [1.0, 4.5],

            # x = 1.5
            [1.5, 0.5],
            [1.5, 1.0],
            [1.5, 1.5],
            [1.5, 2.0],
            [1.5, 2.5],
            [1.5, 3.0],
            [1.5, 3.5],
            [1.5, 4.0],
            [1.5, 4.5],

            # x = 2.0
            [2.0, 0.5],
            [2.0, 1.0],
            [2.0, 1.5],
            [2.0, 2.0],
            [2.0, 2.5],
            [2.0, 3.0],
            [2.0, 3.5],
            [2.0, 4.0],
            [2.0, 4.5],

            # x = 2.5
            [2.5, 0.5],
            [2.5, 1.0],
            [2.5, 1.5],
            [2.5, 2.0],
            [2.5, 2.5],
            [2.5, 3.0],
            [2.5, 3.5],
            [2.5, 4.0],
            [2.5, 4.5]
        ], dtype=tf.float32)


        meanLap_res = 0.
        meanTlimPen = 0.
        for _ in range(45):
            ##compute Laplacian residue here - need to do this once per
            # Use tf.random.uniform to sample an index for the interior
            idx = tf.random.uniform(shape=[], minval=0, maxval=45, dtype
            selected_point = tf.gather(points, idx)
            x_interior, y_interior = selected_point[0], selected_point[1

            # Use tf.GradientTape to compute second derivatives
            with tf.GradientTape(persistent=True) as tape:
                tape.watch(selected_point)
                # Pass the selected (x_interior, y_interior) to the mode
                T_pred_at_interior = model([x_interior, y_interior])  #

            # Compute first derivatives with respect to x and y
```

```
                dT_dx = tape.gradient(T_pred_at_interior, selected_point[0])
                dT_dy = tape.gradient(T_pred_at_interior, selected_point[1])

                # Compute second derivatives (Hessian)
                d2T_dx2 = tape.gradient(dT_dx, selected_point[0])
                d2T_dy2 = tape.gradient(dT_dy, selected_point[1])

                # Calculate the Laplacian (sum of second derivatives)
                laplacian = d2T_dx2 + d2T_dy2

                # Compute the Laplacian residual (sum of squared Laplacian)
                meanLap_res +=  K.square(laplacian)/45.

                # ADD PENALTY IF T below 5 or above 55

                # if (T_pred_at_interior < 5.):
                #     meanTlimPen += 20.*(5. - T_pred_at_interior)*(5. - T_p
                low_penalty = tf.nn.relu(5.0 - T_pred_at_interior)
                meanTlimPen += 20.0 * tf.square(low_penalty) / 45.0

                # if (T_pred_at_interior > 55.):
                #     meanTlimPen += 20.*(T_pred_at_interior - 55.)*(T_pred_
                high_penalty = tf.nn.relu(T_pred_at_interior - 55.0)
                meanTlimPen += 20.0 * tf.square(high_penalty) / 45.0


            #Combine three losses
            total_loss = boundary_loss + kappa * meanLap_res + meanTlimPen


        return total_loss
    return loss


from tensorflow.keras.callbacks import Callback

class EpochTracker(Callback):
    def __init__(self, N_epInit):
        super().__init__()
        self.N_epInit = N_epInit
        self.epochs_trained = 0

    def on_epoch_end(self, epoch, logs=None):
        self.epochs_trained = epoch

# Set the loss function with the required arguments
epoch_tracker = EpochTracker(N_epInit=2)
custom_loss = custom_loss_function(kappa=1.3, x_max=3.0, y_max=5.0, epoch_tr



# Compile the model with the custom loss
model.compile(optimizer='adam', loss=custom_loss)
```

In [1]:
```
#### After the compilation of the model, we'll try the fit method with 100 e
```

```
#The fit method takes three parameters; namely, x, y, and number of epochs.
#During model training, if all the batches of data are seen by the model onc
#we say that one epoch has been completed.


##### RUN FIT

# Assuming `boundary_points` and `boundary_T` are TensorFlow tensors

# Now, fit the model and pass the callback to `fit()`
historyData = model.fit(boundary_points_xtrain, boundary_T_ytrain,
                epochs=100000, batch_size=32,
                callbacks=[epoch_tracker])


loss_hist = historyData.history['loss']
#The above line will return a dictionary, access it's info like this:
best_epoch = np.argmin(historyData.history['loss']) + 1
print ('best epoch = ', best_epoch)
print('smallest loss =', np.min(loss_hist))

predictions = model.predict(data_inputs)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[1], line 14
      1 #### After the compilation of the model, we'll try the fit method wi
th 100 epochs.
      2
      3
   (...)
     12
     13 # Now, fit the model and pass the callback to `fit()`
---> 14 historyData = model.fit(boundary_points_xtrain, boundary_T_ytrain,
     15                 epochs=2000, batch_size=32,
     16                 callbacks=[epoch_tracker])
     19 loss_hist = historyData.history['loss']
     20 #The above line will return a dictionary, access it's info like thi
s:

NameError: name 'model' is not defined
```

In [6]:

```
predictions = model.predict(data_inputs)

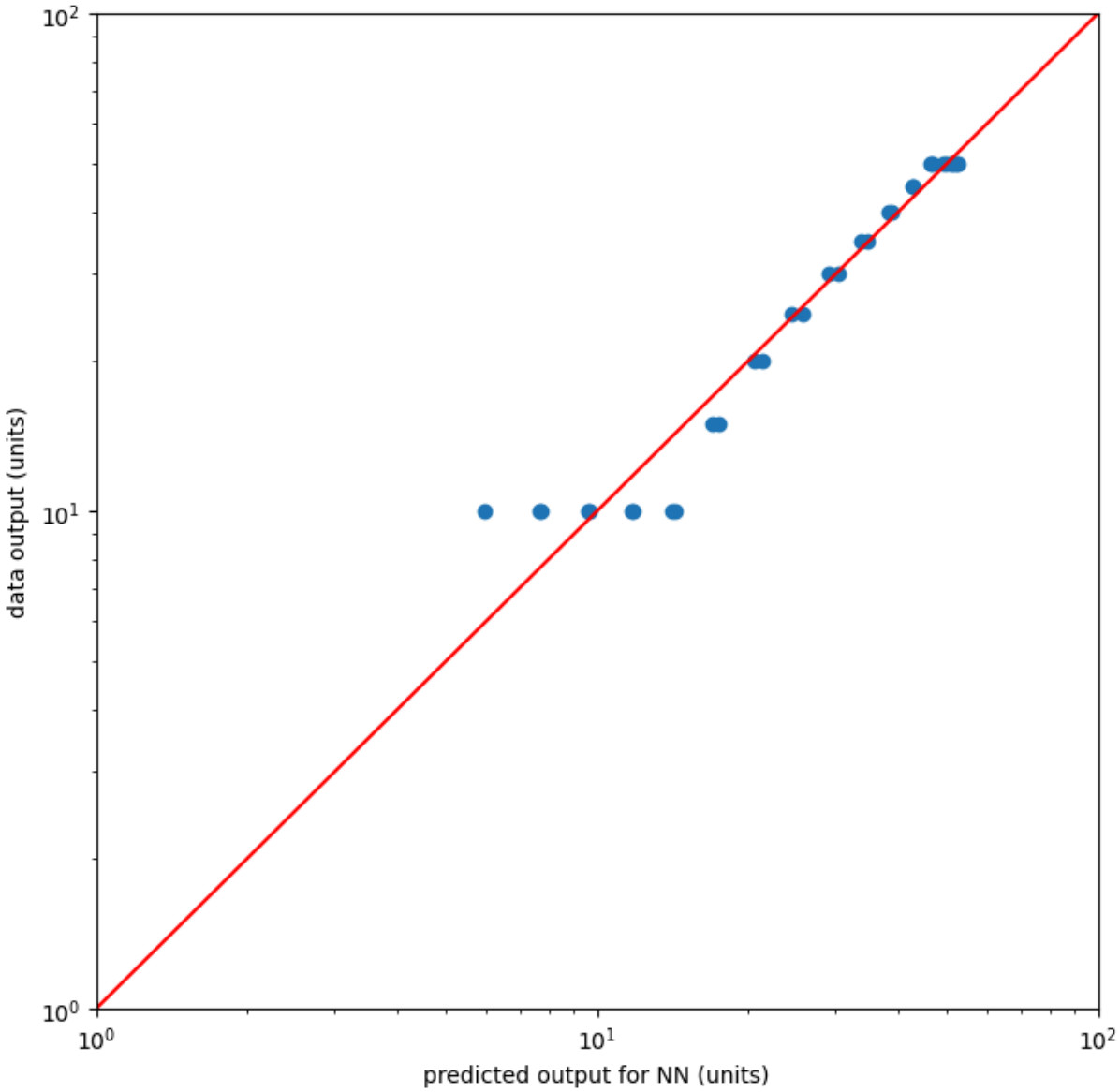


#SETTING UP PLOT
# matplotlib inline
# importing the required module
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [8, 8] # for square canvas
#========

'''CALCULATE PREDICTED VALUES AND RETRIEVE DATA VALUES TO PLOT'''
```

```
plt.scatter(predictions[:, 0], data_outputs[:, 0])
#plt.title('Genetic Algorithm training of Neural Network ==> output = alpha'
plt.xlabel('predicted output for NN (units)')
plt.ylabel('data output (units)')
plt.loglog()
plt.xlim(xmax = 100, xmin = 1.)
plt.ylim(ymax = 100, ymin = 1.)
# Generate red y=x line
x_data = np.linspace(1.0, 100.0, num=3)
y_data = x_data
plt.plot(x_data, y_data, color='red')
plt.savefig("log-log_plot_3.4_model_2.png", dpi=300, bbox_inches='tight')
plt.show()
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step
```



In [7]:
```
test = []
outpt=[]
```

```
predictions = model.predict(data_inputs)

#first point (row [0])comparison of data and prediction
# put in a loop to print comparion for all data points

xa=[]
ya=[]
za=[]

xao = np.linspace(0, 3, 7)  # 1D array for x values
yao = np.linspace(0, 5, 11)    # 1D array for y values
xa = np.linspace(0, 10, 77)    # 1D array for x values
ya = np.linspace(0, 10, 77)    # 1D array for y values
za = np.linspace(0, 10, 77)    # 1D array for z values

x =-0.50
ia=-1
while (x < 2.51):
    x = x + 0.5
    y = -0.5
    while (y < 4.51):
        ia=ia+1
        y = y + 0.50
        xa[ia]=x
        ya[ia]=y

        test = [[ x , y ]]
        testarray = np.array(test)
        outpt = model.predict(testarray)
        za[ia] = outpt
        print ('x= ', testarray[0][0], ', y= ', testarray[0][1],'T= ', outpt



X, Y, Z = xa, ya, za

import sys
#import csv
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import axes3d

# Plot X,Y,Z

# Creating figure
#fig = plt.figure(figsize =(16, 9))
#ax = plt.axes(projection ='3d')

fig = plt.figure()
#ax = fig.add_subplot(111, projection='3d')
ax = plt.axes(projection ='3d')
#ax.plot_trisurf(X, Y, Z, color='white', edgecolors='grey', alpha=0.5)
ax.plot_trisurf(X, Y, Z,
                linewidth = 0.2,
                antialiased = True
                , cmap='viridis');
```

```
surf = ax.plot_trisurf(
    X, Y, Z,
    linewidth=0.2,
    antialiased=True,
    cmap='viridis'
)


#  ADD COLORBAR
fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)



# fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)

ax.invert_yaxis()
ax.invert_xaxis()

plt.savefig("trisurf_plot_model_2.png", dpi=300, bbox_inches='tight')
plt.show()



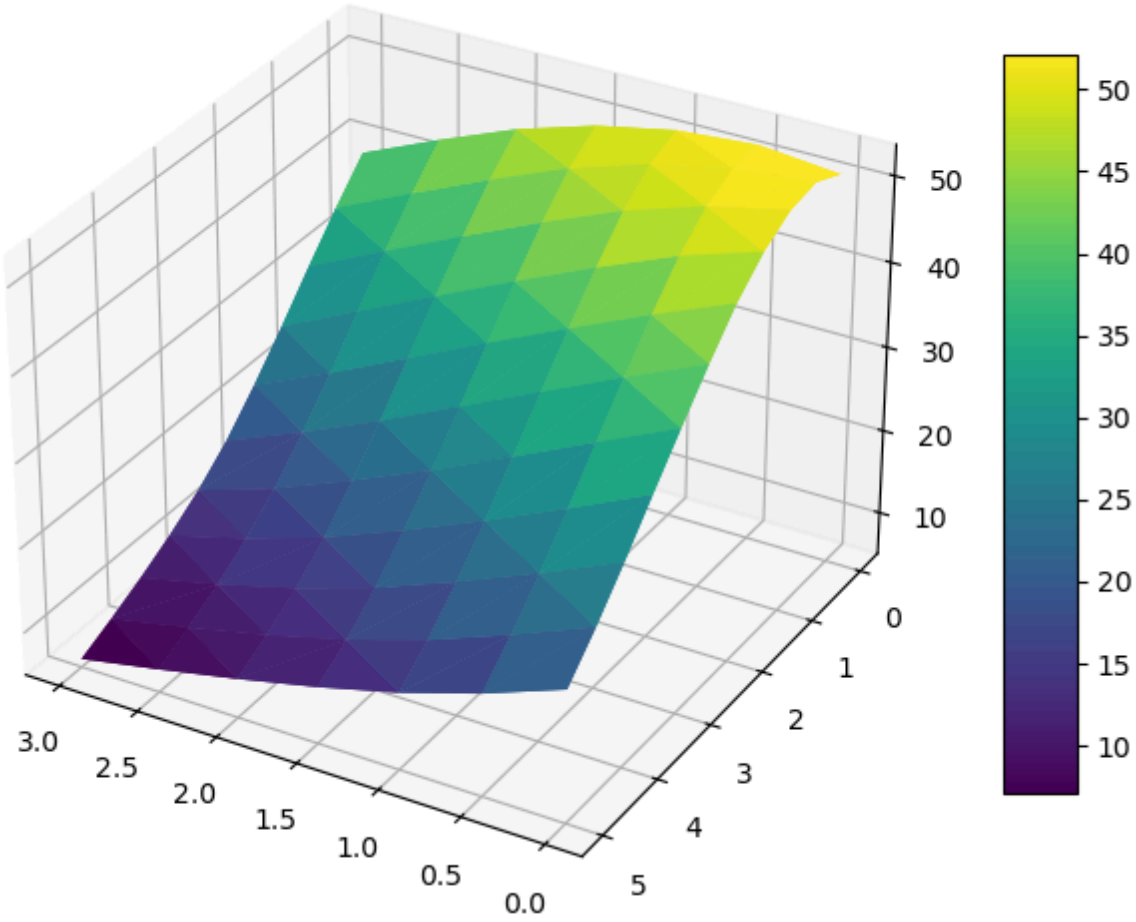
# Plot the data
plt.figure(figsize=(5, 6))
plt.tripcolor(xa, ya, za, shading='flat', cmap='viridis')
plt.savefig("Heat_map_model_2.png", dpi=300, bbox_inches='tight')
plt.colorbar(label='Z value')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Heat Map of Z values over X and Y domain')
plt.show()
```

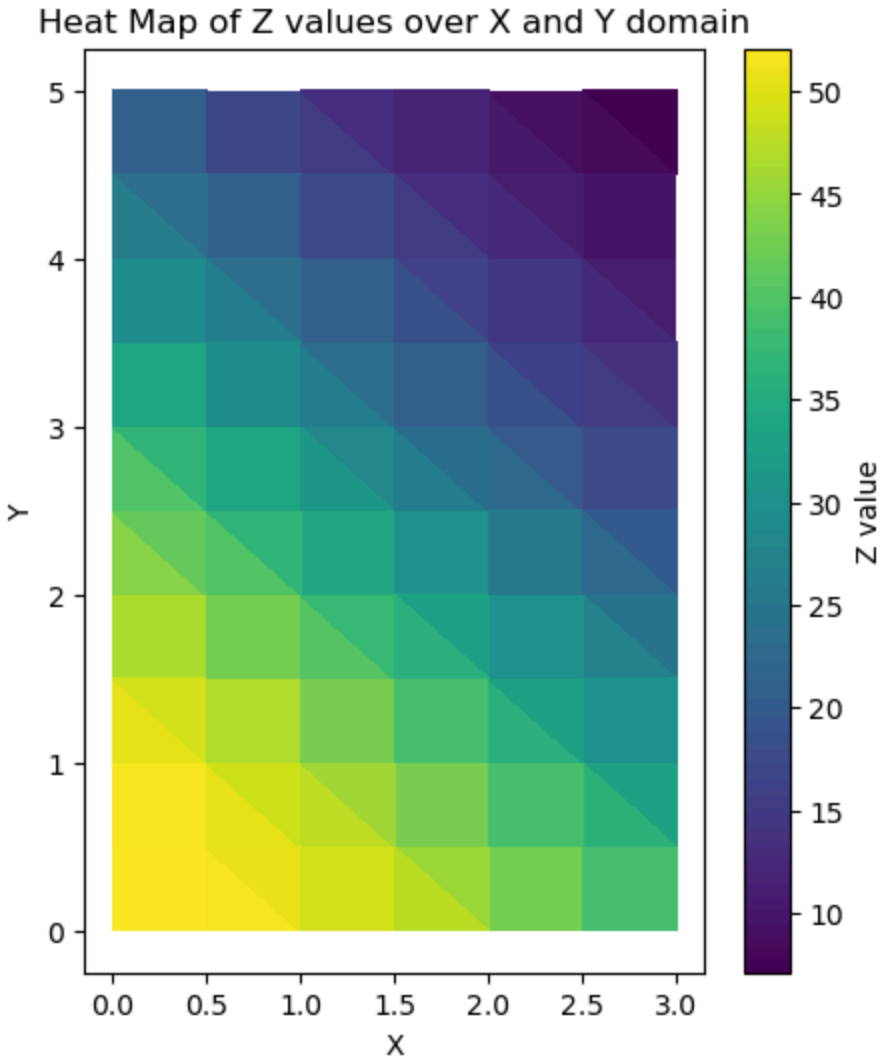
```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 61ms/step
x=  0.0 , y=  0.0 T=  50.96584
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.0 , y=  0.5 T=  52.60945
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  1.0 T=  51.99128
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
/tmp/ipykernel_4180339/3841726189.py:32: DeprecationWarning: Conversion of a
n array with ndim > 0 to a scalar is deprecated, and will error in future. E
nsure you extract a single element from your array before performing this op
eration. (Deprecated NumPy 1.25.)
  za[ia] = outpt
```

```
x=  0.0 , y=  1.5 T=  49.951176
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  2.0 T=  46.81934
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.0 , y=  2.5 T=  42.72145
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.0 , y=  3.0 T=  38.271023
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  3.5 T=  33.658096
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  4.0 T=  28.951752
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  4.5 T=  24.545918
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  5.0 T=  20.576769
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  0.0 T=  51.88528
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  0.5 T=  51.742115
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  1.0 T=  49.95255
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  1.5 T=  46.953724
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  2.0 T=  42.93047
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  2.5 T=  38.521248
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  3.0 T=  33.929714
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  0.5 , y=  3.5 T=  29.225576
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  4.0 T=  24.737137
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  4.5 T=  20.698338
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  5.0 T=  17.037195
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  0.0 T=  51.156948
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  0.5 T=  49.760662
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  1.0 T=  46.974392
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  1.5 T=  43.073265
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.0 , y=  2.0 T=  38.73373
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  2.5 T=  34.17967
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.0 , y=  3.0 T=  29.495737
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.0 , y=  3.5 T=  24.934422
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.0 , y=  4.0 T=  20.826593
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
```

```
x=  1.0 , y=  4.5 T=  17.108381
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.0 , y=  5.0 T=  14.078999
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
x=  1.5 , y=  0.0 T=  49.300545
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  0.5 T=  46.83856
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  1.0 T=  43.126102
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  1.5 T=  38.895535
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  2.0 T=  34.40091
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  2.5 T=  29.749546
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  3.0 T=  25.136122
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  3.5 T=  20.960773
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  4.0 T=  17.186451
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  4.5 T=  14.130054
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  5.0 T=  11.6887
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  0.0 T=  46.481613
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.0 , y=  0.5 T=  43.053326
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.0 , y=  1.0 T=  38.98711
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  1.5 T=  34.582752
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  2.0 T=  29.981144
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  2.5 T=  25.339499
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  3.0 T=  21.099504
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  3.5 T=  17.270689
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  4.0 T=  14.186641
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  4.5 T=  11.719135
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  5.0 T=  9.606077
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.5 , y=  0.0 T=  42.80212
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.5 , y=  0.5 T=  38.97941
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.5 , y=  1.0 T=  34.70923
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.5 , y=  1.5 T=  30.181751
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
```

```
x=  2.5 , y=  2.0 T=  25.540197
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  2.5 T=  21.24052
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  3.0 T=  17.359924
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  3.5 T=  14.248255
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  4.0 T=  11.75411
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.5 , y=  4.5 T=  9.616588
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.5 , y=  5.0 T=  7.712599
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  0.0 T=  38.829678
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  0.5 T=  34.75678
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  1.0 T=  30.338398
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  3.0 , y=  1.5 T=  25.731554
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  2.0 T=  21.380287
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  2.5 T=  17.452333
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  3.0 T=  14.314087
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  3.5 T=  11.793255
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  4.0 T=  9.63088
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  4.5 T=  7.704802
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  5.0 T=  5.94899
```


50
40
30
20
10
0
1
2
3
4
5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
50
45
40
35
30
25
20
15
10


Heat Map of Z values over X and Y domain
Y
X
Z value
0
1
2
3
4
5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
10
15
20
25
30
35
40
45
50

In [ ]:

In [ ]:

In [5]:
```
# CodeP3.4F25 – V.P. Carey, ME249 F25

#import useful packages
import keras
import pandas as pd
from keras.models import Sequential
import numpy as np
import keras.backend as kb
import tensorflow as tf
#the following 2 lines are only needed for Mac OS machines
import os
os.environ['KMP_DUPLICATE_LIB_OK']='True'

#create input data array

#convert to:
# BC's for PINN conduction model with HE enforcement in interior points

# x(cm), y(cm)   These are (x,y) coordinates of boundary location where temp
xdata = []
ND = 32
xdata =  [  [ 0.0, 0.0 ], [ 0.0, 0.5 ], [ 0.0, 1.0 ], [ 0.0, 1.5 ],[ 0.0, 2.
            [ 0.0, 3.5 ], [ 0.0, 4.0 ], [ 0.0, 4.5 ], [ 0.0, 5.0 ],
            [ 3.0, 0.0 ], [ 3.0, 0.5 ], [ 3.0, 1.0 ], [ 3.0, 1.5 ],[ 3.0, 2.
            [ 3.0, 3.5 ], [ 3.0, 4.0 ], [ 3.0, 4.5 ], [ 3.0, 5.0 ]]

xdata.append([  0.5, 0.0 ])
xdata.append([  1.0, 0.0 ])
xdata.append([  1.5, 0.0 ])
xdata.append([  2.0, 0.0 ])
xdata.append([  2.5, 0.0 ])

xdata.append([  0.5, 5.0 ])
xdata.append([  1.0, 5.0 ])
xdata.append([  1.5, 5.0 ])
xdata.append([  2.0, 5.0 ])
xdata.append([  2.5, 5.0 ])
# 6 more points from interior
xdata.append([  0.5, 0.5 ])
xdata.append([  0.5, 1.0 ])
xdata.append([  1.0, 0.5])
xdata.append([  2.0, 4.5])
xdata.append([  2.5, 4.0])
xdata.append([  2.5, 4.5])




# add more here if needed



xarray= np.array(xdata)
```

```
print (xdata)
print (xarray)



#convert to:
# meadian values of output variables

# T (deg C)   These are specified (measured) temperatures for each pair of (

ydata = []

ydata = [ [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 50. ], [ 45. ], [ 40. ], [ 3
         [ 40. ], [ 35. ],  [ 30. ], [ 25. ], [ 20. ], [ 15. ], [ 10. ], [ 1

ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 45. ])

ydata.append([ 15. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 50. ])
ydata.append([ 10. ])
ydata.append([ 10. ])
ydata.append([ 10. ])

# add more here if needed

yarray= np.array(ydata)
print (ydata)
print (yarray)

data_inputs = np.array(xdata)
data_outputs = np.array(ydata)

#boundary_x = ...  # x-coordinates for the boundary data
#boundary_y = ...  # y-coordinates for the boundary data
#boundary_T = ...  # Boundary temperature data (T_data)

# Copy the first column to a 1D numpy array
#vector = arr[:, 0].copy()
boundary_xnp = xarray[:, 0].copy()
# Copy the first column to a 1D numpy array
boundary_ynp = xarray[:, 1].copy()
boundary_Tnp = yarray[:, 0].copy()

# Convert the arrays to a TensorFlow tensors
boundary_x = tf.convert_to_tensor(boundary_xnp, dtype=tf.float32)
boundary_y = tf.convert_to_tensor(boundary_ynp, dtype=tf.float32)
```

```
boundary_T_ytrain = tf.convert_to_tensor(boundary_Tnp, dtype=tf.float32)

boundary_points_xtrain = tf.stack([boundary_x, boundary_y], axis=1)

print(boundary_points_xtrain)
print(boundary_T_ytrain)
```

```
[[0.0, 0.0], [0.0, 0.5], [0.0, 1.0], [0.0, 1.5], [0.0, 2.0], [0.0, 2.5], [0.
0, 3.0], [0.0, 3.5], [0.0, 4.0], [0.0, 4.5], [0.0, 5.0], [3.0, 0.0], [3.0,
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5], [3.0, 4.0], [3.0, 4.5], [3.0, 5.0], [0.5, 0.0], [1.0, 0.0], [1.5, 0.0],
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5, 5.0], [0.5, 0.5], [0.5, 1.0], [1.0, 0.5], [2.0, 4.5], [2.5, 4.0], [2.5,
4.5]]
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]
 [0.5 0.5]
 [0.5 1. ]
 [1.  0.5]
 [2.  4.5]
 [2.5 4. ]
 [2.5 4.5]]
[[50.0], [50.0], [50.0], [50.0], [50.0], [45.0], [40.0], [35.0], [30.0], [2
5.0], [20.0], [40.0], [35.0], [30.0], [25.0], [20.0], [15.0], [10.0], [10.
0], [10.0], [10.0], [10.0], [50.0], [50.0], [50.0], [50.0], [45.0], [15.0],
[10.0], [10.0], [10.0], [10.0], [50.0], [50.0], [50.0], [10.0], [10.0], [10.
0]]
[[50.]
 [50.]
 [50.]
 [50.]
 [50.]
 [45.]
```

```
 [40.]
 [35.]
 [30.]
 [25.]
 [20.]
 [40.]
 [35.]
 [30.]
 [25.]
 [20.]
 [15.]
 [10.]
 [10.]
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 [50.]
 [45.]
 [15.]
 [10.]
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 [50.]
 [50.]
 [50.]
 [10.]
 [10.]
 [10.]]
tf.Tensor(
[[0.  0. ]
 [0.  0.5]
 [0.  1. ]
 [0.  1.5]
 [0.  2. ]
 [0.  2.5]
 [0.  3. ]
 [0.  3.5]
 [0.  4. ]
 [0.  4.5]
 [0.  5. ]
 [3.  0. ]
 [3.  0.5]
 [3.  1. ]
 [3.  1.5]
 [3.  2. ]
 [3.  2.5]
 [3.  3. ]
 [3.  3.5]
 [3.  4. ]
 [3.  4.5]
 [3.  5. ]
 [0.5 0. ]
```

```
 [1.  0. ]
 [1.5 0. ]
 [2.  0. ]
 [2.5 0. ]
 [0.5 5. ]
 [1.  5. ]
 [1.5 5. ]
 [2.  5. ]
 [2.5 5. ]
 [0.5 0.5]
 [0.5 1. ]
 [1.  0.5]
 [2.  4.5]
 [2.5 4. ]
 [2.5 4.5]], shape=(38, 2), dtype=float32)
tf.Tensor(
[50. 50. 50. 50. 50. 45. 40. 35. 30. 25. 20. 40. 35. 30. 25. 20. 15. 10.
 10. 10. 10. 10. 50. 50. 50. 50. 45. 15. 10. 10. 10. 10. 50. 50. 50. 10.
 10. 10.], shape=(38,), dtype=float32)
```

In [6]:
```
# define neural network model

    #As seen below, we have created four dense layers.
    #A dense layer is a layer in neural network that's fully connected.
    #In other words, all the neurons in one layer are connected to all other
    #In the first layer, we need to provide the input shape, which is 1 in c
    #The activation function we have chosen is elu, which stands for exponer

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Create your model (for example, a simple fully connected feedforward NN)
# Assume 2D input (x, y),  Output temperature T
model = Sequential([
        Dense(5, input_dim=2, activation='elu'),
        Dense(10, activation='elu'),
        Dense(1)
    ])

# Set the kappa value (this can be adjusted as per your needs)
kappa = 1.3

# Set the x_max and y_max values of your domain ( can adjust these according

x_max = 3.0
y_max = 5.0

#print summary of model features
model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape	Par
dense_3 (Dense)	(None, 5)	
dense_4 (Dense)	(None, 10)	
dense_5 (Dense)	(None, 1)	

Total params: 86 (344.00 B)
Trainable params: 86 (344.00 B)
Non-trainable params: 0 (0.00 B)

In [7]:
```
# ADD CODE HERE for custom_loss_function <<========
# PINN CUSTOM LOSS FUNCTION - V.P. Carey, ME249 F25
#import useful packages

import tensorflow as tf
from tensorflow.keras import backend as K


def custom_loss_function(kappa, x_max, y_max, epoch_tracker):
    """ Custom loss function for the neural network model.
        - kappa: weight for the Laplace residual term.
        - x_max, y_max: maximum x and y coordinates of the domain.
        - epoch_tracker: callback object to keep track of the current epoch.

    def loss(y_true, y_pred):
        # Define pointsBoundaryChk as a constant tensor
        # === COMPLETE 32 BOUNDARY POINTS FROM THE PROJECT FIGURE ===
        pointsBoundaryChk = tf.constant([
            # Bottom edge (y = 0)
            [0.0, 0.0],
            [0.5, 0.0],
            [1.0, 0.0],
            [1.5, 0.0],
            [2.0, 0.0],
            [2.5, 0.0],
            [3.0, 0.0],

            # Right edge (x = 3)
            [3.0, 0.5],
            [3.0, 1.0],
            [3.0, 1.5],
            [3.0, 2.0],
            [3.0, 2.5],
            [3.0, 3.0],
            [3.0, 3.5],
            [3.0, 4.0],
            [3.0, 4.5],
            [3.0, 5.0],

            # Top edge (y = 5)
            [0.5, 5.0],
            [1.0, 5.0],
```

```
            [1.5, 5.0],
            [2.0, 5.0],
            [2.5, 5.0],
            [3.0, 5.0],

            # Left edge (x = 0)
            [0.0, 0.5],
            [0.0, 1.0],
            [0.0, 1.5],
            [0.0, 2.0],
            [0.0, 2.5],
            [0.0, 3.0],
            [0.0, 3.5],
            [0.0, 4.0],
            [0.0, 4.5],
            [0.0, 5.0],
            # 6 more points from interior
            [  0.5, 0.5 ],
            [  0.5, 1.0 ],
            [  1.0, 0.5],
            [  2.0, 4.5],
            [  2.5, 4.0],
            [  2.5, 4.5]

        ], dtype=tf.float32)


        # Boundary loss: Calculate squared error for the boundary condition
        boundary_loss = K.mean(K.square(y_pred - y_true))

        ESbdsum = 0.
        for _ in range(32):
            # Sample a random index and gather point
            idx = tf.random.uniform(shape=[], minval=0, maxval=32, dtype=tf.
            selected_point = tf.gather(pointsBoundaryChk, idx)

            # Stack selected_point as input for the model (shape becomes (1,
            input_tensor = tf.expand_dims(selected_point, axis=0)
            # Pass input_tensor as the input to the model
            T_pred_at_boundxy = model(input_tensor)  # Model prediction
            # add penalty if temperature at flux location is OUTSIDE RANGE 5
        #     if (T_pred_at_boundxy < 5.):
        #         ESbdsum += 20.*(5. - T_pred_at_boundxy)*(5. - T_pred_at_bo
        #     if (T_pred_at_boundxy > 55.):
        #         ESbdsum += 20.*(T_pred_at_boundxy - 55.)*(T_pred_at_boundx
        # boundary_loss += ESbdsum/32.
        # Compute lower and upper penalties without Python if-statements
            low_penalty  = tf.nn.relu(5.0  - T_pred_at_boundxy)
            high_penalty = tf.nn.relu(T_pred_at_boundxy - 55.0)

            # Square and scale (20×), then average over 32 samples
            ESbdsum += (20.0 * tf.square(low_penalty)  +
                        20.0 * tf.square(high_penalty)) / 32.0

        boundary_loss += ESbdsum
```

```
        # Check epoch from the callback
        epoch = epoch_tracker.epochs_trained

        # If current epoch is less than N_epInit, ignore the Laplace residua
        if epoch < epoch_tracker.N_epInit:
            total_loss = boundary_loss
        else:
            # 3. Randomly select an interior point from this list using Tens
            points = tf.constant([
                # x = 0.5
                # [0.5, 0.5],
                # [0.5, 1.0],
                [0.5, 1.5],
                [0.5, 2.0],
                [0.5, 2.5],
                [0.5, 3.0],
                [0.5, 3.5],
                [0.5, 4.0],
                [0.5, 4.5],

                # x = 1.0
                # [1.0, 0.5],
                [1.0, 1.0],
                [1.0, 1.5],
                [1.0, 2.0],
                [1.0, 2.5],
                [1.0, 3.0],
                [1.0, 3.5],
                [1.0, 4.0],
                [1.0, 4.5],

                # x = 1.5
                [1.5, 0.5],
                [1.5, 1.0],
                [1.5, 1.5],
                [1.5, 2.0],
                [1.5, 2.5],
                [1.5, 3.0],
                [1.5, 3.5],
                [1.5, 4.0],
                [1.5, 4.5],

                # x = 2.0
                [2.0, 0.5],
                [2.0, 1.0],
                [2.0, 1.5],
                [2.0, 2.0],
                [2.0, 2.5],
                [2.0, 3.0],
                [2.0, 3.5],
                [2.0, 4.0],
                # [2.0, 4.5],

                # x = 2.5
                [2.5, 0.5],
```

```
            [2.5, 1.0],
            [2.5, 1.5],
            [2.5, 2.0],
            [2.5, 2.5],
            [2.5, 3.0],
            [2.5, 3.5],
            # [2.5, 4.0],
            # [2.5, 4.5]
        ], dtype=tf.float32)


        meanLap_res = 0.
        meanTlimPen = 0.
        for _ in range(45):
            ##compute Laplacian residue here - need to do this once per
            # Use tf.random.uniform to sample an index for the interior
            idx = tf.random.uniform(shape=[], minval=0, maxval=45, dtype
            selected_point = tf.gather(points, idx)
            x_interior, y_interior = selected_point[0], selected_point[1

            # Use tf.GradientTape to compute second derivatives
            with tf.GradientTape(persistent=True) as tape:
                tape.watch(selected_point)
                # Pass the selected (x_interior, y_interior) to the mode
                T_pred_at_interior = model([x_interior, y_interior])  #

            # Compute first derivatives with respect to x and y
            dT_dx = tape.gradient(T_pred_at_interior, selected_point[0])
            dT_dy = tape.gradient(T_pred_at_interior, selected_point[1])

            # Compute second derivatives (Hessian)
            d2T_dx2 = tape.gradient(dT_dx, selected_point[0])
            d2T_dy2 = tape.gradient(dT_dy, selected_point[1])

            # Calculate the Laplacian (sum of second derivatives)
            laplacian = d2T_dx2 + d2T_dy2

            # Compute the Laplacian residual (sum of squared Laplacian)
            meanLap_res +=  K.square(laplacian)/45.

            # ADD PENALTY IF T below 5 or above 55

            # if (T_pred_at_interior < 5.):
            #     meanTlimPen += 20.*(5. - T_pred_at_interior)*(5. - T_p
            low_penalty = tf.nn.relu(5.0 - T_pred_at_interior)
            meanTlimPen += 20.0 * tf.square(low_penalty) / 45.0

            # if (T_pred_at_interior > 55.):
            #     meanTlimPen += 20.*(T_pred_at_interior - 55.)*(T_pred_
            high_penalty = tf.nn.relu(T_pred_at_interior - 55.0)
            meanTlimPen += 20.0 * tf.square(high_penalty) / 45.0


        #Combine three losses
        total_loss = boundary_loss + kappa * meanLap_res + meanTlimPen
```

```
        return total_loss
    return loss


from tensorflow.keras.callbacks import Callback

class EpochTracker(Callback):
    def __init__(self, N_epInit):
        super().__init__()
        self.N_epInit = N_epInit
        self.epochs_trained = 0

    def on_epoch_end(self, epoch, logs=None):
        self.epochs_trained = epoch

# Set the loss function with the required arguments
epoch_tracker = EpochTracker(N_epInit=2)
custom_loss = custom_loss_function(kappa=1.3, x_max=3.0, y_max=5.0, epoch_tr



# Compile the model with the custom loss
model.compile(optimizer='adam', loss=custom_loss)
```

In [ ]:

```
#### After the compilation of the model, we'll try the fit method with 100 e


#The fit method takes three parameters; namely, x, y, and number of epochs.
#During model training, if all the batches of data are seen by the model onc
#we say that one epoch has been completed.


##### RUN FIT

# Assuming `boundary_points` and `boundary_T` are TensorFlow tensors

# Now, fit the model and pass the callback to `fit()`
historyData = model.fit(boundary_points_xtrain, boundary_T_ytrain,
                epochs=100000, batch_size=32,
                callbacks=[epoch_tracker])


loss_hist = historyData.history['loss']
#The above line will return a dictionary, access it's info like this:
best_epoch = np.argmin(historyData.history['loss']) + 1
print ('best epoch = ', best_epoch)
print('smallest loss =', np.min(loss_hist))

predictions = model.predict(data_inputs)
```

```
Epoch 1/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0766
Epoch 2/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0778
Epoch 3/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1014
Epoch 4/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0834
Epoch 5/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0801
Epoch 6/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0958
Epoch 7/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0780
Epoch 8/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0773
Epoch 9/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1058
Epoch 10/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0918
Epoch 11/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 12/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1043
Epoch 13/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1011
Epoch 14/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 15/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0849
Epoch 16/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0799
Epoch 17/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0823
Epoch 18/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0847
Epoch 19/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 20/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0853
Epoch 21/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0825
Epoch 22/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0747
Epoch 23/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0745
Epoch 24/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0747
Epoch 25/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0786
Epoch 26/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 27/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 28/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
```

```
Epoch 29/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0775
Epoch 30/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0754
Epoch 31/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0801
Epoch 32/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877
Epoch 33/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0771
Epoch 34/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0812
Epoch 35/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0874
Epoch 36/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0849
Epoch 37/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 38/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1003
Epoch 39/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1100
Epoch 40/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 41/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0920
Epoch 42/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0873
Epoch 43/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0753
Epoch 44/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 45/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877
Epoch 46/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0768
Epoch 47/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0778
Epoch 48/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0815
Epoch 49/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 50/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 51/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 52/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 53/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 54/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 55/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0790
Epoch 56/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0753
```

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Epoch 57/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 58/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0787
Epoch 59/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0852
Epoch 60/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 61/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
Epoch 62/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0957
Epoch 63/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0825
Epoch 64/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0910
Epoch 65/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0888
Epoch 66/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0827
Epoch 67/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 68/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 69/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 70/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 71/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0856
Epoch 72/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0813
Epoch 73/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 74/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0847
Epoch 75/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0812
Epoch 76/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 77/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0812
Epoch 78/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0865
Epoch 79/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 80/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 81/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0830
Epoch 82/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0836
Epoch 83/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 84/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0851
```

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Epoch 85/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 86/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0763
Epoch 87/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 88/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0734
Epoch 89/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0782
Epoch 90/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0828
Epoch 91/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0790
Epoch 92/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 93/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 94/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 95/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0823
Epoch 96/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 97/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0938
Epoch 98/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0870
Epoch 99/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0770
Epoch 100/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0835
Epoch 101/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 102/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0784
Epoch 103/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0997
Epoch 104/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1009
Epoch 105/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0787
Epoch 106/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 107/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1080
Epoch 108/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1076
Epoch 109/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 110/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0853
Epoch 111/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0964
Epoch 112/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0722
```

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Epoch 113/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0929
Epoch 114/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1044
Epoch 115/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0804
Epoch 116/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 117/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0795
Epoch 118/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 119/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0816
Epoch 120/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0784
Epoch 121/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 122/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0803
Epoch 123/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 124/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0754
Epoch 125/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 126/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 127/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 128/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 129/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 130/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0771
Epoch 131/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 132/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 133/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0776
Epoch 134/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0789
Epoch 135/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 136/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 137/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0851
Epoch 138/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1079
Epoch 139/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0942
Epoch 140/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
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Epoch 141/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0934
Epoch 142/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0878
Epoch 143/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 144/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0917
Epoch 145/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1042
Epoch 146/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0797
Epoch 147/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 148/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0910
Epoch 149/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0813
Epoch 150/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0772
Epoch 151/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1076
Epoch 152/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0942
Epoch 153/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0841
Epoch 154/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 155/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 156/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 157/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0775
Epoch 158/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0764
Epoch 159/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0778
Epoch 160/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0720
Epoch 161/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0873
Epoch 162/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0943
Epoch 163/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 164/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0896
Epoch 165/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0898
Epoch 166/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0752
Epoch 167/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0956
Epoch 168/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0858
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Epoch 169/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0775
Epoch 170/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1007
Epoch 171/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0843
Epoch 172/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0801
Epoch 173/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0871
Epoch 174/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0791
Epoch 175/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 176/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 177/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 178/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0752
Epoch 179/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
Epoch 180/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0783
Epoch 181/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 182/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 183/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0782
Epoch 184/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 185/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0761
Epoch 186/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0782
Epoch 187/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 188/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0737
Epoch 189/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 190/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 191/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 192/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 193/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0769
Epoch 194/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0767
Epoch 195/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 196/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
```

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Epoch 197/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 198/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 199/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 200/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0816
Epoch 201/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0815
Epoch 202/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0762
Epoch 203/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0797
Epoch 204/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0809
Epoch 205/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 206/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0891
Epoch 207/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0944
Epoch 208/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0788
Epoch 209/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 210/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0802
Epoch 211/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0755
Epoch 212/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 213/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0948
Epoch 214/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0812
Epoch 215/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0794
Epoch 216/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0850
Epoch 217/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0831
Epoch 218/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 219/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 220/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1032
Epoch 221/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0916
Epoch 222/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0793
Epoch 223/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0941
Epoch 224/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0786
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Epoch 225/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0840
Epoch 226/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0845
Epoch 227/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 228/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 229/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 230/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0778
Epoch 231/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 232/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 233/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0838
Epoch 234/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0887
Epoch 235/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0782
Epoch 236/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 237/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 238/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0770
Epoch 239/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 240/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 241/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0794
Epoch 242/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 243/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 244/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 245/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 246/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0853
Epoch 247/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1029
Epoch 248/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0815
Epoch 249/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0817
Epoch 250/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 251/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0731
Epoch 252/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0969
```

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Epoch 253/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1028
Epoch 254/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 255/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0908
Epoch 256/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1151
Epoch 257/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0897
Epoch 258/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0815
Epoch 259/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0944
Epoch 260/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0921
Epoch 261/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
Epoch 262/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0767
Epoch 263/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 264/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 265/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0828
Epoch 266/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0846
Epoch 267/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 268/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 269/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 270/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 271/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0782
Epoch 272/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 273/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0818
Epoch 274/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0857
Epoch 275/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 276/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0870
Epoch 277/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 278/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 279/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0758
Epoch 280/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0816
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Epoch 281/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0942
Epoch 282/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0823
Epoch 283/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 284/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0787
Epoch 285/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0748
Epoch 286/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0791
Epoch 287/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
Epoch 288/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0811
Epoch 289/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0764
Epoch 290/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 291/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 292/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 293/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 294/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 295/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 296/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 297/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 298/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 299/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 300/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0862
Epoch 301/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1074
Epoch 302/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0882
Epoch 303/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0742
Epoch 304/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0806
Epoch 305/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0831
Epoch 306/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 307/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0797
Epoch 308/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0817
```

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Epoch 309/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 310/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 311/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0795
Epoch 312/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 313/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0728
Epoch 314/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0844
Epoch 315/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0836
Epoch 316/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0762
Epoch 317/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0948
Epoch 318/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0829
Epoch 319/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 320/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0822
Epoch 321/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 322/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 323/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0782
Epoch 324/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 325/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 326/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0736
Epoch 327/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 328/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 329/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 330/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 331/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0799
Epoch 332/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 333/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 334/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0764
Epoch 335/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0919
Epoch 336/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
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Epoch 337/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0802
Epoch 338/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0801
Epoch 339/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0738
Epoch 340/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 341/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 342/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 343/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 344/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 345/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 346/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0847
Epoch 347/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0884
Epoch 348/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0733
Epoch 349/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0881
Epoch 350/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0840
Epoch 351/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 352/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0904
Epoch 353/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0880
Epoch 354/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 355/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0834
Epoch 356/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1003
Epoch 357/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0863
Epoch 358/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 359/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 360/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0898
Epoch 361/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0849
Epoch 362/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 363/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0740
Epoch 364/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
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Epoch 365/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 366/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0724
Epoch 367/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0728
Epoch 368/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 369/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 370/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0741
Epoch 371/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 372/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 373/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0717
Epoch 374/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0822
Epoch 375/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 376/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0789
Epoch 377/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0843
Epoch 378/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0750
Epoch 379/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 380/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
Epoch 381/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 382/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 383/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0879
Epoch 384/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 385/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.0729
Epoch 386/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 387/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0844
Epoch 388/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 389/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
Epoch 390/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0801
Epoch 391/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 392/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0810
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Epoch 393/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0787
Epoch 394/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 395/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0787
Epoch 396/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0811
Epoch 397/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 398/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0835
Epoch 399/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 400/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 401/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0965
Epoch 402/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0908
Epoch 403/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 404/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 405/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0780
Epoch 406/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 407/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 408/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0909
Epoch 409/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0827
Epoch 410/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0749
Epoch 411/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1066
Epoch 412/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1023
Epoch 413/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 414/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877
Epoch 415/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.1052
Epoch 416/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0829
Epoch 417/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 418/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0900
Epoch 419/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 420/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
```

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Epoch 421/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 422/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 423/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 424/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0847
Epoch 425/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0796
Epoch 426/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 427/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 428/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 429/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 430/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0894
Epoch 431/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0961
Epoch 432/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 433/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0841
Epoch 434/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1025
Epoch 435/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0862
Epoch 436/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 437/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 438/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 439/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0819
Epoch 440/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0748
Epoch 441/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 442/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0786
Epoch 443/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 444/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0832
Epoch 445/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 446/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0768
Epoch 447/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0965
Epoch 448/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0828
```

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Epoch 449/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0762
Epoch 450/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1094
Epoch 451/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0937
Epoch 452/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 453/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 454/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 455/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0760
Epoch 456/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0831
Epoch 457/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 458/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 459/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0929
Epoch 460/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0861
Epoch 461/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 462/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0853
Epoch 463/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0786
Epoch 464/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 465/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 466/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0734
Epoch 467/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0837
Epoch 468/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0820
Epoch 469/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 470/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0896
Epoch 471/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0921
Epoch 472/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 473/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0920
Epoch 474/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0942
Epoch 475/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 476/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
```

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Epoch 477/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0803
Epoch 478/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 479/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0847
Epoch 480/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 481/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0840
Epoch 482/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1285
Epoch 483/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1003
Epoch 484/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 485/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0942
Epoch 486/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0861
Epoch 487/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 488/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0771
Epoch 489/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 490/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 491/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 492/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 493/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0758
Epoch 494/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0818
Epoch 495/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 496/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 497/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 498/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0754
Epoch 499/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0740
Epoch 500/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 501/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 502/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 503/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 504/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
```

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Epoch 505/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0795
Epoch 506/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0819
Epoch 507/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 508/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 509/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0745
Epoch 510/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0755
Epoch 511/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 512/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 513/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0708
Epoch 514/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0740
Epoch 515/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0845
Epoch 516/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 517/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 518/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0899
Epoch 519/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0786
Epoch 520/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 521/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0829
Epoch 522/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0745
Epoch 523/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 524/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 525/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 526/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0745
Epoch 527/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 528/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 529/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 530/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 531/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 532/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0728
```

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Epoch 533/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 534/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 535/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 536/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 537/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 538/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0729
Epoch 539/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 540/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0735
Epoch 541/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 542/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0754
Epoch 543/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 544/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 545/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0738
Epoch 546/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0756
Epoch 547/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 548/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 549/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 550/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 551/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 552/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0737
Epoch 553/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0787
Epoch 554/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 555/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0767
Epoch 556/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 557/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 558/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0862
Epoch 559/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0821
Epoch 560/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
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Epoch 561/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 562/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 563/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 564/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0734
Epoch 565/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0731
Epoch 566/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0719
Epoch 567/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 568/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 569/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 570/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 571/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 572/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 573/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0835
Epoch 574/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0724
Epoch 575/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0830
Epoch 576/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0822
Epoch 577/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0702
Epoch 578/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 579/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0809
Epoch 580/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 581/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 582/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0790
Epoch 583/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 584/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0825
Epoch 585/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0876
Epoch 586/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 587/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0956
Epoch 588/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
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Epoch 589/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0841
Epoch 590/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1051
Epoch 591/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0850
Epoch 592/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0874
Epoch 593/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0882
Epoch 594/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 595/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877
Epoch 596/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0921
Epoch 597/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0771
Epoch 598/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0877
Epoch 599/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1027
Epoch 600/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0904
Epoch 601/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 602/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0835
Epoch 603/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 604/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0791
Epoch 605/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0813
Epoch 606/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 607/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0732
Epoch 608/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0807
Epoch 609/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0757
Epoch 610/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0719
Epoch 611/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0927
Epoch 612/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0931
Epoch 613/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 614/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0920
Epoch 615/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0929
Epoch 616/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
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Epoch 617/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1100
Epoch 618/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1109
Epoch 619/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 620/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0895
Epoch 621/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0928
Epoch 622/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 623/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0823
Epoch 624/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0842
Epoch 625/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0751
Epoch 626/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 627/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0755
Epoch 628/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 629/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0724
Epoch 630/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0726
Epoch 631/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 632/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0771
Epoch 633/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0760
Epoch 634/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0752
Epoch 635/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0742
Epoch 636/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0739
Epoch 637/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0726
Epoch 638/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 639/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 640/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 641/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0800
Epoch 642/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0747
Epoch 643/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0743
Epoch 644/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
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Epoch 645/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 646/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 647/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0855
Epoch 648/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 649/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0776
Epoch 650/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0863
Epoch 651/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0739
Epoch 652/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0784
Epoch 653/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0792
Epoch 654/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 655/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0786
Epoch 656/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 657/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 658/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 659/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 660/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 661/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0738
Epoch 662/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 663/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 664/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 665/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0730
Epoch 666/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 667/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0820
Epoch 668/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 669/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 670/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 671/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 672/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
```

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Epoch 673/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 674/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 675/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0788
Epoch 676/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0755
Epoch 677/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 678/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0715
Epoch 679/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0729
Epoch 680/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0842
Epoch 681/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 682/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0890
Epoch 683/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1122
Epoch 684/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0831
Epoch 685/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0778
Epoch 686/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0936
Epoch 687/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0775
Epoch 688/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
Epoch 689/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0940
Epoch 690/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 691/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0729
Epoch 692/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0750
Epoch 693/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 694/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 695/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0734
Epoch 696/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0820
Epoch 697/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0775
Epoch 698/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0783
Epoch 699/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0822
Epoch 700/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
```

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Epoch 701/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0852
Epoch 702/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0869
Epoch 703/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0796
Epoch 704/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0816
Epoch 705/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 706/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0729
Epoch 707/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 708/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0925
Epoch 709/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0872
Epoch 710/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 711/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1023
Epoch 712/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0819
Epoch 713/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 714/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0966
Epoch 715/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0909
Epoch 716/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 717/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 718/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0855
Epoch 719/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 720/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 721/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 722/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 723/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 724/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 725/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 726/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0805
Epoch 727/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 728/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
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Epoch 729/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0844
Epoch 730/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 731/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0800
Epoch 732/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 733/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 734/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 735/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 736/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 737/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 738/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 739/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0723
Epoch 740/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 741/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 742/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 743/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 744/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0799
Epoch 745/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 746/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 747/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0789
Epoch 748/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 749/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 750/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 751/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 752/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 753/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877
Epoch 754/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 755/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 756/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
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Epoch 757/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 758/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 759/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 760/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 761/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 762/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 763/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 764/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 765/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 766/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 767/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 768/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0768
Epoch 769/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 770/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 771/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 772/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 773/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 774/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 775/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0845
Epoch 776/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0894
Epoch 777/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 778/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 779/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0833
Epoch 780/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0775
Epoch 781/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 782/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 783/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 784/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
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Epoch 785/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 786/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 787/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 788/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 789/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 790/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0806
Epoch 791/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0725
Epoch 792/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0820
Epoch 793/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 794/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 795/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 796/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 797/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 798/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 799/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 800/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0779
Epoch 801/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0800
Epoch 802/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 803/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 804/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0777
Epoch 805/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 806/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0757
Epoch 807/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 808/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0720
Epoch 809/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 810/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 811/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 812/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
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Epoch 813/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 814/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 815/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 816/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 817/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 818/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 819/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 820/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 821/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 822/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 823/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 824/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 825/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 826/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 827/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 828/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 829/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 830/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 831/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 832/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 833/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 834/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 835/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 836/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 837/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 838/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 839/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 840/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0746
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Epoch 841/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 842/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 843/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0736
Epoch 844/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 845/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 846/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 847/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 848/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 849/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0779
Epoch 850/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0787
Epoch 851/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 852/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0984
Epoch 853/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0843
Epoch 854/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 855/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0889
Epoch 856/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 857/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0898
Epoch 858/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1245
Epoch 859/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0840
Epoch 860/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 861/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1178
Epoch 862/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0967
Epoch 863/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 864/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0746
Epoch 865/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 866/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0727
Epoch 867/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 868/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
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Epoch 869/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0727
Epoch 870/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0901
Epoch 871/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0848
Epoch 872/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 873/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0776
Epoch 874/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0842
Epoch 875/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 876/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 877/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 878/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0787
Epoch 879/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0756
Epoch 880/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0718
Epoch 881/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 882/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 883/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 884/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0716
Epoch 885/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0774
Epoch 886/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 887/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0726
Epoch 888/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 889/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 890/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 891/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 892/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 893/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 894/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 895/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 896/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
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Epoch 897/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0797
Epoch 898/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 899/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0803
Epoch 900/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0957
Epoch 901/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0782
Epoch 902/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 903/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 904/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0729
Epoch 905/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 906/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 907/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 908/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 909/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0736
Epoch 910/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 911/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 912/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 913/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0762
Epoch 914/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 915/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0716
Epoch 916/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 917/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 918/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0718
Epoch 919/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 920/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 921/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 922/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 923/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 924/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
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Epoch 925/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 926/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0714
Epoch 927/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0742
Epoch 928/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0744
Epoch 929/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 930/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0793
Epoch 931/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0880
Epoch 932/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0842
Epoch 933/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 934/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0797
Epoch 935/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0836
Epoch 936/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0772
Epoch 937/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 938/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 939/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0771
Epoch 940/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 941/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 942/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 943/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 944/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0929
Epoch 945/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0830
Epoch 946/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0892
Epoch 947/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0973
Epoch 948/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 949/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0822
Epoch 950/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 951/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 952/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
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Epoch 953/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 954/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 955/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 956/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 957/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 958/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0864
Epoch 959/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 960/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0935
Epoch 961/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1078
Epoch 962/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0873
Epoch 963/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 964/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0880
Epoch 965/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 966/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0805
Epoch 967/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1034
Epoch 968/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0613
Epoch 969/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1297
Epoch 970/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1582
Epoch 971/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 972/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 973/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0931
Epoch 974/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 975/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0869
Epoch 976/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1165
Epoch 977/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 978/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0816
Epoch 979/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1130
Epoch 980/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
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Epoch 981/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1010
Epoch 982/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1146
Epoch 983/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 984/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0958
Epoch 985/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1012
Epoch 986/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 987/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 988/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 989/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 990/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 991/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 992/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 993/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 994/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0852
Epoch 995/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0722
Epoch 996/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0806
Epoch 997/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0987
Epoch 998/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0927
Epoch 999/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 1000/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0786
Epoch 1001/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0897
Epoch 1002/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 1003/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 1004/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0860
Epoch 1005/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0872
Epoch 1006/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0741
Epoch 1007/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0716
Epoch 1008/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0806
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Epoch 1009/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 1010/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
Epoch 1011/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0747
Epoch 1012/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0827
Epoch 1013/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 1014/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 1015/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0820
Epoch 1016/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 1017/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 1018/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 1019/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 1020/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 1021/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0805
Epoch 1022/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 1023/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 1024/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0904
Epoch 1025/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0764
Epoch 1026/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 1027/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0856
Epoch 1028/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0662
Epoch 1029/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0942
Epoch 1030/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1354
Epoch 1031/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0899
Epoch 1032/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0858
Epoch 1033/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1055
Epoch 1034/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0809
Epoch 1035/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0827
Epoch 1036/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0910
```

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Epoch 1037/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 1038/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0862
Epoch 1039/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1105
Epoch 1040/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0988
Epoch 1041/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0794
Epoch 1042/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 1043/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 1044/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0724
Epoch 1045/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 1046/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0748
Epoch 1047/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 1048/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 1049/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 1050/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 1051/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 1052/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1053/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0766
Epoch 1054/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1055/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 1056/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 1057/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 1058/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0767
Epoch 1059/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 1060/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 1061/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 1062/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 1063/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 1064/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0815
```

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Epoch 1065/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0846
Epoch 1066/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 1067/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 1068/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 1069/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 1070/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0899
Epoch 1071/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 1072/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0771
Epoch 1073/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0947
Epoch 1074/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 1075/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0725
Epoch 1076/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0850
Epoch 1077/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 1078/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 1079/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 1080/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0707
Epoch 1081/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0727
Epoch 1082/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0690
Epoch 1083/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 1084/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 1085/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 1086/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 1087/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 1088/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 1089/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 1090/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1091/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 1092/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
```

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Epoch 1093/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1094/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0956
Epoch 1095/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0934
Epoch 1096/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 1097/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877
Epoch 1098/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0868
Epoch 1099/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 1100/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 1101/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 1102/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 1103/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 1104/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1105/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1106/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 1107/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 1108/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 1109/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0852
Epoch 1110/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0862
Epoch 1111/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 1112/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 1113/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 1114/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 1115/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 1116/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 1117/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0741
Epoch 1118/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 1119/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 1120/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
```

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Epoch 1121/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0700
Epoch 1122/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 1123/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0727
Epoch 1124/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 1125/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 1126/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 1127/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 1128/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 1129/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0775
Epoch 1130/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 1131/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0705
Epoch 1132/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 1133/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 1134/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1135/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0726
Epoch 1136/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 1137/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0851
Epoch 1138/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0831
Epoch 1139/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 1140/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0943
Epoch 1141/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0872
Epoch 1142/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1143/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0848
Epoch 1144/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0801
Epoch 1145/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0691
Epoch 1146/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0714
Epoch 1147/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0716
Epoch 1148/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
```

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Epoch 1149/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0772
Epoch 1150/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0815
Epoch 1151/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 1152/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 1153/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 1154/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 1155/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 1156/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 1157/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1158/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1159/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 1160/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1161/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0722
Epoch 1162/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1163/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1164/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 1165/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 1166/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 1167/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 1168/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 1169/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0804
Epoch 1170/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 1171/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
Epoch 1172/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 1173/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0696
Epoch 1174/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 1175/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 1176/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
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Epoch 1177/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 1178/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 1179/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 1180/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0776
Epoch 1181/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0813
Epoch 1182/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0708
Epoch 1183/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1184/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0860
Epoch 1185/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1186/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0811
Epoch 1187/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0855
Epoch 1188/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
Epoch 1189/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0965
Epoch 1190/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0993
Epoch 1191/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 1192/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0883
Epoch 1193/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0803
Epoch 1194/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 1195/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 1196/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0784
Epoch 1197/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0830
Epoch 1198/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0841
Epoch 1199/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 1200/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0695
Epoch 1201/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1202/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 1203/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 1204/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
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Epoch 1205/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 1206/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 1207/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1208/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1209/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 1210/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1211/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 1212/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0799
Epoch 1213/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0875
Epoch 1214/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0688
Epoch 1215/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 1216/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0821
Epoch 1217/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1218/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0840
Epoch 1219/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0786
Epoch 1220/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 1221/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1222/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 1223/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 1224/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 1225/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0730
Epoch 1226/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 1227/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 1228/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 1229/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 1230/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 1231/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0792
Epoch 1232/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0764
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Epoch 1233/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0732
Epoch 1234/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 1235/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 1236/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 1237/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 1238/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 1239/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0714
Epoch 1240/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 1241/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 1242/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 1243/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 1244/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 1245/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1246/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0837
Epoch 1247/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0969
Epoch 1248/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0802
Epoch 1249/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 1250/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 1251/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0891
Epoch 1252/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 1253/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0922
Epoch 1254/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0887
Epoch 1255/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1256/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0983
Epoch 1257/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 1258/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 1259/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0942
Epoch 1260/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0794
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Epoch 1261/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1262/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0763
Epoch 1263/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0751
Epoch 1264/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0692
Epoch 1265/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1266/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 1267/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 1268/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 1269/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0876
Epoch 1270/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
Epoch 1271/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 1272/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 1273/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0746
Epoch 1274/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1275/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 1276/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0755
Epoch 1277/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1278/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0822
Epoch 1279/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0916
Epoch 1280/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1281/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 1282/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0873
Epoch 1283/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 1284/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 1285/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.1145
Epoch 1286/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0934
Epoch 1287/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0813
Epoch 1288/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0964
```

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Epoch 1289/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1290/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0942
Epoch 1291/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1183
Epoch 1292/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0793
Epoch 1293/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 1294/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0915
Epoch 1295/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 1296/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 1297/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 1298/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1299/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 1300/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 1301/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 1302/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1303/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1304/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 1305/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 1306/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 1307/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 1308/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1309/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0695
Epoch 1310/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0693
Epoch 1311/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 1312/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 1313/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 1314/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0732
Epoch 1315/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0700
Epoch 1316/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0737
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Epoch 1317/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 1318/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0720
Epoch 1319/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0695
Epoch 1320/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 1321/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1322/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0736
Epoch 1323/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 1324/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1325/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0894
Epoch 1326/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0970
Epoch 1327/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 1328/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 1329/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1015
Epoch 1330/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 1331/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0822
Epoch 1332/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1275
Epoch 1333/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0903
Epoch 1334/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 1335/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1171
Epoch 1336/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0878
Epoch 1337/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0820
Epoch 1338/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1003
Epoch 1339/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0745
Epoch 1340/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0757
Epoch 1341/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0943
Epoch 1342/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0816
Epoch 1343/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0708
Epoch 1344/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0964
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Epoch 1345/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0844
Epoch 1346/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1347/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0793
Epoch 1348/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1349/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1350/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0788
Epoch 1351/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1352/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 1353/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0735
Epoch 1354/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1355/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 1356/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0870
Epoch 1357/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 1358/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0835
Epoch 1359/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0820
Epoch 1360/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0662
Epoch 1361/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0815
Epoch 1362/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0954
Epoch 1363/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 1364/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 1365/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1366/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 1367/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 1368/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1369/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 1370/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 1371/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 1372/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0940
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Epoch 1373/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0804
Epoch 1374/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0721
Epoch 1375/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0786
Epoch 1376/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 1377/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 1378/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1379/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1380/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 1381/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1382/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 1383/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 1384/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1385/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 1386/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1387/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 1388/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0839
Epoch 1389/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 1390/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 1391/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 1392/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0677
Epoch 1393/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0833
Epoch 1394/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0860
Epoch 1395/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 1396/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0823
Epoch 1397/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0797
Epoch 1398/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 1399/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0745
Epoch 1400/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
```

```
Epoch 1401/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 1402/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 1403/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 1404/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 1405/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 1406/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1407/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1408/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1409/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0745
Epoch 1410/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0769
Epoch 1411/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0810
Epoch 1412/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 1413/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1414/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 1415/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1416/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1417/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1418/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1419/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 1420/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0762
Epoch 1421/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 1422/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 1423/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 1424/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0702
Epoch 1425/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 1426/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1427/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 1428/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0780
```

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Epoch 1429/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 1430/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 1431/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 1432/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1433/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1434/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 1435/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 1436/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 1437/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0925
Epoch 1438/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0643
Epoch 1439/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0923
Epoch 1440/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1464
Epoch 1441/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0823
Epoch 1442/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0973
Epoch 1443/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1432
Epoch 1444/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0877
Epoch 1445/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0786
Epoch 1446/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1300
Epoch 1447/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0971
Epoch 1448/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 1449/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1157
Epoch 1450/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0997
Epoch 1451/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 1452/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0831
Epoch 1453/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0806
Epoch 1454/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0678
Epoch 1455/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 1456/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0841
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Epoch 1457/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0836
Epoch 1458/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0720
Epoch 1459/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0694
Epoch 1460/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1461/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 1462/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 1463/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 1464/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0686
Epoch 1465/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 1466/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 1467/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0784
Epoch 1468/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0748
Epoch 1469/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1470/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 1471/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0729
Epoch 1472/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 1473/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 1474/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 1475/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0665
Epoch 1476/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 1477/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 1478/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 1479/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1087
Epoch 1480/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0885
Epoch 1481/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 1482/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1082
Epoch 1483/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0986
Epoch 1484/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
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Epoch 1485/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 1486/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 1487/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1488/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0810
Epoch 1489/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0844
Epoch 1490/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 1491/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 1492/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0829
Epoch 1493/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
Epoch 1494/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0716
Epoch 1495/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1496/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1497/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 1498/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 1499/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0721
Epoch 1500/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0829
Epoch 1501/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0769
Epoch 1502/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 1503/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 1504/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 1505/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 1506/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 1507/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 1508/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 1509/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1510/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 1511/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1512/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0697
```

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Epoch 1513/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 1514/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0706
Epoch 1515/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 1516/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 1517/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 1518/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 1519/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0889
Epoch 1520/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0754
Epoch 1521/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0758
Epoch 1522/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0762
Epoch 1523/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0670
Epoch 1524/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 1525/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0884
Epoch 1526/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0738
Epoch 1527/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0726
Epoch 1528/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0964
Epoch 1529/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0793
Epoch 1530/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0702
Epoch 1531/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0836
Epoch 1532/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0856
Epoch 1533/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0770
Epoch 1534/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 1535/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0827
Epoch 1536/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0867
Epoch 1537/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 1538/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 1539/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0799
Epoch 1540/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
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Epoch 1541/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 1542/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 1543/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 1544/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1545/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 1546/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 1547/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 1548/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 1549/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 1550/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 1551/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 1552/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 1553/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 1554/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0780
Epoch 1555/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 1556/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0693
Epoch 1557/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1558/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1559/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 1560/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1561/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 1562/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1563/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1564/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1565/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 1566/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 1567/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 1568/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0699
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Epoch 1569/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0729
Epoch 1570/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1571/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 1572/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0774
Epoch 1573/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0731
Epoch 1574/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 1575/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0705
Epoch 1576/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1577/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1578/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0740
Epoch 1579/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0801
Epoch 1580/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 1581/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0749
Epoch 1582/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 1583/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1584/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1585/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 1586/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0688
Epoch 1587/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1588/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1589/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 1590/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1591/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0894
Epoch 1592/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0749
Epoch 1593/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 1594/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0860
Epoch 1595/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0782
Epoch 1596/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
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Epoch 1597/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 1598/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 1599/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 1600/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 1601/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 1602/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0829
Epoch 1603/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0721
Epoch 1604/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1605/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1606/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1607/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 1608/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0767
Epoch 1609/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0812
Epoch 1610/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 1611/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0800
Epoch 1612/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0809
Epoch 1613/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 1614/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0848
Epoch 1615/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0838
Epoch 1616/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 1617/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0900
Epoch 1618/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0775
Epoch 1619/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 1620/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0871
Epoch 1621/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 1622/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0796
Epoch 1623/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0901
Epoch 1624/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
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Epoch 1625/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 1626/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 1627/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 1628/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1003
Epoch 1629/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 1630/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 1631/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1021
Epoch 1632/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0961
Epoch 1633/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 1634/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0984
Epoch 1635/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0922
Epoch 1636/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0772
Epoch 1637/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0999
Epoch 1638/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 1639/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 1640/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0842
Epoch 1641/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 1642/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 1643/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 1644/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 1645/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1646/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1647/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1648/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 1649/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 1650/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 1651/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 1652/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0684
```

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Epoch 1653/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0688
Epoch 1654/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1655/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 1656/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1657/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1658/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 1659/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 1660/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 1661/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 1662/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1663/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1664/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1665/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1666/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1667/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1668/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0721
Epoch 1669/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0704
Epoch 1670/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1671/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1672/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 1673/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0693
Epoch 1674/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 1675/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 1676/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1677/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0663
Epoch 1678/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0794
Epoch 1679/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 1680/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
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Epoch 1681/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 1682/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 1683/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 1684/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 1685/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0700
Epoch 1686/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 1687/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 1688/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 1689/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 1690/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0767
Epoch 1691/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0656
Epoch 1692/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0875
Epoch 1693/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0922
Epoch 1694/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 1695/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 1696/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 1697/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 1698/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0907
Epoch 1699/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1045
Epoch 1700/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 1701/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0838
Epoch 1702/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0881
Epoch 1703/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 1704/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 1705/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0896
Epoch 1706/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0696
Epoch 1707/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0734
Epoch 1708/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0914
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Epoch 1709/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0704
Epoch 1710/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0783
Epoch 1711/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0890
Epoch 1712/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 1713/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 1714/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 1715/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1716/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 1717/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 1718/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 1719/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1720/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 1721/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1722/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 1723/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 1724/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1725/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0758
Epoch 1726/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1727/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 1728/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 1729/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 1730/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 1731/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0806
Epoch 1732/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 1733/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1734/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 1735/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 1736/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
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Epoch 1737/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1738/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0697
Epoch 1739/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1740/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 1741/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 1742/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1743/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 1744/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0835
Epoch 1745/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 1746/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 1747/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 1748/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 1749/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 1750/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 1751/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0793
Epoch 1752/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 1753/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 1754/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 1755/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 1756/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 1757/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 1758/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1759/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 1760/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 1761/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 1762/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 1763/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0734
Epoch 1764/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0656
```

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Epoch 1765/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0824
Epoch 1766/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1026
Epoch 1767/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 1768/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 1769/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0784
Epoch 1770/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0701
Epoch 1771/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 1772/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0961
Epoch 1773/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0983
Epoch 1774/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 1775/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0832
Epoch 1776/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1008
Epoch 1777/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0785
Epoch 1778/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0673
Epoch 1779/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0853
Epoch 1780/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0852
Epoch 1781/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 1782/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 1783/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0818
Epoch 1784/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1785/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 1786/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 1787/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 1788/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 1789/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1790/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1791/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1066
Epoch 1792/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0897
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Epoch 1793/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 1794/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0939
Epoch 1795/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 1796/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 1797/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0776
Epoch 1798/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 1799/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0863
Epoch 1800/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0761
Epoch 1801/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 1802/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0855
Epoch 1803/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 1804/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 1805/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1005
Epoch 1806/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0801
Epoch 1807/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 1808/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
Epoch 1809/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 1810/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0713
Epoch 1811/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 1812/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 1813/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 1814/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1815/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 1816/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 1817/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 1818/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0816
Epoch 1819/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 1820/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
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Epoch 1821/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0765
Epoch 1822/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0762
Epoch 1823/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 1824/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 1825/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 1826/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 1827/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 1828/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0863
Epoch 1829/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 1830/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0729
Epoch 1831/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 1832/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 1833/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0724
Epoch 1834/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 1835/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1836/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 1837/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 1838/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1839/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1840/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0704
Epoch 1841/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0735
Epoch 1842/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0690
Epoch 1843/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1844/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 1845/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 1846/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0654
Epoch 1847/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 1848/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0787
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Epoch 1849/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 1850/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 1851/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0677
Epoch 1852/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 1853/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0662
Epoch 1854/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 1855/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 1856/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 1857/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 1858/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 1859/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 1860/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 1861/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 1862/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 1863/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0803
Epoch 1864/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0853
Epoch 1865/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1866/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0740
Epoch 1867/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 1868/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 1869/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0965
Epoch 1870/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 1871/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 1872/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 1873/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0796
Epoch 1874/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 1875/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1876/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
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Epoch 1877/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 1878/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 1879/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 1880/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1881/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0768
Epoch 1882/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0719
Epoch 1883/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0664
Epoch 1884/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 1885/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 1886/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 1887/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0803
Epoch 1888/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0920
Epoch 1889/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 1890/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 1891/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0918
Epoch 1892/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0764
Epoch 1893/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 1894/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0950
Epoch 1895/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 1896/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 1897/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 1898/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 1899/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 1900/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 1901/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1902/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0732
Epoch 1903/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0708
Epoch 1904/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
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Epoch 1905/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 1906/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0667
Epoch 1907/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 1908/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 1909/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 1910/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 1911/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 1912/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1913/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 1914/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0750
Epoch 1915/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1916/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 1917/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 1918/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 1919/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0688
Epoch 1920/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0695
Epoch 1921/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1922/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 1923/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0705
Epoch 1924/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 1925/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 1926/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0729
Epoch 1927/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 1928/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 1929/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
Epoch 1930/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 1931/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0664
Epoch 1932/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
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Epoch 1933/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0681
Epoch 1934/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 1935/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 1936/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 1937/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 1938/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 1939/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 1940/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 1941/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 1942/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 1943/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0648
Epoch 1944/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 1945/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0841
Epoch 1946/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 1947/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1948/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 1949/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1950/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0665
Epoch 1951/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0665
Epoch 1952/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0671
Epoch 1953/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0660
Epoch 1954/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0666
Epoch 1955/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0684
Epoch 1956/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 1957/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0747
Epoch 1958/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0847
Epoch 1959/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 1960/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0698
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Epoch 1961/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 1962/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 1963/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 1964/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 1965/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 1966/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 1967/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0902
Epoch 1968/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0882
Epoch 1969/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
Epoch 1970/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 1971/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 1972/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 1973/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0794
Epoch 1974/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1089
Epoch 1975/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0868
Epoch 1976/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 1977/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0791
Epoch 1978/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 1979/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 1980/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 1981/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 1982/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 1983/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 1984/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 1985/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0905
Epoch 1986/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 1987/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0698
Epoch 1988/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0780
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Epoch 1989/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0767
Epoch 1990/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 1991/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0654
Epoch 1992/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 1993/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
Epoch 1994/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 1995/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 1996/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0728
Epoch 1997/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 1998/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
Epoch 1999/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0700
Epoch 2000/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 2001/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 2002/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0841
Epoch 2003/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 2004/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 2005/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 2006/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 2007/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 2008/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 2009/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 2010/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 2011/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 2012/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0648
Epoch 2013/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2014/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 2015/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 2016/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
```

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Epoch 2017/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 2018/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 2019/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 2020/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0807
Epoch 2021/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0789
Epoch 2022/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 2023/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 2024/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0654
Epoch 2025/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 2026/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0797
Epoch 2027/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 2028/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0708
Epoch 2029/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 2030/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2031/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2032/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0787
Epoch 2033/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 2034/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 2035/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 2036/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 2037/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2038/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 2039/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 2040/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2041/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 2042/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2043/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 2044/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0641
```

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Epoch 2045/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 2046/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 2047/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 2048/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 2049/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0731
Epoch 2050/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 2051/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 2052/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 2053/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2054/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2055/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 2056/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 2057/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 2058/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 2059/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0759
Epoch 2060/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 2061/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 2062/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 2063/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 2064/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0765
Epoch 2065/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 2066/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0685
Epoch 2067/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 2068/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 2069/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0664
Epoch 2070/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 2071/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 2072/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
```

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Epoch 2073/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2074/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671
Epoch 2075/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 2076/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 2077/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 2078/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 2079/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 2080/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 2081/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 2082/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 2083/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0697
Epoch 2084/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0703
Epoch 2085/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2086/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 2087/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0731
Epoch 2088/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 2089/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
Epoch 2090/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 2091/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 2092/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 2093/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2094/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0671
Epoch 2095/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 2096/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 2097/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0800
Epoch 2098/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 2099/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 2100/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0837
```

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Epoch 2101/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
Epoch 2102/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0664
Epoch 2103/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0803
Epoch 2104/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0765
Epoch 2105/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0652
Epoch 2106/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 2107/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 2108/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 2109/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 2110/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 2111/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2112/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 2113/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2114/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 2115/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 2116/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 2117/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 2118/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0870
Epoch 2119/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0729
Epoch 2120/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 2121/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0803
Epoch 2122/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 2123/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2124/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 2125/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 2126/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 2127/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0780
Epoch 2128/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
```

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Epoch 2129/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2130/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0662
Epoch 2131/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671
Epoch 2132/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 2133/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0663
Epoch 2134/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0657
Epoch 2135/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0649
Epoch 2136/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0692
Epoch 2137/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2138/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 2139/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0829
Epoch 2140/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 2141/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0677
Epoch 2142/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 2143/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 2144/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2145/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 2146/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2147/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0664
Epoch 2148/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 2149/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0649
Epoch 2150/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 2151/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0812
Epoch 2152/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 2153/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0781
Epoch 2154/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0790
Epoch 2155/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0688
Epoch 2156/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0916
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Epoch 2157/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 2158/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 2159/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0971
Epoch 2160/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0840
Epoch 2161/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0656
Epoch 2162/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1003
Epoch 2163/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1033
Epoch 2164/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0724
Epoch 2165/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 2166/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 2167/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 2168/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 2169/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 2170/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0651
Epoch 2171/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2172/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 2173/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 2174/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0641
Epoch 2175/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0764
Epoch 2176/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 2177/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 2178/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0815
Epoch 2179/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0725
Epoch 2180/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 2181/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 2182/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 2183/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0762
Epoch 2184/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0879
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Epoch 2185/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 2186/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 2187/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0860
Epoch 2188/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2189/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0763
Epoch 2190/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0834
Epoch 2191/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 2192/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 2193/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 2194/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 2195/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 2196/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0663
Epoch 2197/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0650
Epoch 2198/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0679
Epoch 2199/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0675
Epoch 2200/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 2201/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 2202/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 2203/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 2204/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 2205/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
Epoch 2206/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0682
Epoch 2207/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0694
Epoch 2208/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2209/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 2210/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0838
Epoch 2211/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0750
Epoch 2212/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0692
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Epoch 2213/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 2214/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 2215/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 2216/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 2217/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0707
Epoch 2218/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 2219/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 2220/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0650
Epoch 2221/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 2222/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0804
Epoch 2223/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 2224/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 2225/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0661
Epoch 2226/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0649
Epoch 2227/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2228/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0742
Epoch 2229/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0736
Epoch 2230/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 2231/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 2232/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 2233/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0882
Epoch 2234/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 2235/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 2236/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0944
Epoch 2237/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2238/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 2239/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1001
Epoch 2240/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0830
```

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Epoch 2241/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 2242/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 2243/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 2244/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 2245/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0737
Epoch 2246/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 2247/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0646
Epoch 2248/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 2249/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0895
Epoch 2250/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 2251/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2252/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0664
Epoch 2253/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 2254/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 2255/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2256/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0656
Epoch 2257/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 2258/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0654
Epoch 2259/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0644
Epoch 2260/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 2261/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 2262/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 2263/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 2264/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 2265/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 2266/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2267/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0770
Epoch 2268/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
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Epoch 2269/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 2270/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 2271/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 2272/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0762
Epoch 2273/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0814
Epoch 2274/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 2275/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0647
Epoch 2276/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0819
Epoch 2277/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0794
Epoch 2278/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2279/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 2280/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 2281/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 2282/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 2283/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 2284/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 2285/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 2286/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 2287/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0862
Epoch 2288/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0701
Epoch 2289/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 2290/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 2291/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 2292/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 2293/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2294/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0701
Epoch 2295/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0649
Epoch 2296/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0679
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Epoch 2297/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 2298/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 2299/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0876
Epoch 2300/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0833
Epoch 2301/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 2302/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2303/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0647
Epoch 2304/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2305/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 2306/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0641
Epoch 2307/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 2308/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 2309/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0652
Epoch 2310/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2311/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 2312/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2313/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
Epoch 2314/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 2315/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0938
Epoch 2316/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0896
Epoch 2317/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2318/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0844
Epoch 2319/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 2320/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0698
Epoch 2321/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0838
Epoch 2322/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0726
Epoch 2323/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 2324/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0749
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Epoch 2325/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0694
Epoch 2326/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 2327/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 2328/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0635
Epoch 2329/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0755
Epoch 2330/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0932
Epoch 2331/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 2332/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0786
Epoch 2333/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 2334/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0649
Epoch 2335/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0766
Epoch 2336/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0688
Epoch 2337/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 2338/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0720
Epoch 2339/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2340/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0712
Epoch 2341/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 2342/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0667
Epoch 2343/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 2344/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0647
Epoch 2345/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
Epoch 2346/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 2347/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 2348/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2349/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 2350/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 2351/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0745
Epoch 2352/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0882
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Epoch 2353/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 2354/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 2355/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0866
Epoch 2356/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 2357/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0772
Epoch 2358/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 2359/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 2360/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0869
Epoch 2361/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 2362/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2363/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 2364/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2365/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 2366/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 2367/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2368/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 2369/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 2370/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0693
Epoch 2371/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2372/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2373/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 2374/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 2375/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 2376/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2377/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0663
Epoch 2378/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0655
Epoch 2379/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0641
Epoch 2380/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
```

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Epoch 2381/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0648
Epoch 2382/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 2383/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 2384/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 2385/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0734
Epoch 2386/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 2387/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0655
Epoch 2388/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 2389/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0793
Epoch 2390/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 2391/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0822
Epoch 2392/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0872
Epoch 2393/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 2394/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 2395/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0809
Epoch 2396/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 2397/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 2398/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0715
Epoch 2399/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0634
Epoch 2400/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 2401/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0990
Epoch 2402/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0790
Epoch 2403/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0716
Epoch 2404/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0833
Epoch 2405/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 2406/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 2407/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
Epoch 2408/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0685
```

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Epoch 2409/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 2410/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0648
Epoch 2411/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0654
Epoch 2412/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 2413/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 2414/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 2415/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0646
Epoch 2416/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0662
Epoch 2417/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 2418/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 2419/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 2420/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 2421/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0772
Epoch 2422/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0634
Epoch 2423/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0760
Epoch 2424/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1068
Epoch 2425/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0779
Epoch 2426/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 2427/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0826
Epoch 2428/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 2429/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 2430/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0817
Epoch 2431/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 2432/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 2433/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0747
Epoch 2434/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 2435/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 2436/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
```

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Epoch 2437/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0647
Epoch 2438/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0805
Epoch 2439/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0706
Epoch 2440/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 2441/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0799
Epoch 2442/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 2443/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
Epoch 2444/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0813
Epoch 2445/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671
Epoch 2446/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0640
Epoch 2447/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0662
Epoch 2448/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0719
Epoch 2449/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 2450/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
Epoch 2451/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 2452/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 2453/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 2454/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 2455/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0673
Epoch 2456/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 2457/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 2458/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 2459/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0640
Epoch 2460/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0645
Epoch 2461/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0654
Epoch 2462/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 2463/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
Epoch 2464/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0982
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Epoch 2465/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1018
Epoch 2466/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0739
Epoch 2467/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0800
Epoch 2468/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 2469/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0644
Epoch 2470/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 2471/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 2472/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0769
Epoch 2473/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0726
Epoch 2474/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0722
Epoch 2475/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0981
Epoch 2476/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0818
Epoch 2477/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 2478/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0797
Epoch 2479/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0706
Epoch 2480/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 2481/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 2482/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 2483/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 2484/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0635
Epoch 2485/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 2486/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 2487/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0625
Epoch 2488/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0785
Epoch 2489/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0830
Epoch 2490/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0643
Epoch 2491/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 2492/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
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Epoch 2493/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 2494/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 2495/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0684
Epoch 2496/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0658
Epoch 2497/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0641
Epoch 2498/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 2499/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0664
Epoch 2500/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0629
Epoch 2501/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2502/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0721
Epoch 2503/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 2504/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0709
Epoch 2505/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 2506/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2507/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 2508/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0643
Epoch 2509/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0637
Epoch 2510/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 2511/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0730
Epoch 2512/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 2513/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2514/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0663
Epoch 2515/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 2516/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 2517/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 2518/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0620
Epoch 2519/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0727
Epoch 2520/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0799
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Epoch 2521/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0627
Epoch 2522/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0788
Epoch 2523/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0968
Epoch 2524/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0683
Epoch 2525/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 2526/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0843
Epoch 2527/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 2528/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0703
Epoch 2529/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0769
Epoch 2530/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0657
Epoch 2531/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0667
Epoch 2532/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0736
Epoch 2533/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 2534/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0864
Epoch 2535/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0754
Epoch 2536/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0744
Epoch 2537/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0818
Epoch 2538/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0662
Epoch 2539/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0876
Epoch 2540/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 2541/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0791
Epoch 2542/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 2543/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0656
Epoch 2544/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0832
Epoch 2545/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0855
Epoch 2546/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 2547/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0673
Epoch 2548/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
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Epoch 2549/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0712
Epoch 2550/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0709
Epoch 2551/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0691
Epoch 2552/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0661
Epoch 2553/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0637
Epoch 2554/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 2555/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679
Epoch 2556/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 2557/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 2558/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 2559/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 2560/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0669
Epoch 2561/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 2562/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0658
Epoch 2563/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0761
Epoch 2564/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0785
Epoch 2565/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2566/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0723
Epoch 2567/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 2568/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0752
Epoch 2569/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0955
Epoch 2570/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0803
Epoch 2571/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
Epoch 2572/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0770
Epoch 2573/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0661
Epoch 2574/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0706
Epoch 2575/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0801
Epoch 2576/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
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Epoch 2577/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 2578/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 2579/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 2580/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0648
Epoch 2581/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 2582/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 2583/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 2584/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 2585/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0633
Epoch 2586/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 2587/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 2588/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 2589/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 2590/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 2591/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0644
Epoch 2592/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0637
Epoch 2593/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 2594/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 2595/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 2596/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2597/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0740
Epoch 2598/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 2599/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 2600/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 2601/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2602/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0717
Epoch 2603/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0694
Epoch 2604/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
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Epoch 2605/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 2606/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0638
Epoch 2607/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 2608/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 2609/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 2610/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0625
Epoch 2611/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 2612/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 2613/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 2614/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 2615/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0656
Epoch 2616/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0655
Epoch 2617/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0649
Epoch 2618/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679
Epoch 2619/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 2620/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 2621/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 2622/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2623/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0678
Epoch 2624/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0727
Epoch 2625/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 2626/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 2627/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 2628/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 2629/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0627
Epoch 2630/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 2631/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 2632/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0673
```

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Epoch 2633/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0756
Epoch 2634/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0686
Epoch 2635/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0628
Epoch 2636/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0709
Epoch 2637/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0813
Epoch 2638/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0784
Epoch 2639/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0778
Epoch 2640/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 2641/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0634
Epoch 2642/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 2643/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0707
Epoch 2644/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0660
Epoch 2645/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0733
Epoch 2646/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 2647/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0705
Epoch 2648/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0708
Epoch 2649/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0606
Epoch 2650/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0753
Epoch 2651/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0874
Epoch 2652/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0690
Epoch 2653/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0638
Epoch 2654/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0745
Epoch 2655/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0729
Epoch 2656/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0666
Epoch 2657/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0669
Epoch 2658/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 2659/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 2660/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
```

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Epoch 2661/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2662/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 2663/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 2664/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 2665/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 2666/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 2667/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0646
Epoch 2668/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 2669/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 2670/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 2671/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 2672/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 2673/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0695
Epoch 2674/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0658
Epoch 2675/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 2676/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 2677/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 2678/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0710
Epoch 2679/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0663
Epoch 2680/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
Epoch 2681/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0774
Epoch 2682/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0723
Epoch 2683/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0754
Epoch 2684/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1041
Epoch 2685/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0859
Epoch 2686/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 2687/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0834
Epoch 2688/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0731
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Epoch 2689/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
Epoch 2690/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0725
Epoch 2691/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0741
Epoch 2692/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0669
Epoch 2693/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2694/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 2695/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0656
Epoch 2696/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 2697/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0622
Epoch 2698/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 2699/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 2700/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0656
Epoch 2701/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0645
Epoch 2702/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 2703/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 2704/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0666
Epoch 2705/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0645
Epoch 2706/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0694
Epoch 2707/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 2708/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 2709/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 2710/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0657
Epoch 2711/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2712/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 2713/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
Epoch 2714/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0615
Epoch 2715/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0857
Epoch 2716/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0836
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Epoch 2717/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0608
Epoch 2718/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0781
Epoch 2719/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0998
Epoch 2720/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0692
Epoch 2721/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0795
Epoch 2722/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0971
Epoch 2723/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
Epoch 2724/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0778
Epoch 2725/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0864
Epoch 2726/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 2727/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0808
Epoch 2728/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0759
Epoch 2729/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 2730/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0937
Epoch 2731/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1113
Epoch 2732/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0768
Epoch 2733/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0725
Epoch 2734/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0894
Epoch 2735/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0681
Epoch 2736/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0738
Epoch 2737/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0769
Epoch 2738/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0710
Epoch 2739/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0771
Epoch 2740/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0670
Epoch 2741/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0674
Epoch 2742/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0781
Epoch 2743/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0764
Epoch 2744/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
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```
Epoch 2745/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 2746/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 2747/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 2748/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 2749/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 2750/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
Epoch 2751/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 2752/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 2753/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0667
Epoch 2754/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 2755/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0714
Epoch 2756/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0759
Epoch 2757/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 2758/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0686
Epoch 2759/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0691
Epoch 2760/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 2761/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0780
Epoch 2762/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0728
Epoch 2763/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0653
Epoch 2764/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0714
Epoch 2765/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0701
Epoch 2766/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 2767/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0664
Epoch 2768/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0645
Epoch 2769/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0631
Epoch 2770/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0704
Epoch 2771/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 2772/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
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Epoch 2773/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0734
Epoch 2774/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0773
Epoch 2775/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 2776/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 2777/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 2778/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0605
Epoch 2779/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0968
Epoch 2780/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0965
Epoch 2781/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 2782/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 2783/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 2784/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0746
Epoch 2785/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0659
Epoch 2786/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0682
Epoch 2787/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
Epoch 2788/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0645
Epoch 2789/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 2790/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0732
Epoch 2791/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0611
Epoch 2792/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0849
Epoch 2793/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0766
Epoch 2794/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 2795/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 2796/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0612
Epoch 2797/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0868
Epoch 2798/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0863
Epoch 2799/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0610
Epoch 2800/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0808
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Epoch 2801/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1025
Epoch 2802/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0705
Epoch 2803/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0751
Epoch 2804/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0840
Epoch 2805/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0664
Epoch 2806/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 2807/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 2808/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0634
Epoch 2809/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 2810/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 2811/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 2812/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0637
Epoch 2813/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0618
Epoch 2814/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0670
Epoch 2815/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0744
Epoch 2816/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682
Epoch 2817/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0648
Epoch 2818/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2819/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 2820/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0646
Epoch 2821/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 2822/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0648
Epoch 2823/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 2824/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 2825/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 2826/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0699
Epoch 2827/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 2828/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0657
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Epoch 2829/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 2830/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 2831/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 2832/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0638
Epoch 2833/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 2834/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0620
Epoch 2835/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 2836/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 2837/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 2838/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0619
Epoch 2839/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0643
Epoch 2840/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 2841/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 2842/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 2843/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0609
Epoch 2844/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0758
Epoch 2845/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 2846/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 2847/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 2848/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0632
Epoch 2849/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0625
Epoch 2850/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0688
Epoch 2851/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 2852/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 2853/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0640
Epoch 2854/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 2855/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 2856/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
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Epoch 2857/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0674
Epoch 2858/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
Epoch 2859/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0618
Epoch 2860/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 2861/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0675
Epoch 2862/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0676
Epoch 2863/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0705
Epoch 2864/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0669
Epoch 2865/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 2866/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0620
Epoch 2867/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0707
Epoch 2868/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0768
Epoch 2869/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0635
Epoch 2870/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0769
Epoch 2871/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0914
Epoch 2872/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0657
Epoch 2873/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0873
Epoch 2874/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0986
Epoch 2875/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0669
Epoch 2876/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 2877/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0727
Epoch 2878/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0744
Epoch 2879/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 2880/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 2881/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0694
Epoch 2882/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0729
Epoch 2883/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 2884/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
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Epoch 2885/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 2886/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 2887/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0623
Epoch 2888/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 2889/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 2890/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 2891/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 2892/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 2893/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 2894/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0673
Epoch 2895/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 2896/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0605
Epoch 2897/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0720
Epoch 2898/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0701
Epoch 2899/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0648
Epoch 2900/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0699
Epoch 2901/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0652
Epoch 2902/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0655
Epoch 2903/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 2904/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 2905/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0652
Epoch 2906/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0670
Epoch 2907/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 2908/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679
Epoch 2909/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0757
Epoch 2910/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0735
Epoch 2911/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0766
Epoch 2912/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
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Epoch 2913/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0608
Epoch 2914/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0702
Epoch 2915/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0737
Epoch 2916/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2917/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0673
Epoch 2918/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679
Epoch 2919/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0652
Epoch 2920/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 2921/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 2922/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0632
Epoch 2923/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0627
Epoch 2924/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 2925/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0701
Epoch 2926/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 2927/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0715
Epoch 2928/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0792
Epoch 2929/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 2930/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0780
Epoch 2931/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0748
Epoch 2932/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0614
Epoch 2933/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0776
Epoch 2934/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0738
Epoch 2935/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0631
Epoch 2936/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0940
Epoch 2937/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 2938/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0885
Epoch 2939/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1085
Epoch 2940/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
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Epoch 2941/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1277
Epoch 2942/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1350
Epoch 2943/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 2944/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0956
Epoch 2945/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1043
Epoch 2946/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0667
Epoch 2947/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0711
Epoch 2948/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0817
Epoch 2949/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2950/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 2951/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0785
Epoch 2952/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 2953/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0734
Epoch 2954/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0770
Epoch 2955/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 2956/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0697
Epoch 2957/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0738
Epoch 2958/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0648
Epoch 2959/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0712
Epoch 2960/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0731
Epoch 2961/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0647
Epoch 2962/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0626
Epoch 2963/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
Epoch 2964/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0619
Epoch 2965/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 2966/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 2967/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 2968/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
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Epoch 2969/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0791
Epoch 2970/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 2971/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 2972/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0946
Epoch 2973/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0818
Epoch 2974/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0618
Epoch 2975/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0862
Epoch 2976/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0688
Epoch 2977/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0739
Epoch 2978/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0778
Epoch 2979/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0595
Epoch 2980/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0894
Epoch 2981/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0876
Epoch 2982/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
Epoch 2983/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0878
Epoch 2984/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0795
Epoch 2985/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 2986/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0614
Epoch 2987/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0615
Epoch 2988/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0621
Epoch 2989/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0621
Epoch 2990/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 2991/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 2992/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0614
Epoch 2993/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 2994/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 2995/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 2996/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
```

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Epoch 2997/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0652
Epoch 2998/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0643
Epoch 2999/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0795
Epoch 3000/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0709
Epoch 3001/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0752
Epoch 3002/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682
Epoch 3003/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682
Epoch 3004/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0675
Epoch 3005/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 3006/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0691
Epoch 3007/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 3008/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0647
Epoch 3009/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0865
Epoch 3010/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0743
Epoch 3011/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0631
Epoch 3012/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0788
Epoch 3013/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0648
Epoch 3014/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0746
Epoch 3015/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0959
Epoch 3016/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0712
Epoch 3017/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0644
Epoch 3018/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 3019/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0643
Epoch 3020/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0673
Epoch 3021/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0720
Epoch 3022/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 3023/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0630
Epoch 3024/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
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Epoch 3025/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0634
Epoch 3026/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 3027/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3028/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 3029/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0695
Epoch 3030/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 3031/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3032/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 3033/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0715
Epoch 3034/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0719
Epoch 3035/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0661
Epoch 3036/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 3037/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0688
Epoch 3038/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0663
Epoch 3039/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0634
Epoch 3040/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0620
Epoch 3041/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3042/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 3043/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 3044/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3045/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 3046/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0611
Epoch 3047/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0634
Epoch 3048/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0790
Epoch 3049/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 3050/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3051/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 3052/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
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Epoch 3053/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679
Epoch 3054/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3055/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3056/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3057/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3058/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3059/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3060/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0623
Epoch 3061/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0656
Epoch 3062/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 3063/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0627
Epoch 3064/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671
Epoch 3065/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
Epoch 3066/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 3067/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 3068/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 3069/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0681
Epoch 3070/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
Epoch 3071/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0701
Epoch 3072/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0650
Epoch 3073/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0656
Epoch 3074/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0727
Epoch 3075/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0662
Epoch 3076/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 3077/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3078/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0705
Epoch 3079/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0661
Epoch 3080/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
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Epoch 3081/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0720
Epoch 3082/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 3083/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3084/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3085/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 3086/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0616
Epoch 3087/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3088/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3089/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0662
Epoch 3090/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 3091/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 3092/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3093/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0662
Epoch 3094/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0778
Epoch 3095/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0693
Epoch 3096/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 3097/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0656
Epoch 3098/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 3099/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0608
Epoch 3100/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3101/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0733
Epoch 3102/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0697
Epoch 3103/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0614
Epoch 3104/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 3105/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 3106/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682
Epoch 3107/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 3108/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0653
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Epoch 3109/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 3110/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 3111/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3112/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3113/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 3114/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0650
Epoch 3115/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0650
Epoch 3116/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 3117/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3118/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0712
Epoch 3119/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0702
Epoch 3120/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0660
Epoch 3121/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0667
Epoch 3122/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 3123/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0772
Epoch 3124/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0918
Epoch 3125/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0684
Epoch 3126/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 3127/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0904
Epoch 3128/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0750
Epoch 3129/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0577
Epoch 3130/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1111
Epoch 3131/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0973
Epoch 3132/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
Epoch 3133/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0866
Epoch 3134/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 3135/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0677
Epoch 3136/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0757
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Epoch 3137/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 3138/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 3139/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0691
Epoch 3140/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 3141/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3142/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 3143/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 3144/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0588
Epoch 3145/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0704
Epoch 3146/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0801
Epoch 3147/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0612
Epoch 3148/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0785
Epoch 3149/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0844
Epoch 3150/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0614
Epoch 3151/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0824
Epoch 3152/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0832
Epoch 3153/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0649
Epoch 3154/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0915
Epoch 3155/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0848
Epoch 3156/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0586
Epoch 3157/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0897
Epoch 3158/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0982
Epoch 3159/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0659
Epoch 3160/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0760
Epoch 3161/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0825
Epoch 3162/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0640
Epoch 3163/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 3164/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0723
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Epoch 3165/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0686
Epoch 3166/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0744
Epoch 3167/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 3168/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 3169/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0781
Epoch 3170/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0743
Epoch 3171/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 3172/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 3173/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0744
Epoch 3174/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 3175/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 3176/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0845
Epoch 3177/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 3178/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0614
Epoch 3179/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
Epoch 3180/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0724
Epoch 3181/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3182/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 3183/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 3184/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 3185/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0627
Epoch 3186/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3187/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
Epoch 3188/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0652
Epoch 3189/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0681
Epoch 3190/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3191/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 3192/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0653
```

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Epoch 3193/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0648
Epoch 3194/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0623
Epoch 3195/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0598
Epoch 3196/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0657
Epoch 3197/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0656
Epoch 3198/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0609
Epoch 3199/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3200/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 3201/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3202/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0611
Epoch 3203/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0613
Epoch 3204/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0611
Epoch 3205/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3206/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 3207/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3208/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3209/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0653
Epoch 3210/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3211/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 3212/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0622
Epoch 3213/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0624
Epoch 3214/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 3215/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0630
Epoch 3216/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 3217/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0642
Epoch 3218/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 3219/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0658
Epoch 3220/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0597
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Epoch 3221/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0702
Epoch 3222/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0765
Epoch 3223/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3224/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0770
Epoch 3225/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0662
Epoch 3226/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0750
Epoch 3227/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1025
Epoch 3228/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0756
Epoch 3229/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0680
Epoch 3230/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0981
Epoch 3231/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0842
Epoch 3232/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 3233/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0877
Epoch 3234/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
Epoch 3235/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0698
Epoch 3236/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1096
Epoch 3237/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0861
Epoch 3238/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 3239/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0848
Epoch 3240/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0694
Epoch 3241/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3242/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0812
Epoch 3243/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0836
Epoch 3244/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0630
Epoch 3245/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0696
Epoch 3246/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0709
Epoch 3247/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 3248/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
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Epoch 3249/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0646
Epoch 3250/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3251/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3252/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0630
Epoch 3253/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0596
Epoch 3254/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 3255/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 3256/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0648
Epoch 3257/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0604
Epoch 3258/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0683
Epoch 3259/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3260/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0738
Epoch 3261/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 3262/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 3263/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0660
Epoch 3264/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0643
Epoch 3265/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0727
Epoch 3266/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0604
Epoch 3267/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 3268/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0818
Epoch 3269/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
Epoch 3270/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0713
Epoch 3271/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1066
Epoch 3272/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0944
Epoch 3273/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 3274/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1096
Epoch 3275/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1185
Epoch 3276/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0674
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Epoch 3277/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0919
Epoch 3278/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0842
Epoch 3279/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 3280/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0902
Epoch 3281/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0803
Epoch 3282/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0602
Epoch 3283/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0706
Epoch 3284/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0693
Epoch 3285/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0581
Epoch 3286/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0880
Epoch 3287/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0841
Epoch 3288/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0608
Epoch 3289/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0747
Epoch 3290/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0694
Epoch 3291/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 3292/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0779
Epoch 3293/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0763
Epoch 3294/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3295/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0841
Epoch 3296/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0777
Epoch 3297/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 3298/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0787
Epoch 3299/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0753
Epoch 3300/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 3301/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 3302/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0680
Epoch 3303/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0611
Epoch 3304/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0701
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Epoch 3305/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 3306/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 3307/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 3308/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 3309/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3310/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 3311/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 3312/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 3313/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0613
Epoch 3314/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3315/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
Epoch 3316/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 3317/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3318/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 3319/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682
Epoch 3320/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3321/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3322/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0732
Epoch 3323/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0630
Epoch 3324/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0680
Epoch 3325/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0769
Epoch 3326/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 3327/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 3328/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0660
Epoch 3329/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3330/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3331/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 3332/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
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Epoch 3333/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3334/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0602
Epoch 3335/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3336/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3337/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3338/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3339/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0683
Epoch 3340/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 3341/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3342/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
Epoch 3343/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0681
Epoch 3344/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3345/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0662
Epoch 3346/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3347/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3348/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0640
Epoch 3349/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0661
Epoch 3350/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 3351/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 3352/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 3353/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 3354/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3355/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0597
Epoch 3356/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 3357/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0870
Epoch 3358/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0815
Epoch 3359/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0630
Epoch 3360/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
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Epoch 3361/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 3362/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0644
Epoch 3363/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 3364/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3365/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
Epoch 3366/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 3367/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0608
Epoch 3368/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 3369/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0720
Epoch 3370/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0597
Epoch 3371/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 3372/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 3373/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 3374/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0677
Epoch 3375/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 3376/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0641
Epoch 3377/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0625
Epoch 3378/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0662
Epoch 3379/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0647
Epoch 3380/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3381/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
Epoch 3382/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3383/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0618
Epoch 3384/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3385/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 3386/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 3387/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3388/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
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Epoch 3389/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671
Epoch 3390/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 3391/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
Epoch 3392/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0597
Epoch 3393/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3394/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 3395/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3396/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3397/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3398/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0603
Epoch 3399/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0608
Epoch 3400/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3401/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
Epoch 3402/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0605
Epoch 3403/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 3404/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 3405/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 3406/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 3407/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0706
Epoch 3408/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0594
Epoch 3409/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0797
Epoch 3410/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0912
Epoch 3411/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0708
Epoch 3412/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0778
Epoch 3413/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0816
Epoch 3414/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 3415/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0681
Epoch 3416/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
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Epoch 3417/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 3418/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0813
Epoch 3419/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 3420/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 3421/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0776
Epoch 3422/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0762
Epoch 3423/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3424/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 3425/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0646
Epoch 3426/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0733
Epoch 3427/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0882
Epoch 3428/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0707
Epoch 3429/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 3430/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0653
Epoch 3431/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0596
Epoch 3432/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0674
Epoch 3433/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0723
Epoch 3434/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3435/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0648
Epoch 3436/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0822
Epoch 3437/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0674
Epoch 3438/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 3439/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0711
Epoch 3440/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
Epoch 3441/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
Epoch 3442/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 3443/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0611
Epoch 3444/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
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Epoch 3445/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3446/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 3447/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0805
Epoch 3448/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 3449/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 3450/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0994
Epoch 3451/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0866
Epoch 3452/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0721
Epoch 3453/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0698
Epoch 3454/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0658
Epoch 3455/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 3456/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0596
Epoch 3457/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0599
Epoch 3458/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3459/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0697
Epoch 3460/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0650
Epoch 3461/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 3462/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3463/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0625
Epoch 3464/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0614
Epoch 3465/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 3466/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0618
Epoch 3467/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0692
Epoch 3468/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0602
Epoch 3469/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 3470/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0731
Epoch 3471/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 3472/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0676
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Epoch 3473/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 3474/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
Epoch 3475/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3476/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0717
Epoch 3477/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 3478/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0585
Epoch 3479/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 3480/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0709
Epoch 3481/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3482/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0723
Epoch 3483/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0764
Epoch 3484/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671
Epoch 3485/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 3486/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0629
Epoch 3487/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0615
Epoch 3488/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3489/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0622
Epoch 3490/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0618
Epoch 3491/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0627
Epoch 3492/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0631
Epoch 3493/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0630
Epoch 3494/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3495/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0602
Epoch 3496/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0589
Epoch 3497/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0597
Epoch 3498/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0656
Epoch 3499/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682
Epoch 3500/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
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Epoch 3501/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3502/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3503/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0602
Epoch 3504/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0596
Epoch 3505/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0595
Epoch 3506/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0598
Epoch 3507/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
Epoch 3508/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0594
Epoch 3509/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0591
Epoch 3510/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0590
Epoch 3511/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0595
Epoch 3512/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0622
Epoch 3513/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 3514/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0615
Epoch 3515/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 3516/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 3517/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3518/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3519/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3520/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0594
Epoch 3521/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 3522/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3523/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 3524/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0653
Epoch 3525/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0588
Epoch 3526/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 3527/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 3528/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0666
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Epoch 3529/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 3530/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0634
Epoch 3531/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3532/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0602
Epoch 3533/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
Epoch 3534/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 3535/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0588
Epoch 3536/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0688
Epoch 3537/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0862
Epoch 3538/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0680
Epoch 3539/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 3540/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0746
Epoch 3541/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0624
Epoch 3542/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3543/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0726
Epoch 3544/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 3545/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0603
Epoch 3546/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 3547/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0596
Epoch 3548/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 3549/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 3550/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3551/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 3552/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
Epoch 3553/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0618
Epoch 3554/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 3555/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0649
Epoch 3556/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
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Epoch 3557/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3558/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3559/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 3560/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 3561/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3562/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3563/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0671
Epoch 3564/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0684
Epoch 3565/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
Epoch 3566/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3567/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0761
Epoch 3568/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0749
Epoch 3569/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0631
Epoch 3570/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0717
Epoch 3571/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 3572/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 3573/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0771
Epoch 3574/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 3575/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 3576/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0817
Epoch 3577/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 3578/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0614
Epoch 3579/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 3580/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3581/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
Epoch 3582/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0592
Epoch 3583/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0599
Epoch 3584/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
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Epoch 3585/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0590
Epoch 3586/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0710
Epoch 3587/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0696
Epoch 3588/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 3589/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0824
Epoch 3590/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0706
Epoch 3591/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 3592/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0682
Epoch 3593/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3594/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
Epoch 3595/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 3596/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0699
Epoch 3597/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0634
Epoch 3598/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
Epoch 3599/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 3600/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 3601/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 3602/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0618
Epoch 3603/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0608
Epoch 3604/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0627
Epoch 3605/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0612
Epoch 3606/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3607/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 3608/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 3609/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0657
Epoch 3610/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0652
Epoch 3611/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3612/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
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Epoch 3613/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0597
Epoch 3614/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0590
Epoch 3615/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3616/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 3617/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0597
Epoch 3618/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0602
Epoch 3619/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0594
Epoch 3620/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0651
Epoch 3621/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0686
Epoch 3622/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0627
Epoch 3623/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0581
Epoch 3624/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 3625/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 3746/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0736
Epoch 3747/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0710
Epoch 3748/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0586
Epoch 3749/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0652
Epoch 3750/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0754
Epoch 3751/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 3752/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0684
Epoch 3753/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0688
Epoch 3754/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0568
Epoch 3755/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0766
Epoch 3756/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0781
Epoch 3757/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step - loss: 0.0586
Epoch 3758/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0725
Epoch 3759/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0696
Epoch 3760/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0578
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Epoch 3761/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0769
Epoch 3762/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0640
Epoch 3763/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 3764/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0831
Epoch 3765/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0639
Epoch 3766/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0645
Epoch 3767/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0726
Epoch 3768/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3769/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0590
Epoch 3770/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3771/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3772/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 3773/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0594
Epoch 3774/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0599
Epoch 3775/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3776/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 3777/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0572
Epoch 3778/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 3779/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 3780/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3781/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 3782/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0652
Epoch 3783/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0584
Epoch 3784/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0775
Epoch 3785/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0664
Epoch 3786/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0870
Epoch 3787/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0780
Epoch 3788/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0658
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Epoch 3789/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0722
Epoch 3790/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 3791/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0665
Epoch 3792/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0684
Epoch 3793/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0740
Epoch 3794/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0633
Epoch 3795/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0584
Epoch 3796/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 3797/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0719
Epoch 3798/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0587
Epoch 3799/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 3800/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0585
Epoch 3801/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
Epoch 3802/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3803/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 3804/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0631
Epoch 3805/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 3806/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0653
Epoch 3807/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0588
Epoch 3808/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 3809/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0738
Epoch 3810/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0672
Epoch 3811/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0641
Epoch 3812/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0587
Epoch 3813/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0740
Epoch 3814/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0719
Epoch 3815/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0644
Epoch 3816/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0854
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Epoch 3817/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0625
Epoch 3818/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 3819/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0776
Epoch 3820/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0623
Epoch 3821/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0696
Epoch 3822/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0614
Epoch 3823/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0624
Epoch 3824/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0599
Epoch 3825/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3826/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0594
Epoch 3827/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0666
Epoch 3828/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0622
Epoch 3829/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3830/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 3831/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0630
Epoch 3832/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0614
Epoch 3833/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0647
Epoch 3834/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0644
Epoch 3835/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0697
Epoch 3836/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3837/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0580
Epoch 3838/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0631
Epoch 3839/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0624
Epoch 3840/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0580
Epoch 3841/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0584
Epoch 3842/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0583
Epoch 3843/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 3844/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
```

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Epoch 3845/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0584
Epoch 3846/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 3847/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
Epoch 3848/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 3849/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0705
Epoch 3850/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0602
Epoch 3851/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0705
Epoch 3852/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0684
Epoch 3853/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0572
Epoch 3854/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0716
Epoch 3855/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 3856/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679
Epoch 3857/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0964
Epoch 3858/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0767
Epoch 3859/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0591
Epoch 3860/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0662
Epoch 3861/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0660
Epoch 3862/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0592
Epoch 3863/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0621
Epoch 3864/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 3865/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0587
Epoch 3866/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0702
Epoch 3867/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0728
Epoch 3868/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0584
Epoch 3869/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0644
Epoch 3870/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0675
Epoch 3871/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0587
Epoch 3872/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0696
```

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Epoch 3873/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 3874/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 3875/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0672
Epoch 3876/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0665
Epoch 3877/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0589
Epoch 3878/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3879/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0655
Epoch 3880/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0626
Epoch 3881/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 3882/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0928
Epoch 3883/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0752
Epoch 3884/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0647
Epoch 3885/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0825
Epoch 3886/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 3887/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0685
Epoch 3888/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.1146
Epoch 3889/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0702
Epoch 3890/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0871
Epoch 3891/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0954
Epoch 3892/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 3893/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0979
Epoch 3894/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0920
Epoch 3895/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0547
Epoch 3896/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0845
Epoch 3897/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0995
Epoch 3898/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0604
Epoch 3899/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0769
Epoch 3900/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0966
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Epoch 3901/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0619
Epoch 3902/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0859
Epoch 3903/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0967
Epoch 3904/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0773
Epoch 3905/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 3906/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0585
Epoch 3907/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0625
Epoch 3908/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0687
Epoch 3909/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0624
Epoch 3910/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0589
Epoch 3911/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0582
Epoch 3912/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0587
Epoch 3913/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0582
Epoch 3914/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0578
Epoch 3915/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0602
Epoch 3916/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0597
Epoch 3917/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0580
Epoch 3918/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0597
Epoch 3919/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 3920/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0593
Epoch 3921/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0581
Epoch 3922/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0603
Epoch 3923/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 3924/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0589
Epoch 3925/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0589
Epoch 3926/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0591
Epoch 3927/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 3928/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0690
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Epoch 3929/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 3930/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0792
Epoch 3931/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 3932/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3933/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 3934/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 3935/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0637
Epoch 3936/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0641
Epoch 3937/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0686
Epoch 3938/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0613
Epoch 3939/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0654
Epoch 3940/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0737
Epoch 3941/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0677
Epoch 3942/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0594
Epoch 3943/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0591
Epoch 3944/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0578
Epoch 3945/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 3946/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 3947/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 3948/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 3949/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0601
Epoch 3950/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0674
Epoch 3951/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 3952/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0601
Epoch 3953/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0587
Epoch 3954/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0612
Epoch 3955/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
Epoch 3956/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0592
```

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Epoch 3957/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0581
Epoch 3958/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0599
Epoch 3959/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0581
Epoch 3960/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0616
Epoch 3961/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0646
Epoch 3962/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0557
Epoch 3963/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0751
Epoch 3964/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0825
Epoch 3965/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0573
Epoch 3966/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1013
Epoch 3967/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0901
Epoch 3968/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 3969/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.1229
Epoch 3970/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0900
Epoch 3971/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0649
Epoch 3972/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0953
Epoch 3973/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0757
Epoch 3974/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0619
Epoch 3975/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 3976/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0603
Epoch 3977/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0613
Epoch 3978/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0632
Epoch 3979/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0607
Epoch 3980/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0623
Epoch 3981/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0609
Epoch 3982/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0618
Epoch 3983/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0609
Epoch 3984/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0589
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Epoch 3985/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0586
Epoch 3986/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0586
Epoch 3987/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0639
Epoch 3988/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0646
Epoch 3989/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 3990/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 3991/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0583
Epoch 3992/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0592
Epoch 3993/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0602
Epoch 3994/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 3995/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0692
Epoch 3996/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0562
Epoch 3997/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0719
Epoch 3998/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0800
Epoch 3999/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0636
Epoch 4000/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0678
Epoch 4001/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0648
Epoch 4002/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 4003/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 4004/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0586
Epoch 4005/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0603
Epoch 4006/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0679
Epoch 4007/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 4008/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0635
Epoch 4009/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0618
Epoch 4010/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 4011/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 4012/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
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Epoch 4013/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 4014/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0885
Epoch 4015/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0892
Epoch 4016/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0603
Epoch 4017/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0748
Epoch 4018/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0837
Epoch 4019/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0677
Epoch 4020/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0586
Epoch 4021/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0571
Epoch 4022/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0670
Epoch 4023/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0589
Epoch 4024/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0682
Epoch 4025/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0828
Epoch 4026/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0629
Epoch 4027/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0648
Epoch 4028/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0657
Epoch 4029/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0592
Epoch 4030/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0771
Epoch 4031/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0694
Epoch 4032/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0589
Epoch 4033/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 4034/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0582
Epoch 4035/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 4036/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0583
Epoch 4037/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0630
Epoch 4038/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0684
Epoch 4039/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0591
Epoch 4040/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0635
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Epoch 4041/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0667
Epoch 4042/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0619
Epoch 4043/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0900
Epoch 4044/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0733
Epoch 4045/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0588
Epoch 4046/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0777
Epoch 4047/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0709
Epoch 4048/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0703
Epoch 4049/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0715
Epoch 4050/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0559
Epoch 4051/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 4052/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0703
Epoch 4053/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0595
Epoch 4054/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0862
Epoch 4055/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0715
Epoch 4056/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0609
Epoch 4057/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0874
Epoch 4058/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0770
Epoch 4059/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0609
Epoch 4060/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0747
Epoch 4061/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0753
Epoch 4062/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
Epoch 4063/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0753
Epoch 4064/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0699
Epoch 4065/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 4066/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0582
Epoch 4067/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0666
Epoch 4068/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0616
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Epoch 4069/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0708
Epoch 4070/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0638
Epoch 4071/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0606
Epoch 4072/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0666
Epoch 4073/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0668
Epoch 4074/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0620
Epoch 4075/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0572
Epoch 4076/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0641
Epoch 4077/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0649
Epoch 4078/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0581
Epoch 4079/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0619
Epoch 4080/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0630
Epoch 4081/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0592
Epoch 4082/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0596
Epoch 4083/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0591
Epoch 4084/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0594
Epoch 4085/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 4086/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0572
Epoch 4087/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0564
Epoch 4088/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 4089/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0661
Epoch 4090/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0640
Epoch 4091/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0599
Epoch 4092/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0596
Epoch 4093/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0595
Epoch 4094/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0574
Epoch 4095/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 4096/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
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Epoch 4097/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0600
Epoch 4098/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0593
Epoch 4099/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0603
Epoch 4100/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0605
Epoch 4101/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0637
Epoch 4102/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0586
Epoch 4103/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0587
Epoch 4104/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0595
Epoch 4105/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0588
Epoch 4106/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0584
Epoch 4107/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0560
Epoch 4108/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0622
Epoch 4109/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0748
Epoch 4110/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0749
Epoch 4111/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0681
Epoch 4112/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0743
Epoch 4113/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0645
Epoch 4114/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0625
Epoch 4115/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0798
Epoch 4116/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0703
Epoch 4117/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0581
Epoch 4118/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0611
Epoch 4119/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0608
Epoch 4120/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0583
Epoch 4121/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0562
Epoch 4122/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0653
Epoch 4123/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0653
Epoch 4124/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0604
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Epoch 4125/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0700
Epoch 4126/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0712
Epoch 4127/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0597
Epoch 4128/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0624
Epoch 4129/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0630
Epoch 4130/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0567
Epoch 4131/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0594
Epoch 4132/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0689
Epoch 4133/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0708
Epoch 4134/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0583
Epoch 4135/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0615
Epoch 4136/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 4137/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0601
Epoch 4138/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0575
Epoch 4139/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0626
Epoch 4140/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0611
Epoch 4141/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0579
Epoch 4142/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0581
Epoch 4143/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0616
Epoch 4144/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0627
Epoch 4145/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0559
Epoch 4146/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0744
Epoch 4147/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 4148/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0609
Epoch 4149/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0717
Epoch 4150/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0612
Epoch 4151/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0621
Epoch 4152/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
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Epoch 4153/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0621
Epoch 4154/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0571
Epoch 4155/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0612
Epoch 4156/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0597
Epoch 4157/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0582
Epoch 4158/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0586
Epoch 4159/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0572
Epoch 4160/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0587
Epoch 4161/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0597
Epoch 4162/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0594
Epoch 4163/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0563
Epoch 4164/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0606
Epoch 4165/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0654
Epoch 4166/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0604
Epoch 4167/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0609
Epoch 4168/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 4169/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0698
Epoch 4170/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 4171/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0604
Epoch 4172/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 4173/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0574
Epoch 4174/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0607
Epoch 4175/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0687
Epoch 4176/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0625
Epoch 4177/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0577
Epoch 4178/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0582
Epoch 4179/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0589
Epoch 4180/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0584
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Epoch 4181/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0579
Epoch 4182/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0600
Epoch 4183/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0590
Epoch 4184/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0598
Epoch 4185/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0551
Epoch 4186/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0762
Epoch 4187/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0689
Epoch 4188/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0612
Epoch 4189/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0675
Epoch 4190/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0565
Epoch 4191/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0655
Epoch 4192/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0818
Epoch 4193/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0585
Epoch 4194/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0757
Epoch 4195/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0743
Epoch 4196/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0589
Epoch 4197/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0792
Epoch 4198/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0774
Epoch 4199/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0592
Epoch 4200/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0600
Epoch 4201/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0660
Epoch 4202/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0590
Epoch 4203/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0659
Epoch 4204/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0679
Epoch 4205/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0603
Epoch 4206/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0621
Epoch 4207/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0612
Epoch 4208/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0594
```

```
Epoch 4209/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0579
Epoch 4210/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0566
Epoch 4211/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0568
Epoch 4212/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0590
Epoch 4213/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0574
Epoch 4214/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0578
Epoch 4215/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0607
Epoch 4216/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0600
Epoch 4217/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0599
Epoch 4218/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0600
Epoch 4219/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0596
Epoch 4220/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0631
Epoch 4221/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0587
Epoch 4222/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 4223/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0646
Epoch 4224/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0568
Epoch 4225/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0635
Epoch 4226/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0687
Epoch 4227/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0636
Epoch 4228/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0659
Epoch 4229/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0643
Epoch 4230/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0586
Epoch 4231/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0633
Epoch 4232/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0638
Epoch 4233/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 4234/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0617
Epoch 4235/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0599
Epoch 4236/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0615
```

```
Epoch 4237/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0629
Epoch 4238/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0563
Epoch 4239/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0686
Epoch 4240/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0748
Epoch 4241/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0557
Epoch 4242/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0812
Epoch 4243/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0799
Epoch 4244/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0581
Epoch 4245/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0608
Epoch 4246/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0623
Epoch 4247/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0618
Epoch 4248/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0622
Epoch 4249/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0591
Epoch 4250/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0583
Epoch 4251/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0718
Epoch 4252/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0714
Epoch 4253/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0616
Epoch 4254/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0582
Epoch 4255/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0628
Epoch 4256/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0601
Epoch 4257/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0585
Epoch 4258/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 21ms/step - loss: 0.0581
Epoch 4259/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0566
Epoch 4260/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0565
Epoch 4261/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0557
Epoch 4262/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0642
Epoch 4263/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0691
Epoch 4264/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0568
```

```
Epoch 4265/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 25ms/step - loss: 0.0684
Epoch 4266/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0707
Epoch 4267/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0610
Epoch 4268/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0722
Epoch 4269/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0713
Epoch 4270/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0616
Epoch 4271/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 4272/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0581
Epoch 4273/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0606
Epoch 4274/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0714
Epoch 4275/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0659
Epoch 4276/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0556
Epoch 4277/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0671
Epoch 4278/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0650
Epoch 4279/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0588
Epoch 4280/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0735
Epoch 4281/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0668
Epoch 4282/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 20ms/step - loss: 0.0577
Epoch 4283/5000
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 19ms/step - loss: 0.0577
Epoch 4284/5000
```

In [10]:
```
predictions = model.predict(data_inputs)



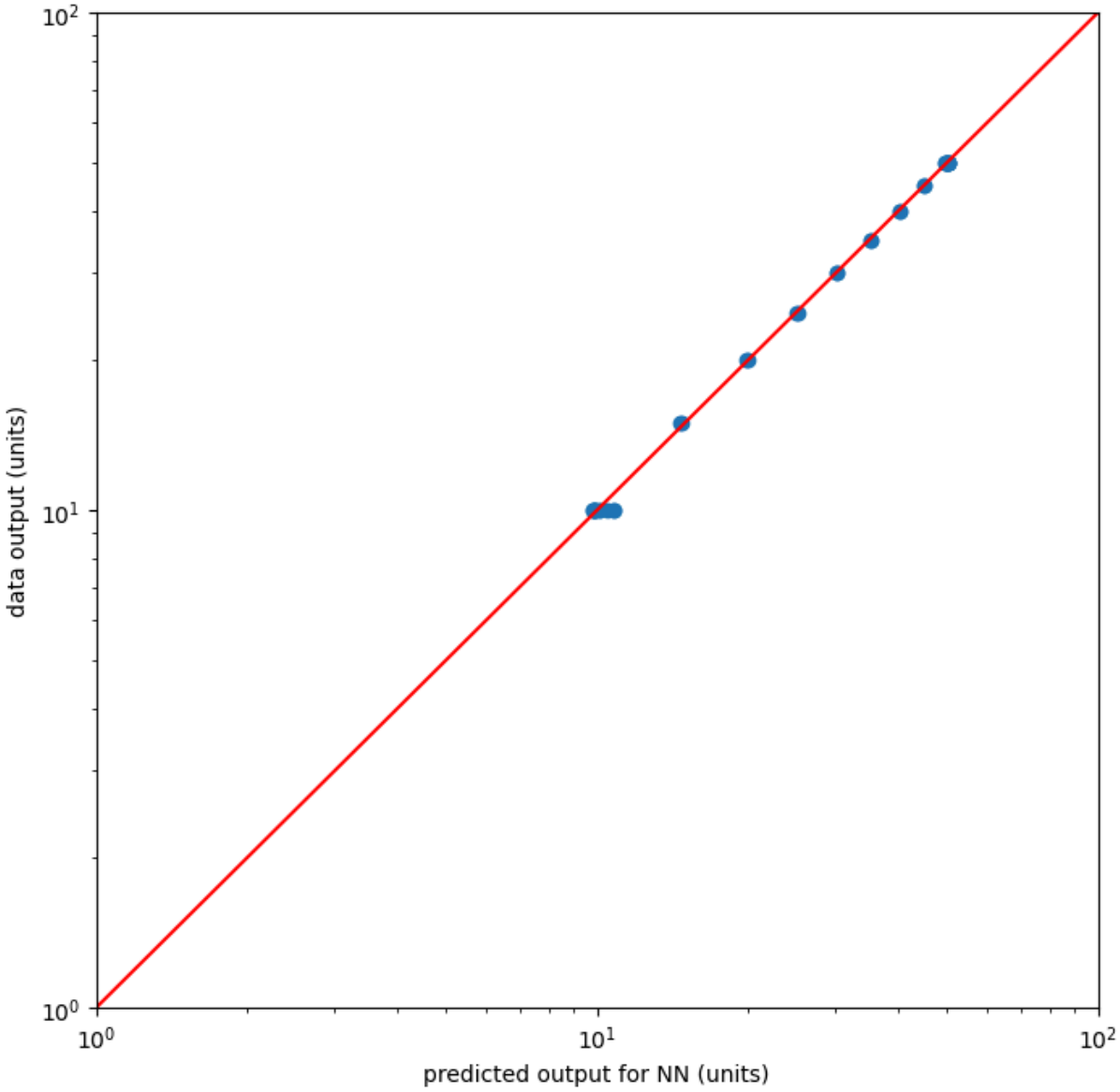
#SETTING UP PLOT
# matplotlib inline
# importing the required module
import matplotlib.pyplot as plt
plt.rcParams['figure.figsize'] = [8, 8] # for square canvas
#========

'''CALCULATE PREDICTED VALUES AND RETRIEVE DATA VALUES TO PLOT'''

plt.scatter(predictions[:, 0], data_outputs[:, 0])
#plt.title('Genetic Algorithm training of Neural Network ==> output = alpha'
plt.xlabel('predicted output for NN (units)')
```

```
plt.ylabel('data output (units)')
plt.loglog()
plt.xlim(xmax = 100, xmin = 1.)
plt.ylim(ymax = 100, ymin = 1.)
# Generate red y=x line
x_data = np.linspace(1.0, 100.0, num=3)
y_data = x_data
plt.plot(x_data, y_data, color='red')
plt.savefig("log-log_plot_3.4_hexagonal_shape.png", dpi=300, bbox_inches='ti
plt.show()
```

```
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step
```



In [11]:
```
test = []
outpt=[]
predictions = model.predict(data_inputs)

#first point (row [0])comparison of data and prediction
```

```
# put in a loop to print comparion for all data points

xa=[]
ya=[]
za=[]

xao = np.linspace(0, 3, 7)  # 1D array for x values
yao = np.linspace(0, 5, 11)    # 1D array for y values
xa = np.linspace(0, 10, 77)    # 1D array for x values
ya = np.linspace(0, 10, 77)    # 1D array for y values
za = np.linspace(0, 10, 77)    # 1D array for z values

x =-0.50
ia=-1
while (x < 2.51):
    x = x + 0.5
    y = -0.5
    while (y < 4.51):
        ia=ia+1
        y = y + 0.50
        xa[ia]=x
        ya[ia]=y

        test = [[ x , y ]]
        testarray = np.array(test)
        outpt = model.predict(testarray)
        za[ia] = outpt
        print ('x= ', testarray[0][0], ', y= ', testarray[0][1],'T= ', outpt



X, Y, Z = xa, ya, za

import sys
#import csv
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import axes3d

# Plot X,Y,Z

# Creating figure
#fig = plt.figure(figsize =(16, 9))
#ax = plt.axes(projection ='3d')

fig = plt.figure()
#ax = fig.add_subplot(111, projection='3d')
ax = plt.axes(projection ='3d')
#ax.plot_trisurf(X, Y, Z, color='white', edgecolors='grey', alpha=0.5)
ax.plot_trisurf(X, Y, Z,
                linewidth = 0.2,
                antialiased = True
                , cmap='viridis');


surf = ax.plot_trisurf(
```

```
    X, Y, Z,
    linewidth=0.2,
    antialiased=True,
    cmap='viridis'
)


#  ADD COLORBAR
fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)



# fig.colorbar(surf, ax=ax, shrink=0.6, aspect=10)

ax.invert_yaxis()
ax.invert_xaxis()

plt.savefig("trisurf_plot_hexagonal_shape.png", dpi=300, bbox_inches='tight'
plt.show()



# Plot the data
plt.figure(figsize=(5, 6))
plt.tripcolor(xa, ya, za, shading='flat', cmap='viridis')
plt.savefig("Heat_map_hexagonal_shape.png", dpi=300, bbox_inches='tight')
plt.colorbar(label='Z value')
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Heat Map of Z values over X and Y domain')
plt.show()
```

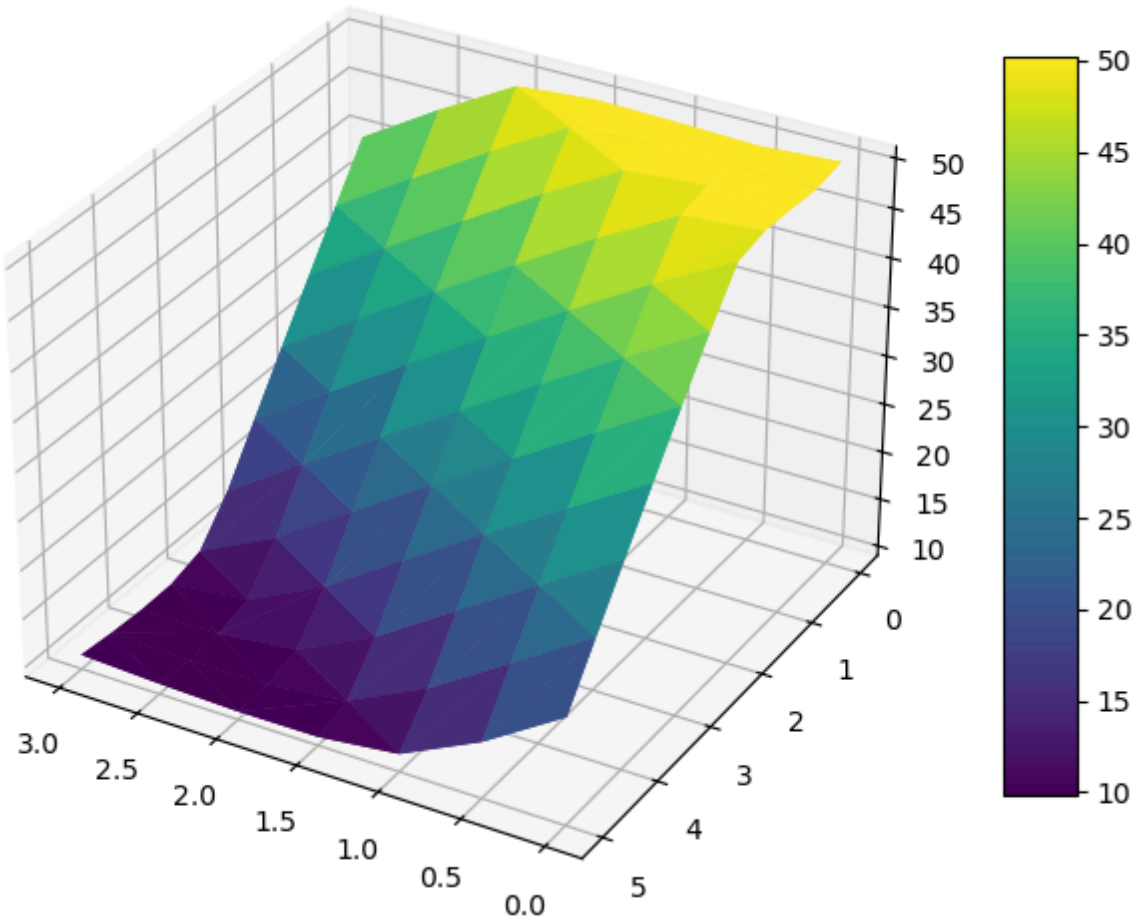
```
2/2 ━━━━━━━━━━━━━━━━━━━━ 0s 18ms/step
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
x=  0.0 , y=  0.0 T=  50.311737
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
x=  0.0 , y=  0.5 T=  49.801617
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.0 , y=  1.0 T=  50.061436
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
```

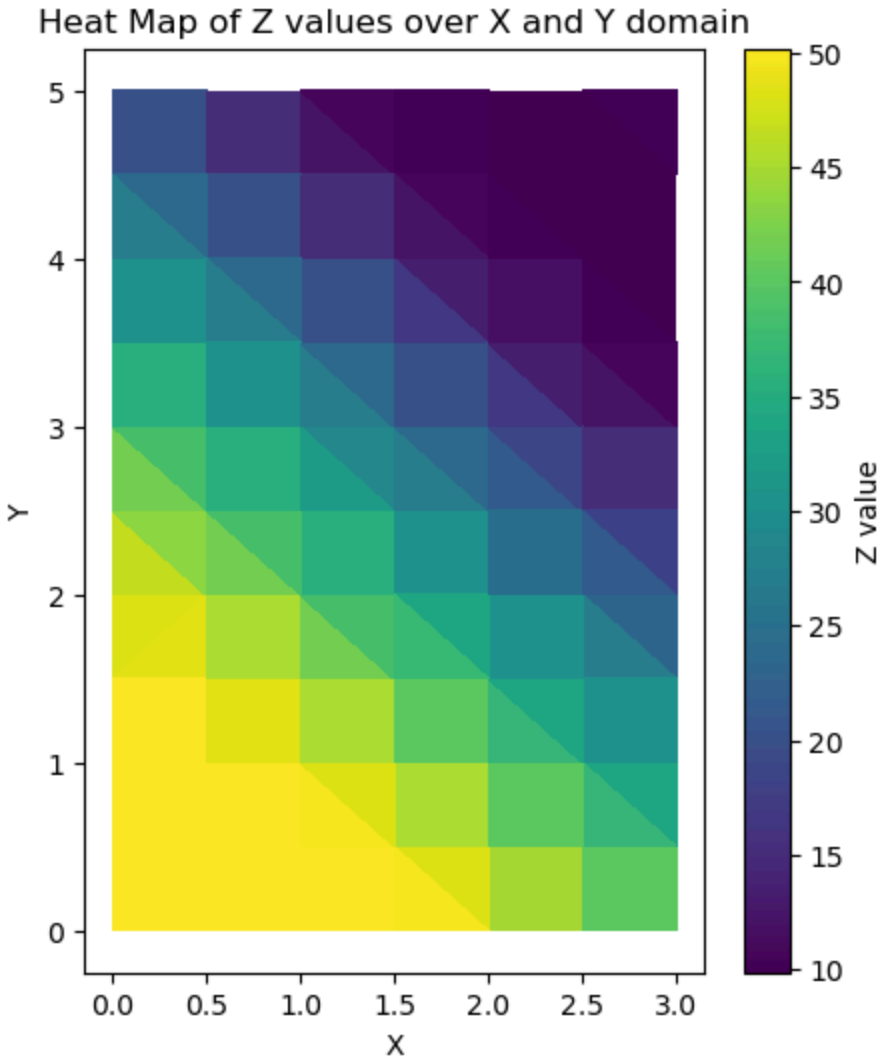
```
/tmp/ipykernel_2771072/3841726189.py:32: DeprecationWarning: Conversion of a
n array with ndim > 0 to a scalar is deprecated, and will error in future. E
nsure you extract a single element from your array before performing this op
eration. (Deprecated NumPy 1.25.)
  za[ia] = outpt
```

```
x=  0.0 , y=  1.5 T=  50.22152
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  2.0 T=  49.554394
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.0 , y=  2.5 T=  45.0829
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.0 , y=  3.0 T=  40.21741
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
x=  0.0 , y=  3.5 T=  35.25771
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  4.0 T=  30.220037
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.0 , y=  4.5 T=  25.117563
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step
x=  0.0 , y=  5.0 T=  19.96097
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  0.0 T=  49.79185
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  0.5 T=  50.072014
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  1.0 T=  50.24686
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  1.5 T=  49.587837
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  2.0 T=  45.11816
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  2.5 T=  40.252293
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  3.0 T=  35.290012
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  3.5 T=  30.248232
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  4.0 T=  25.140593
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  0.5 , y=  4.5 T=  19.978182
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  0.5 , y=  5.0 T=  14.769966
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  0.0 T=  50.05682
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  0.5 T=  50.249855
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  1.0 T=  49.6014
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  1.5 T=  45.1356
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  2.0 T=  40.271477
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.0 , y=  2.5 T=  35.308456
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.0 , y=  3.0 T=  30.26412
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  3.5 T=  25.152683
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  4.0 T=  19.985605
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
```

```
x=  1.0 , y=  4.5 T=  14.772215
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.0 , y=  5.0 T=  10.823002
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  1.5 , y=  0.0 T=  50.230892
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.5 , y=  0.5 T=  49.59552
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  1.0 T=  45.135704
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.5 , y=  1.5 T=  40.275455
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  2.0 T=  35.31349
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  2.5 T=  30.268152
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  3.0 T=  25.154242
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  3.5 T=  19.983673
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  1.5 , y=  4.0 T=  14.766106
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 32ms/step
x=  1.5 , y=  4.5 T=  10.824936
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  1.5 , y=  5.0 T=  9.879528
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  0.0 T=  49.57058
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  0.5 T=  45.118885
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  1.0 T=  40.26464
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  1.5 T=  35.305534
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  2.0 T=  30.260735
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  2.0 , y=  2.5 T=  25.145658
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  3.0 T=  19.972733
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  3.5 T=  14.751966
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  4.0 T=  10.823551
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  4.5 T=  9.889149
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.0 , y=  5.0 T=  9.831032
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  0.0 T=  45.085514
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  0.5 T=  40.23939
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  1.0 T=  35.284943
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  1.5 T=  30.242203
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
```

```
x=  2.5 , y=  2.0 T=  25.12729
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  2.5 T=  19.953117
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  3.0 T=  14.73012
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.5 , y=  3.5 T=  10.8190155
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  4.0 T=  9.896992
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 31ms/step
x=  2.5 , y=  4.5 T=  9.845612
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  2.5 , y=  5.0 T=  10.0781975
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  0.0 T=  40.200027
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  0.5 T=  35.252045
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  1.0 T=  30.212904
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  1.5 T=  25.099426
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  2.0 T=  19.925121
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
x=  3.0 , y=  2.5 T=  14.700851
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  3.0 T=  10.811504
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  3.5 T=  9.903164
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  4.0 T=  9.858688
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  4.5 T=  10.096329
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step
x=  3.0 , y=  5.0 T=  10.473217
```


50
45
40
35
30
25
20
15
10
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
1
2
3
4
5
50
45
40
35
30
25
20
15
10


Heat Map of Z values over X and Y domain
Y
X
Z value
0
1
2
3
4
5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
10
15
20
25
30
35
40
45
50

In [ ]:

In [ ]: